

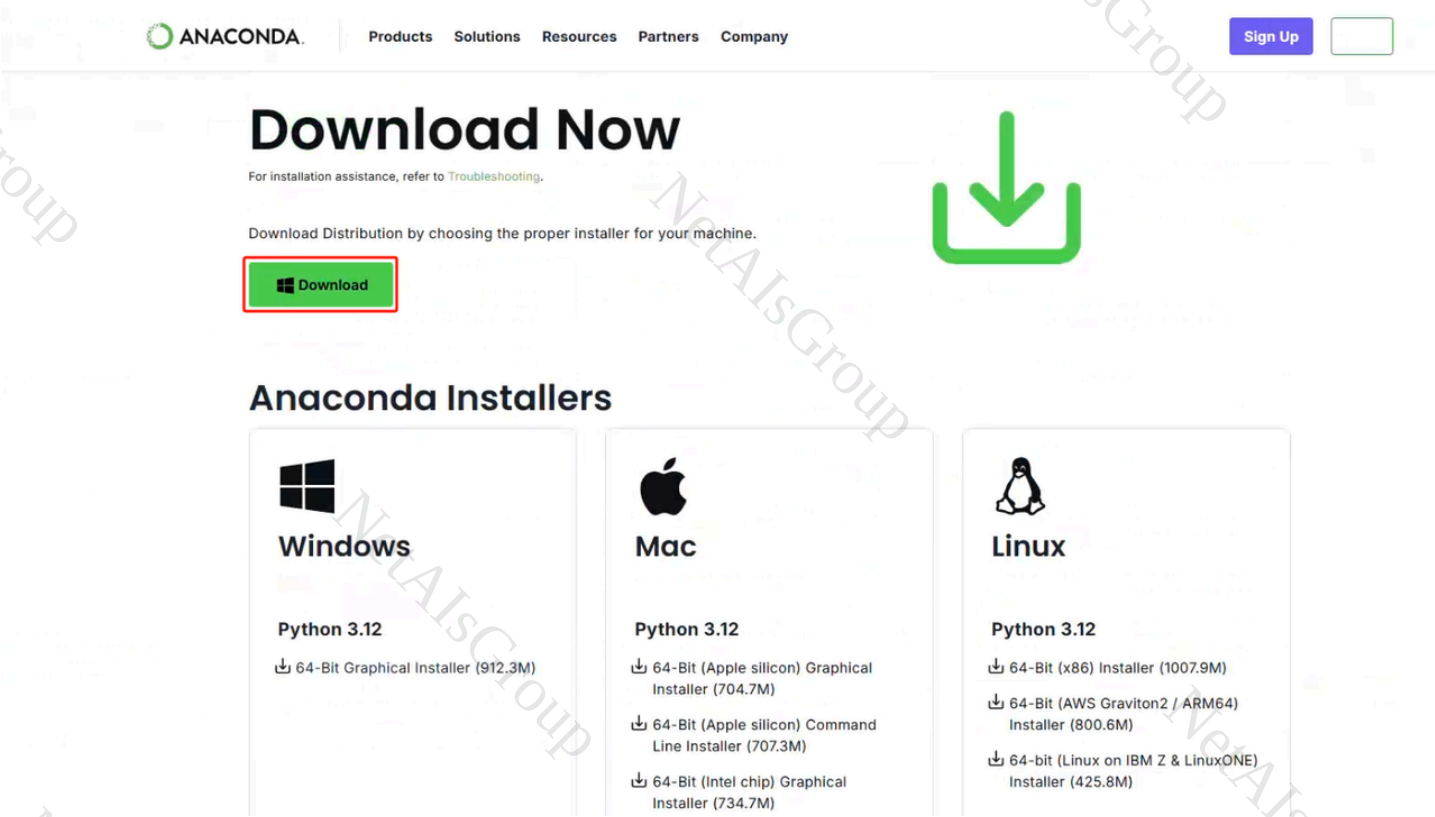
你长大了，该自学成才了

注意：本文的可以随时修改，如果你发现了文档中的某些错误，还请直接反馈。

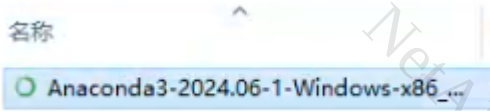
一、Conda（本地）

提示：如果你已经会自己创建在本地的虚拟环境可以移步至第三节。

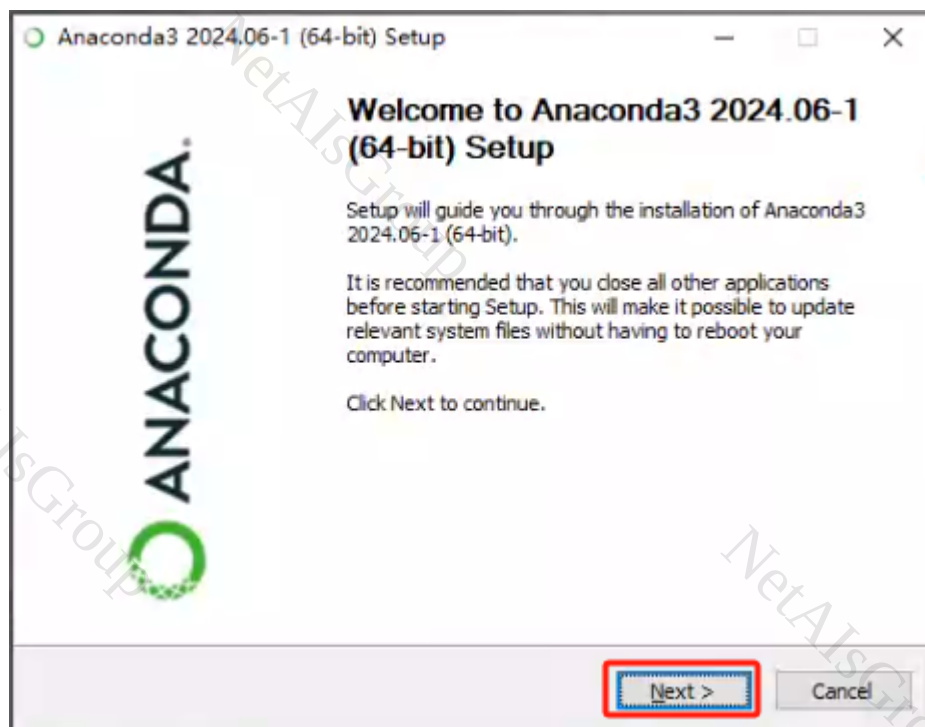
1. 进入<https://www.anaconda.com/download/success>，下载对应操作系统的Anaconda。



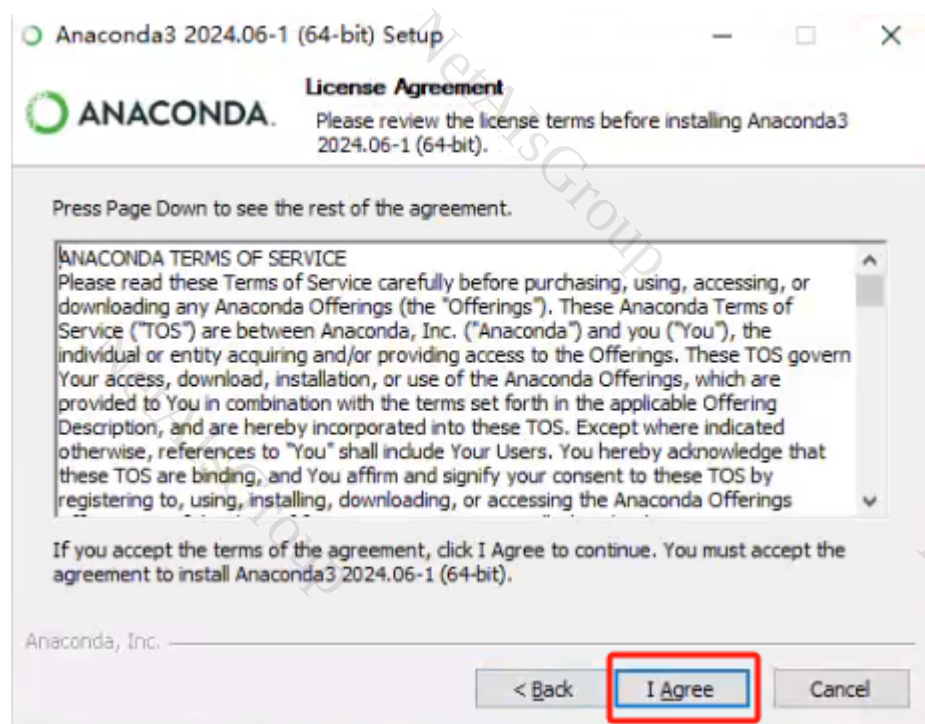
2. 双击安装包。



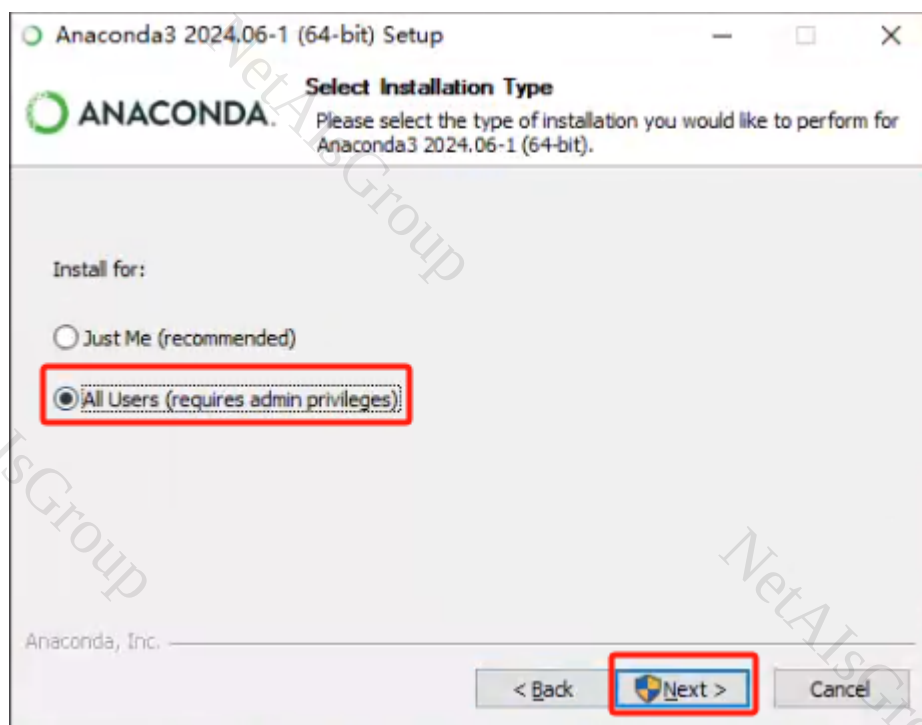
3. 出现如下界面，点击Next。



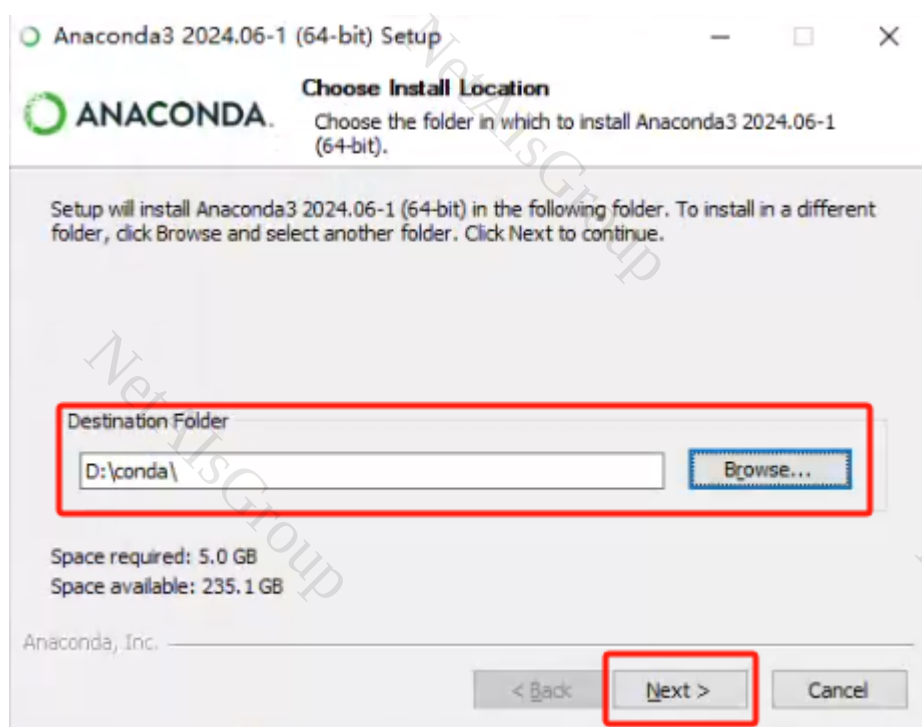
4. 点击I Agree。



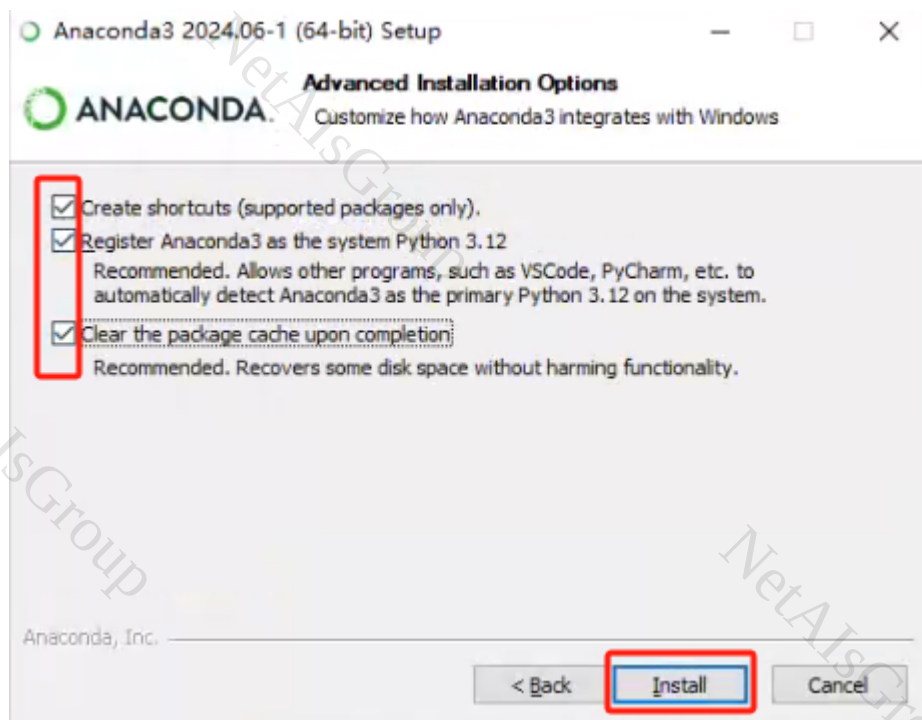
5. 选择 All Users，然后点击Next。



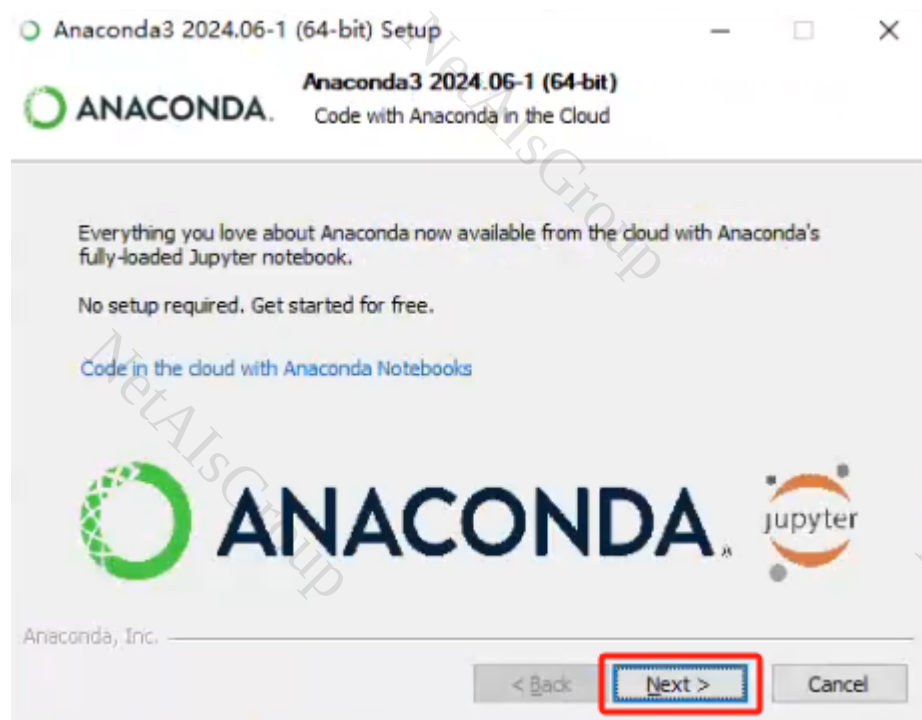
6. 选择合适的安装路径（不要包含中文路径，不建议安装在C盘），点击Next。



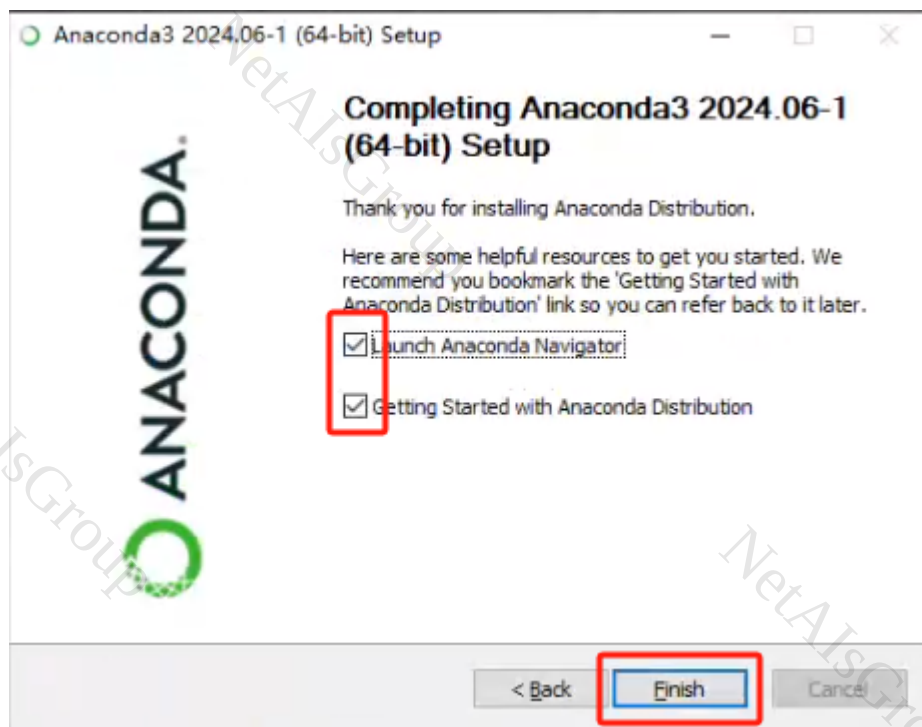
7. 全选，点击安装。



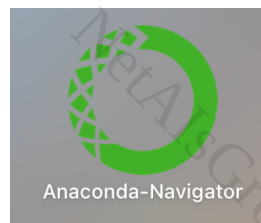
8. 点击next。



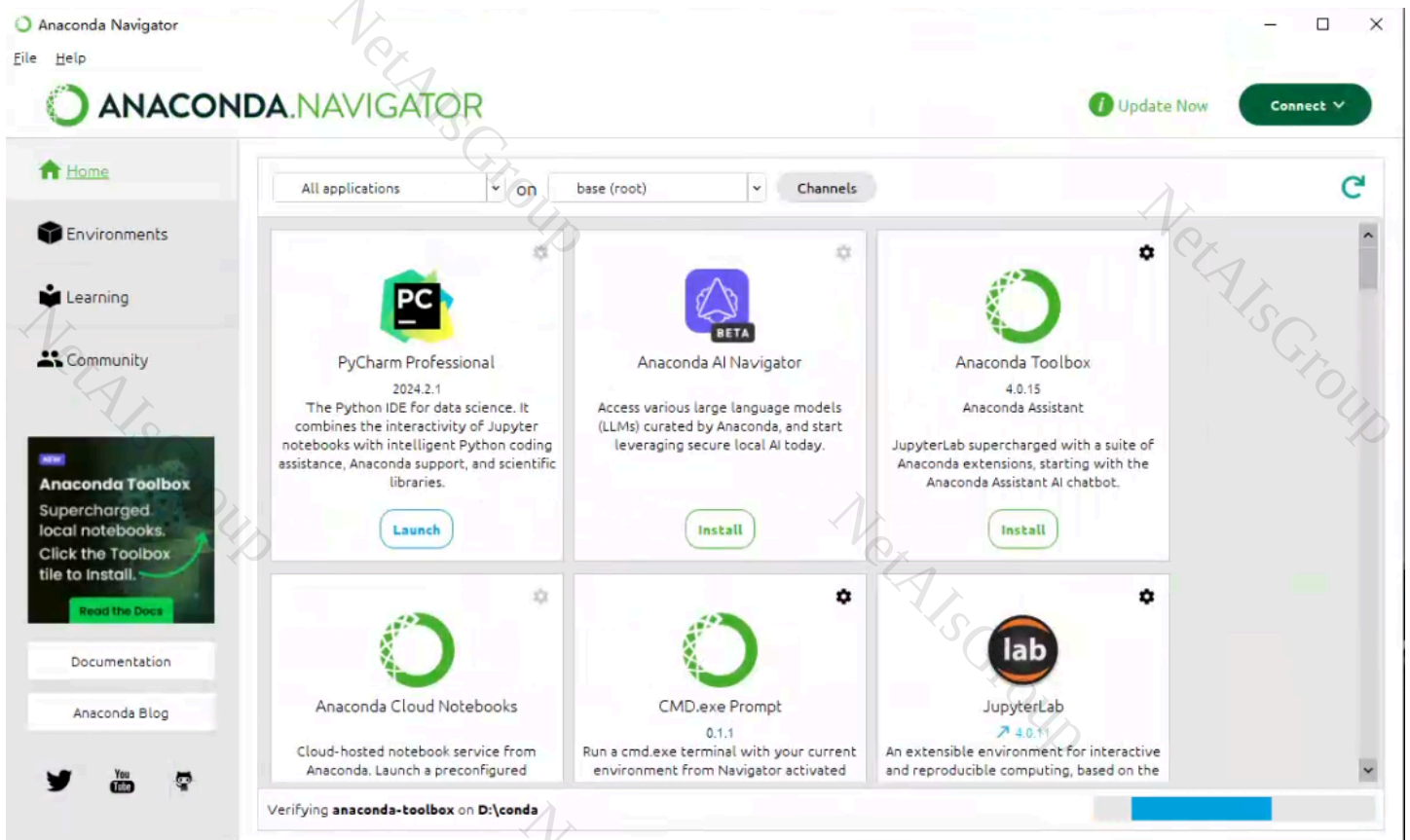
9. 全选，Finish。



10. 下载完成后，打开启动器如下：



11. 安装完会自动打开。



配置环境变量

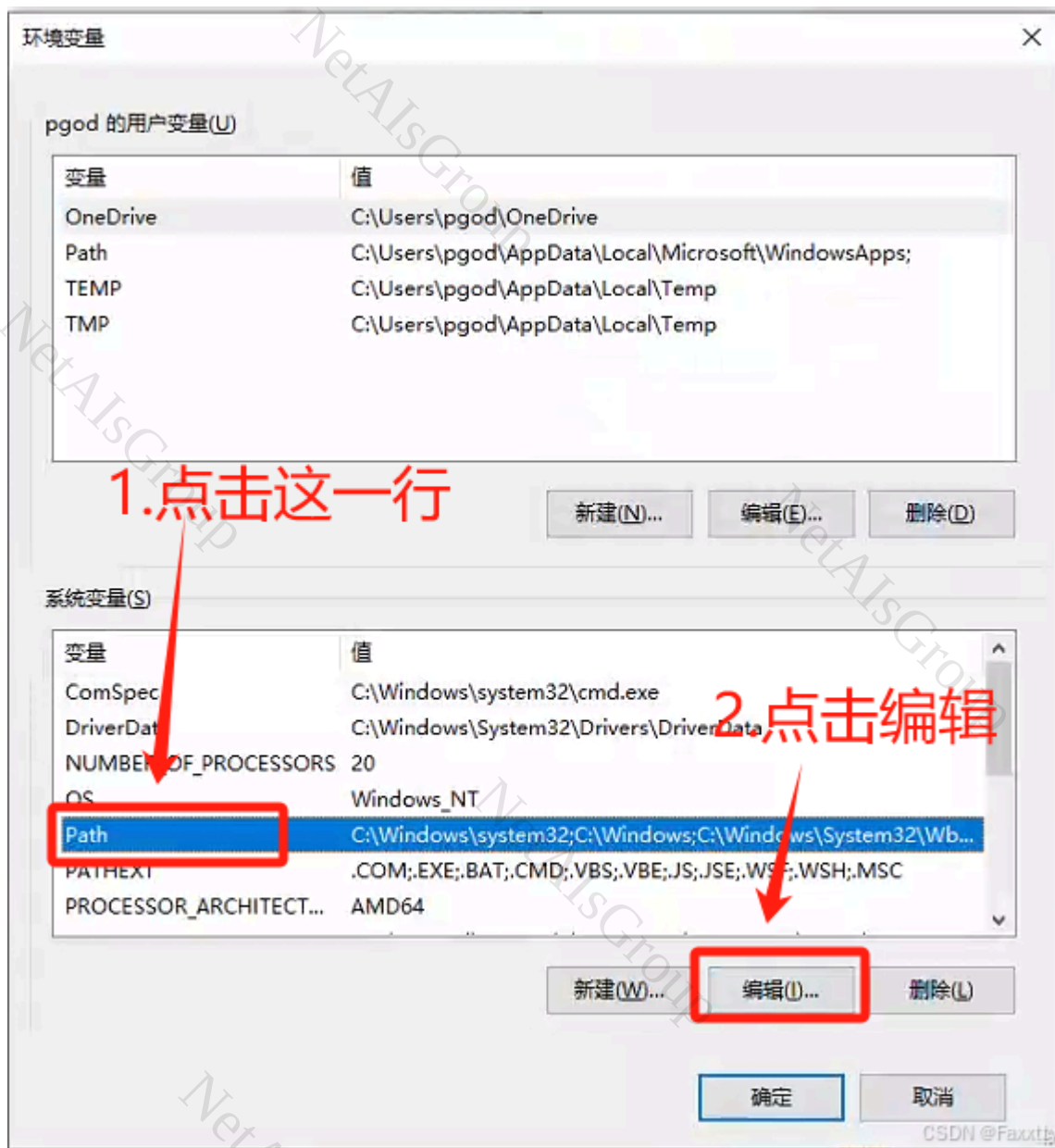
1. 搜索 查看高级系统设置，并点击。



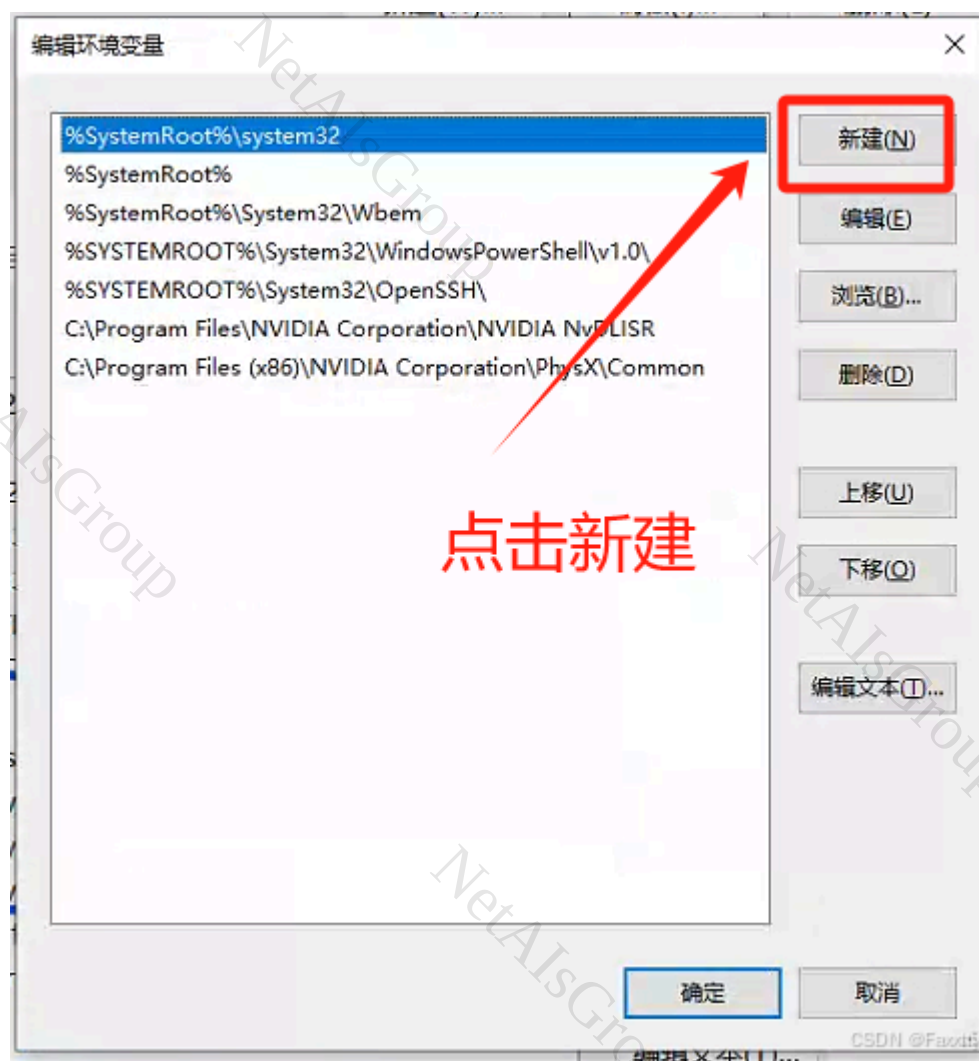
2. 选中 高级，环境变量。



3. 在系统变量中找到Path，选中 点击编辑。

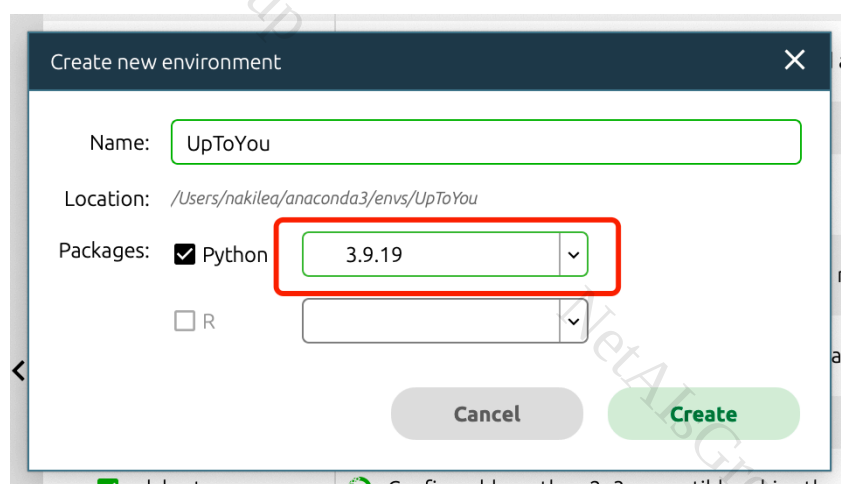
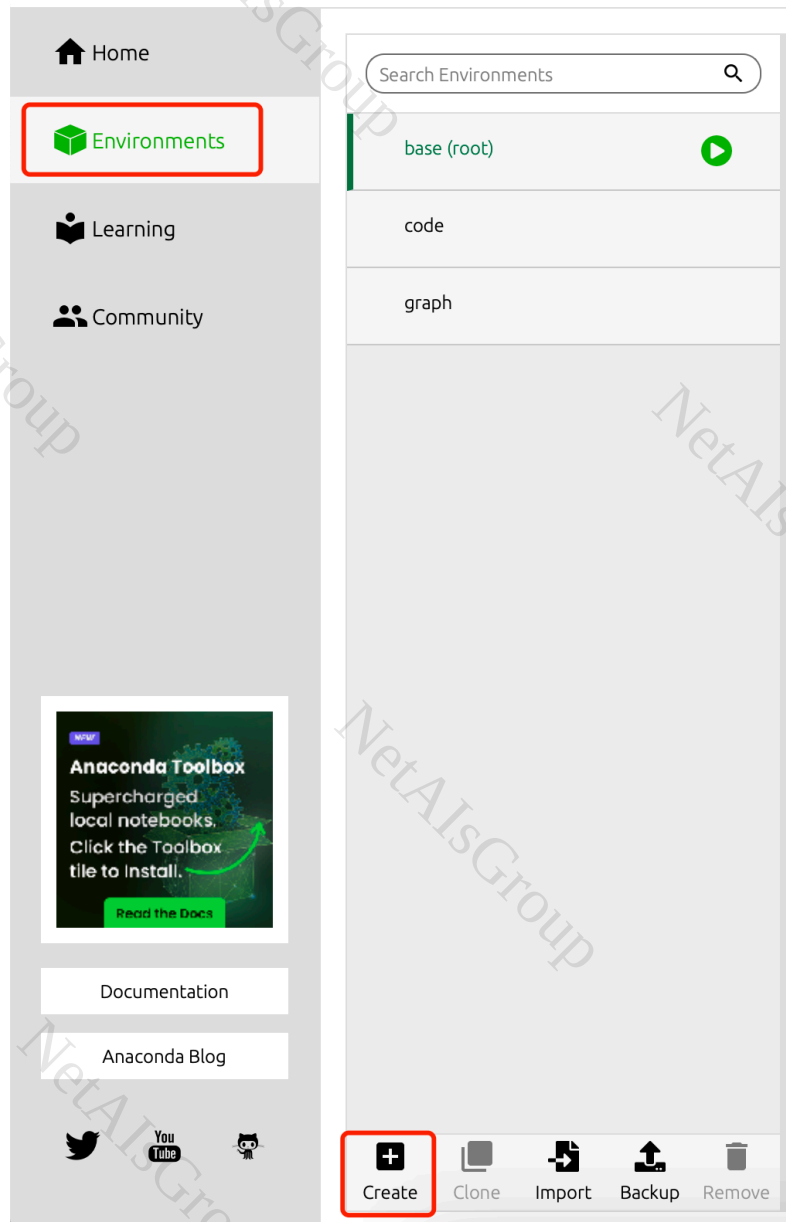


4. 点击新建。

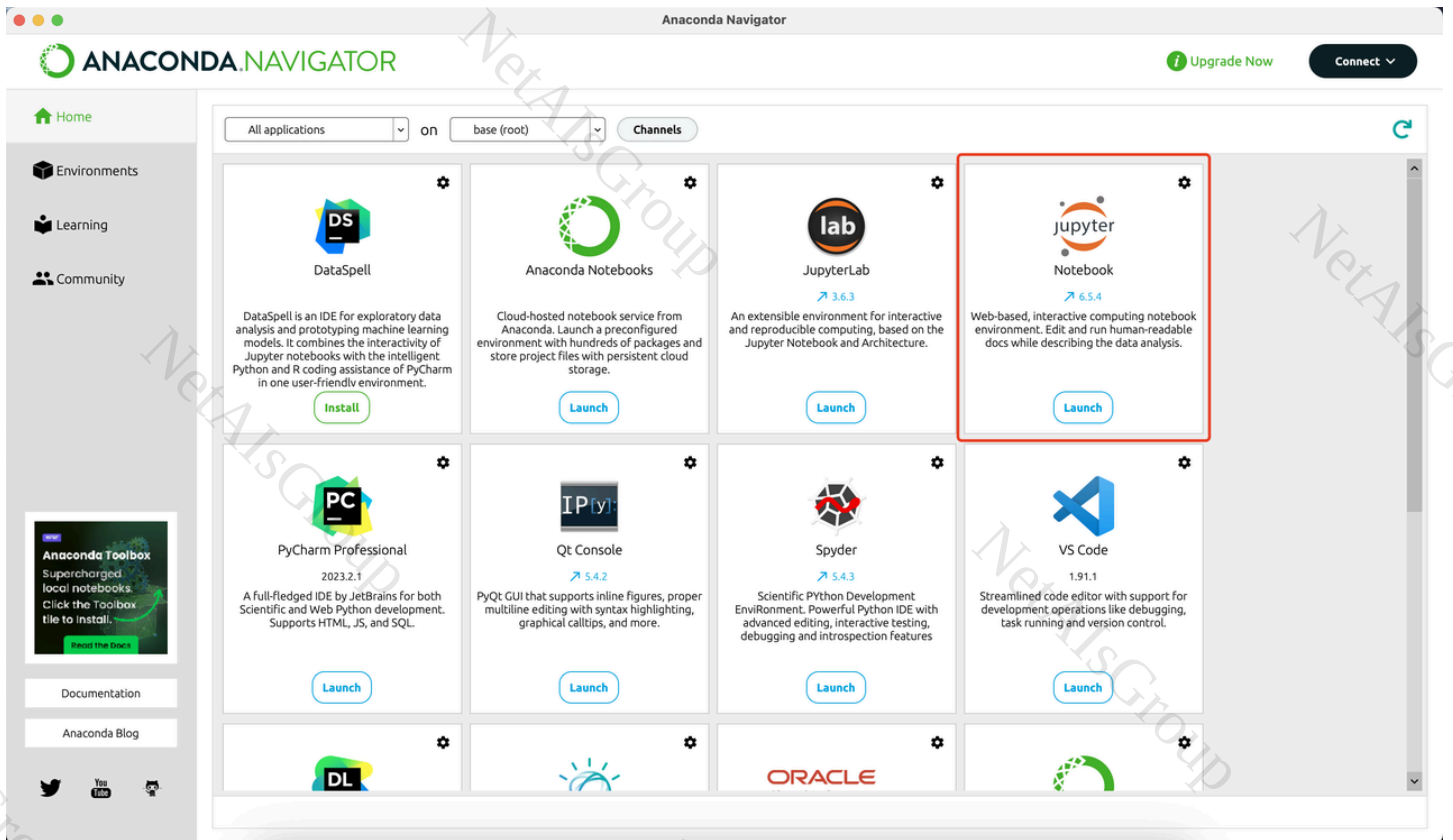


5. 添加路径。注意该路径和自己的安装路径应相对应。看conda安装步骤6中的路径。然后全部确定。

1. 点击Enviroments，选择Create创建Python虚拟环境。Python版本建议选择3.8（这里不重新截图）。



如果你喜欢Python的即时编译，可以使用Jupyter notebook进行代码编写。



2. 使用命令创建环境，进入cmd，输入conda create -n my_conda_env python=3.8
3. 进入cmd，输入conda init指令，重新打开cmd

```
命令提示符
Microsoft Windows [版本 10.0.22631.4037]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\Zhou>conda init
no change      E:\software\anaconda\Scripts\conda.exe
no change      E:\software\anaconda\Scripts\conda-env.exe
no change      E:\software\anaconda\Scripts\conda-script.py
no change      E:\software\anaconda\Scripts\conda-env-script.py
no change      E:\software\anaconda\condabin\conda.bat
no change      E:\software\anaconda\Library\bin\conda.bat
no change      E:\software\anaconda\condabin\_conda_activate.bat
no change      E:\software\anaconda\condabin\rename_tmp.bat
no change      E:\software\anaconda\condabin\conda_auto_activate.bat
no change      E:\software\anaconda\condabin\conda_hook.bat
no change      E:\software\anaconda\Scripts\activate.bat
no change      E:\software\anaconda\condabin\activate.bat
no change      E:\software\anaconda\condabin\deactivate.bat
no change      E:\software\anaconda\Scripts\activate
no change      E:\software\anaconda\Scripts\deactivate
no change      E:\software\anaconda\etc\profile.d\conda.sh
no change      E:\software\anaconda\etc\fish\conf.d\conda.fish
no change      E:\software\anaconda\shell\condabin\Conda.psm1
no change      E:\software\anaconda\shell\condabin\conda-hook.ps1
no change      E:\software\anaconda\Lib\site-packages\xontrib\conda.xsh
no change      E:\software\anaconda\etc\profile.d\conda.csh
no change      C:\Users\Zhou\Documents\WindowsPowerShell\profile.ps1
no change      HKEY_CURRENT_USER\Software\Microsoft\Command Processor\AutoRun
No action taken.

C:\Users\Zhou>
```

4. 激活环境可以使用conda activate base进入base环境。

```
命令提示符 - "E:\software\ana...
Microsoft Windows [版本 10.0.22631.4037]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\Zhou_>conda activate base

(base) C:\Users\Zhou_>|
```

5. 删除名为my_conda_env的环境 `conda env remove -n my_conda_env`

```
C:\Users\Zhou_>conda env remove -n my_conda_env

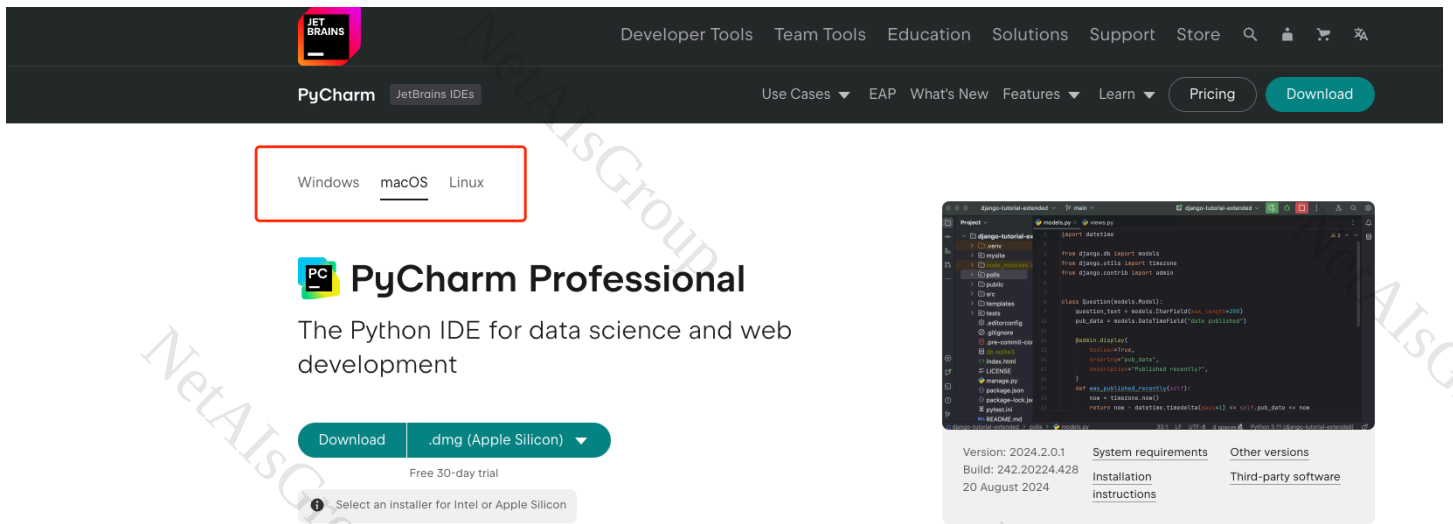
Remove all packages in environment E:\software\anaconda\envs\my_conda_env:

C:\Users\Zhou_>|
```

二、安装你的Pycharm

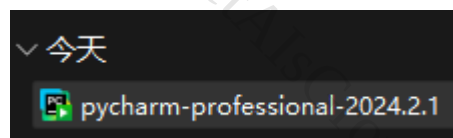
提示：如果你已经安装了，请移步至第二节。

1. 打开地址<https://www.jetbrains.com.cn/en-us/pycharm/download>。请选择你的操作系统进行下载。



We value the vibrant Python community, and that's why we proudly offer the PyCharm Community Edition for free, as our open-source contribution to support the Python ecosystem.

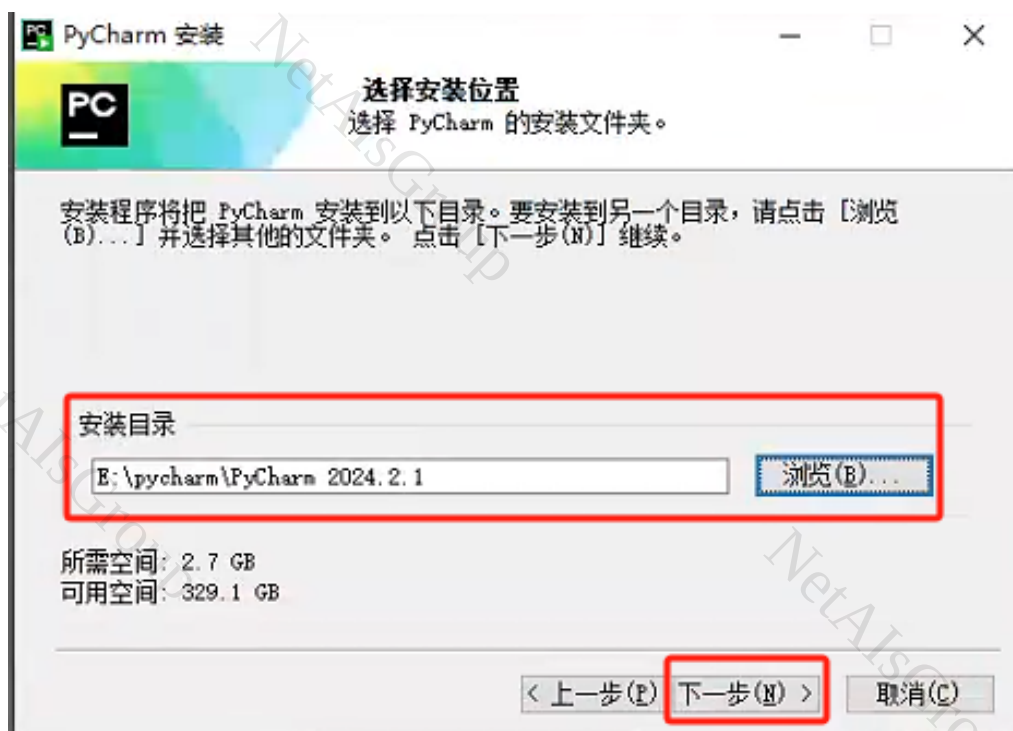
2. 双击安装包。



3. 点击“下一步”。



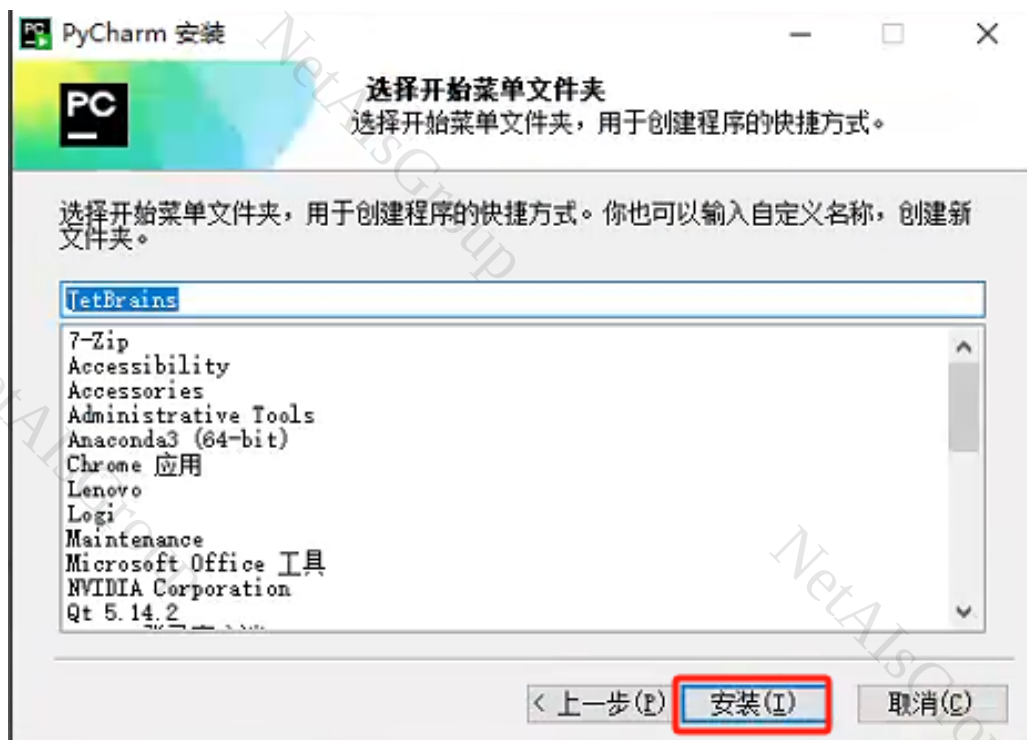
4. 选择合适的安装目录（不要包含中文路径，不建议安装在C盘），点击下一步。



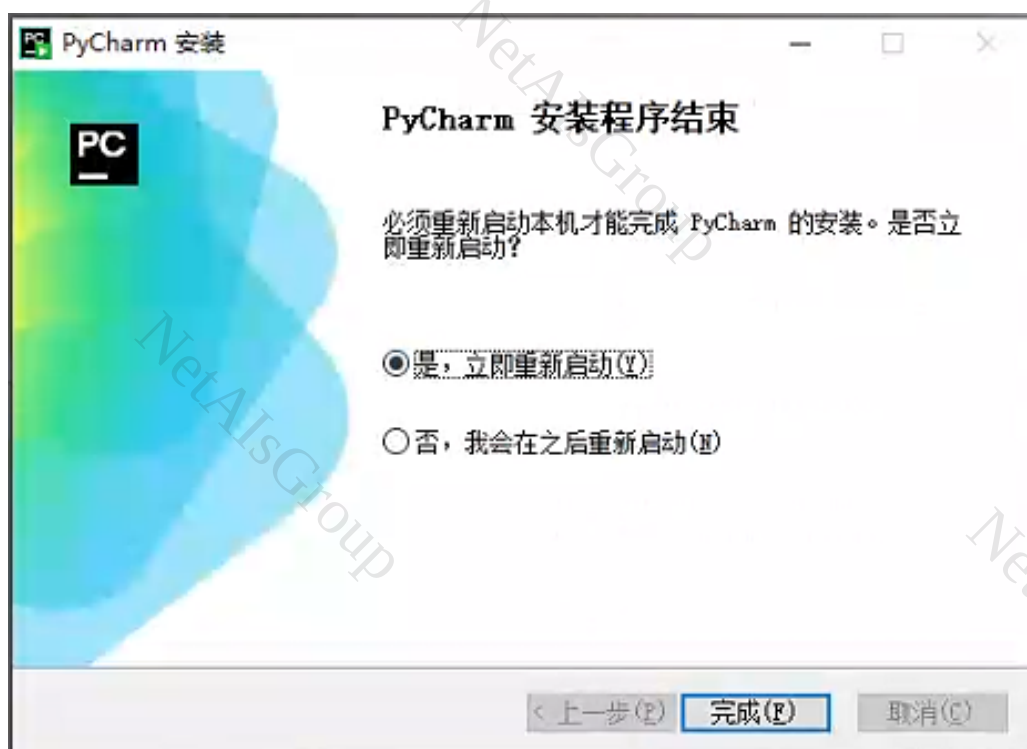
5. 安装选项全勾选，点击下一步。



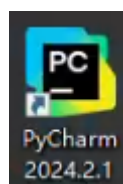
6. 点击安装。



7. 可选 立即重启，也可稍后重启。

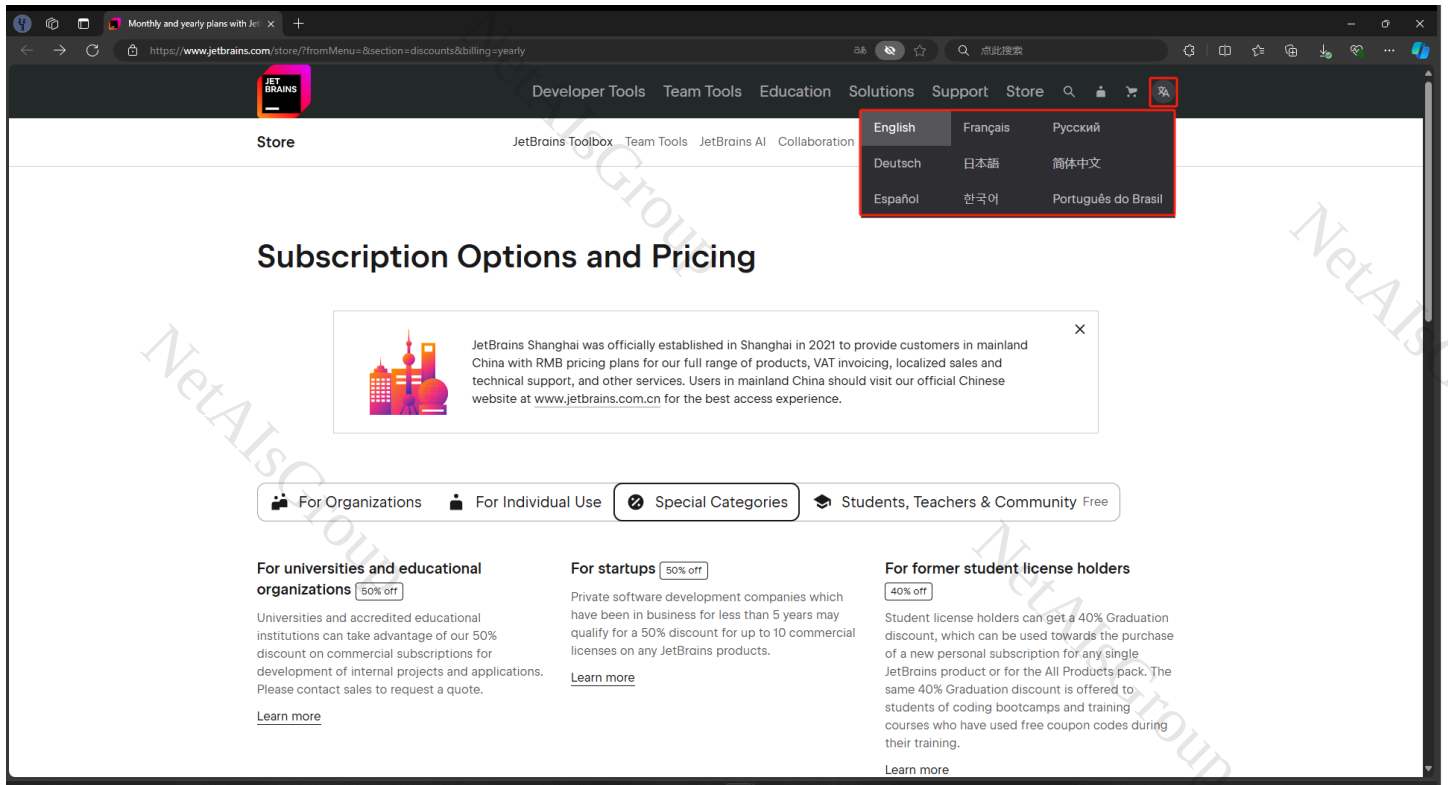


8. 这样pycharm安装算是完成了。



pycharm专业版学生认证

1. 打开网址<https://www.jetbrains.com/store/?fromMenu=§ion=discounts&billing=yearly>



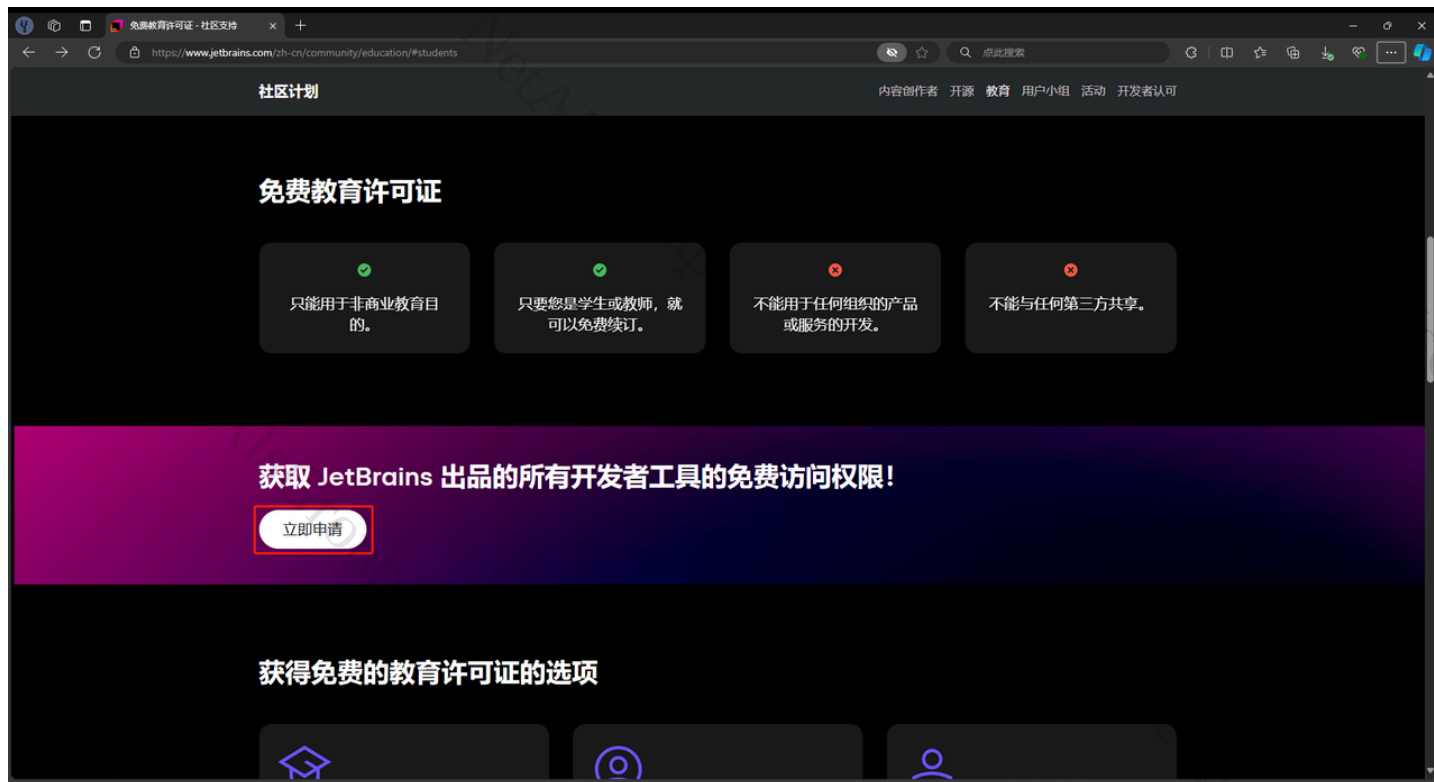
点击右上角红色框框处修改语言，选择合适的语言。（这边使用中文作为范例）

2. 选择学生、教师和社区



点击 适用于学生和教师 了解详情

3. 下滑找到如图界面



点击立即申请。

4.出现如下界面

按照流程填写。

5. 由于学生邮箱已申请，无法继续展示后续步骤。

(参考<https://blog.csdn.net/Colorkiller86/article/details/135457683>)

申请后应收到邮件。

Hi Yulin,

Congratulations! Your JetBrains Educational Pack has been confirmed.

Please [link your free license](#) to a new or an existing JetBrains Account. You will need to use this account whenever you want to access JetBrains tools.

Here are some useful materials to go with your Educational Pack:

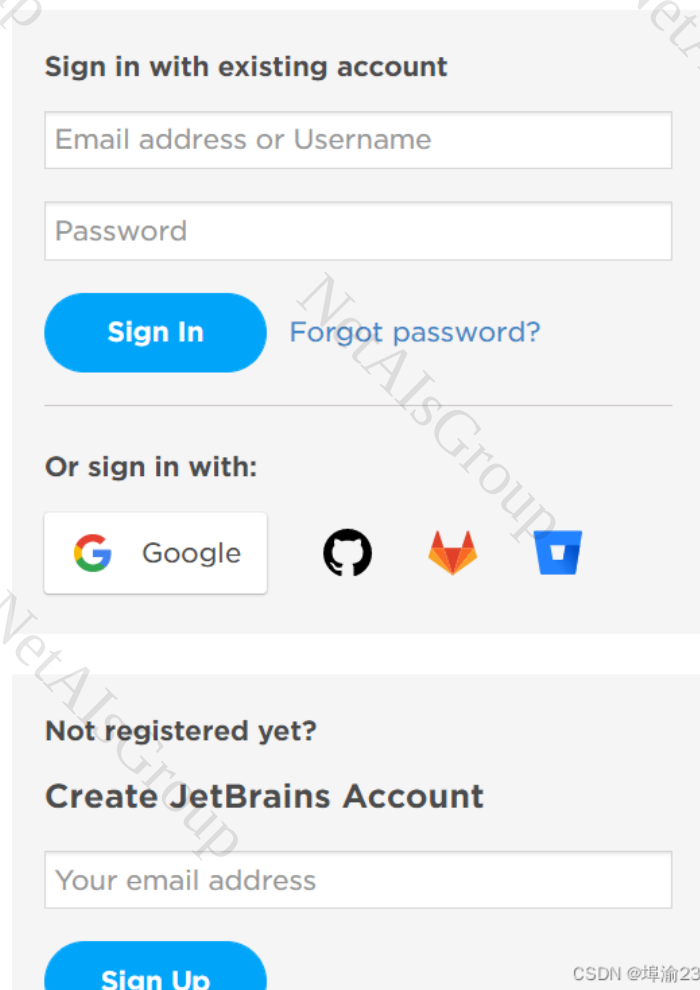
- Build interactive projects with [JetBrains Academy on Hyperskill](#). You can use it for free or get a special 50% discount on a Premium subscription for 3 months with the coupon code **EDUPACK50**.
- Get started by learning basic shortcuts and essential features from right inside IntelliJ IDEA and other JetBrains IDEs with the [IDE Features Trainer](#) plugin.

If you have any questions, please email us and we will be glad to help.

Kind Regards,
The JetBrains team
www.jetbrains.com
The Drive to Develop

CSDN @埠渝23

4. 点击链接，进入下方界面。

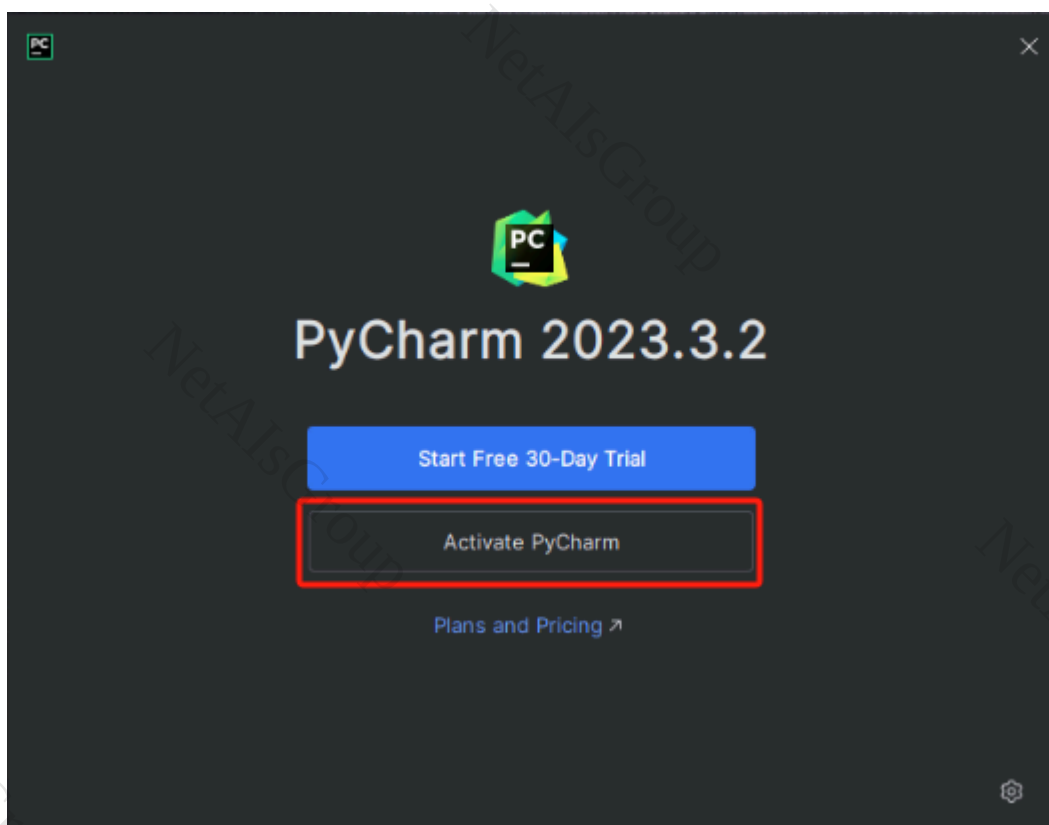


The screenshot shows the JetBrains account management interface. It is divided into two main sections. The top section, titled "Sign in with existing account", contains two input fields for "Email address or Username" and "Password". Below these fields is a blue "Sign In" button and a link for "Forgot password?". The bottom section, titled "Or sign in with:", features four social login options: Google, GitHub, Docker Hub, and Bitbucket. Below this is a section for new users titled "Not registered yet? Create JetBrains Account", which includes an input field for "Your email address" and a blue "Sign Up" button. The text "CSDN @埠渝23" is visible in the bottom right corner of the interface.

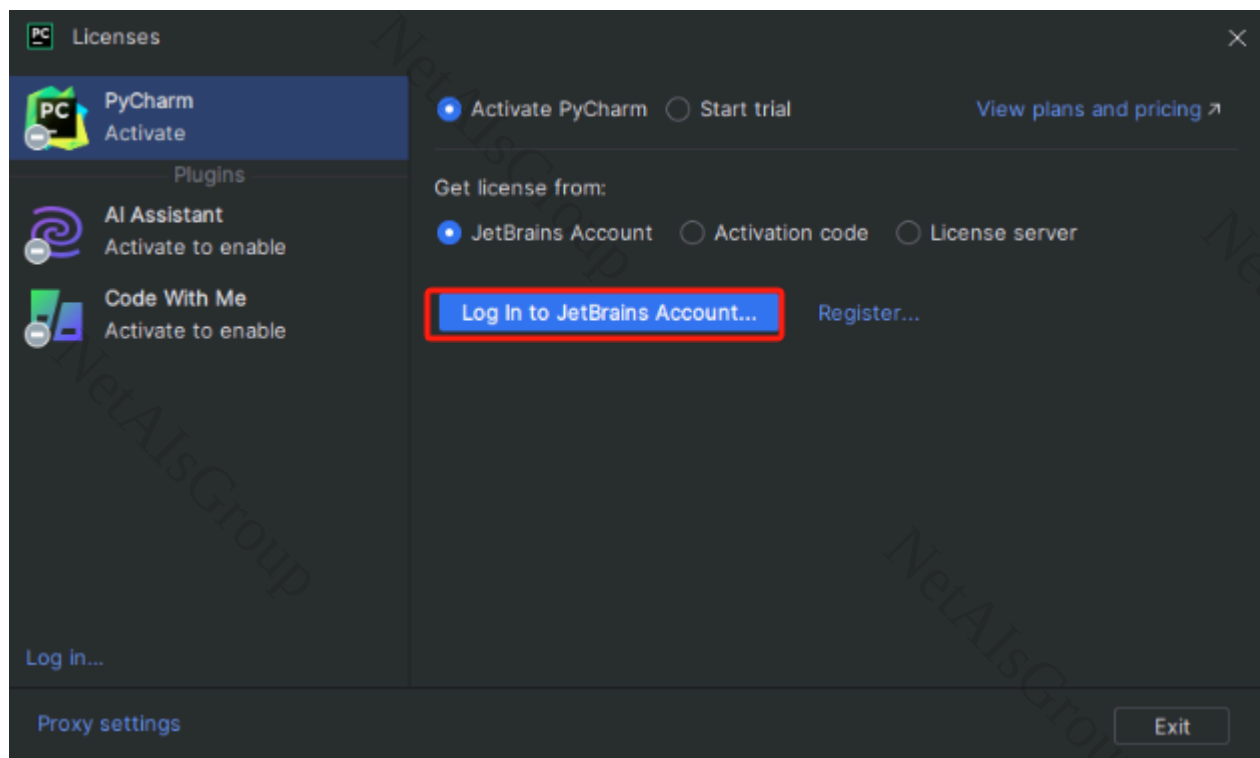
5. 激活账户 双击Pycharm，出现如下界面，点击continue。



6. 激活pycharm（使用学校邮箱注册免费试用专业版，学校邮箱申请到之前可以使用免费的30天）。



7. 进入下方界面，点击log in JetBrains Account...



8. 会有下面界面

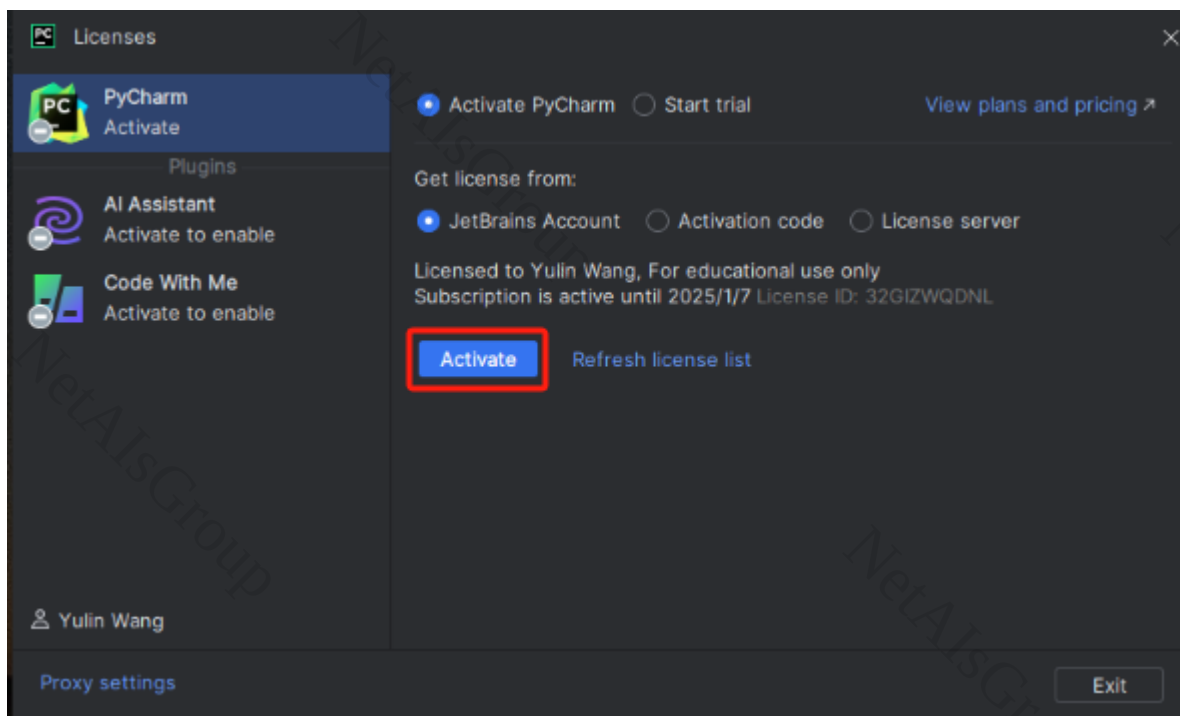
The screenshot shows the JetBrains Account login page. It has a 'Sign In' button and a 'Create Account' link. Below these are input fields for 'Username' (containing 'Buyu23') and 'Password' (masked with dots). There's a 'Sign In' button and a 'Forgot password?' link. At the bottom, it says 'Or sign in with:' followed by icons for Google, Apple, GitHub, and JetBrains.

9. 登录成功后会出现下面界面。

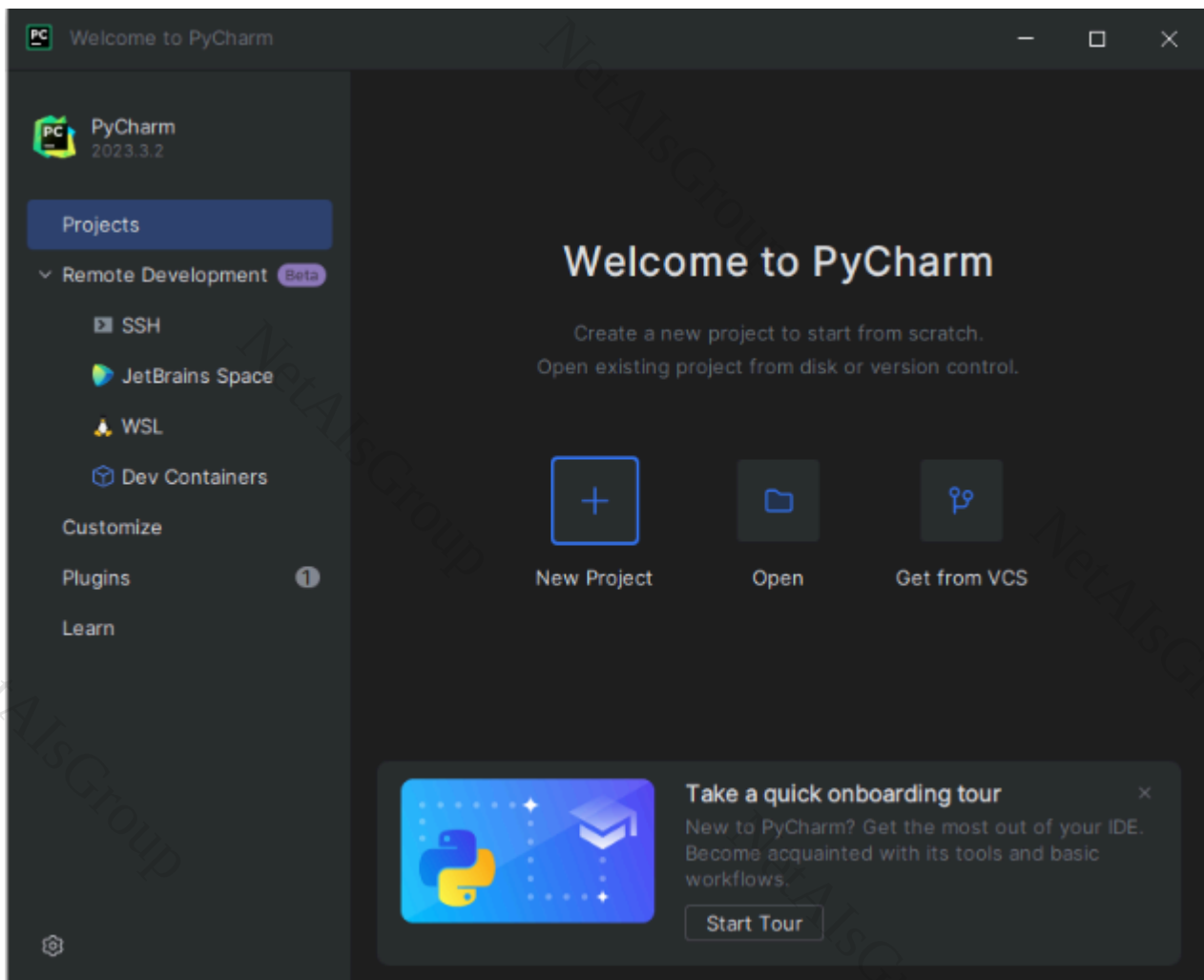
Authorization Successful

You can close this page and return to the IDE

10. 回到软件界面，点击activate。

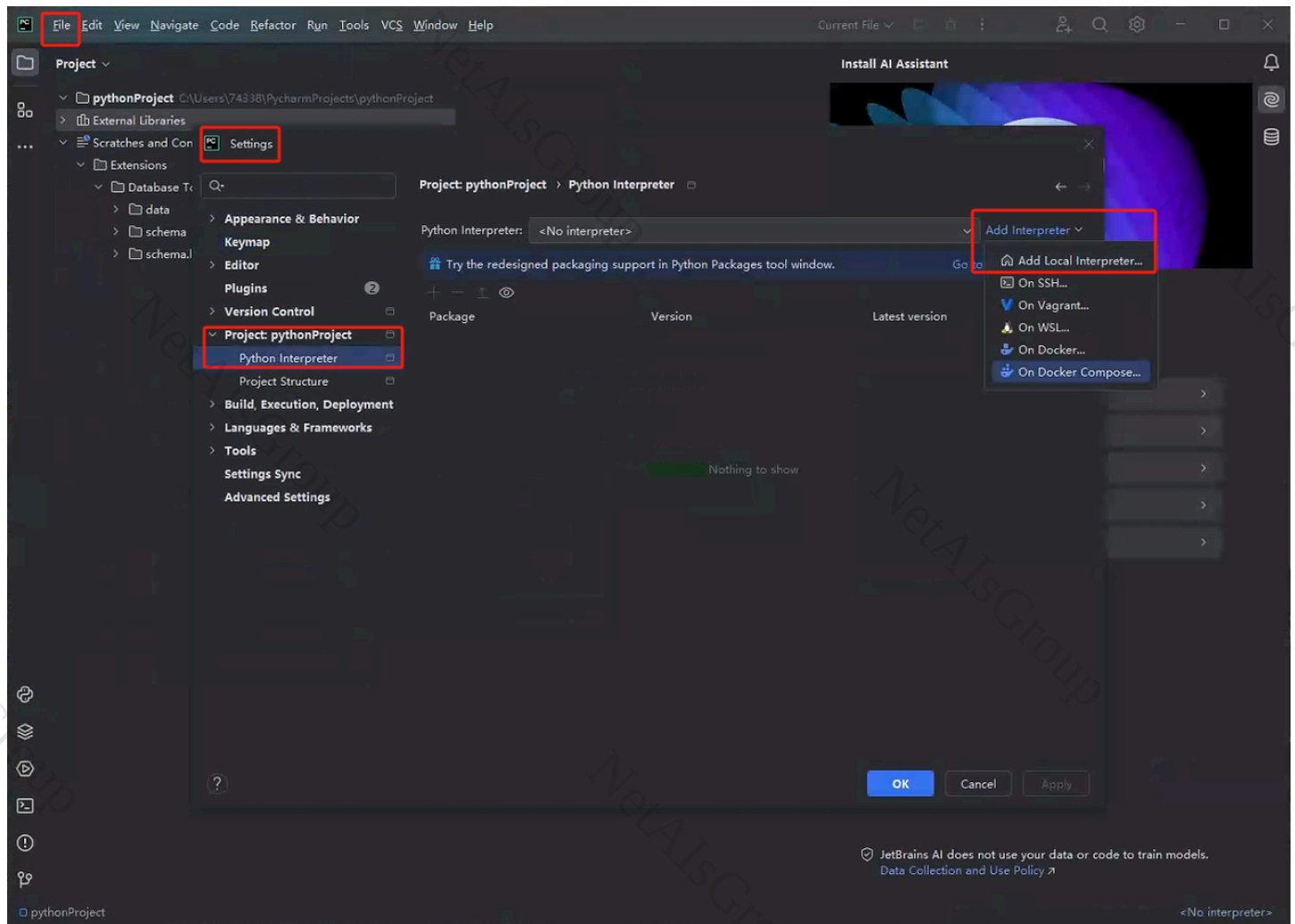


11. 然后就进入了pycharm初始界面。

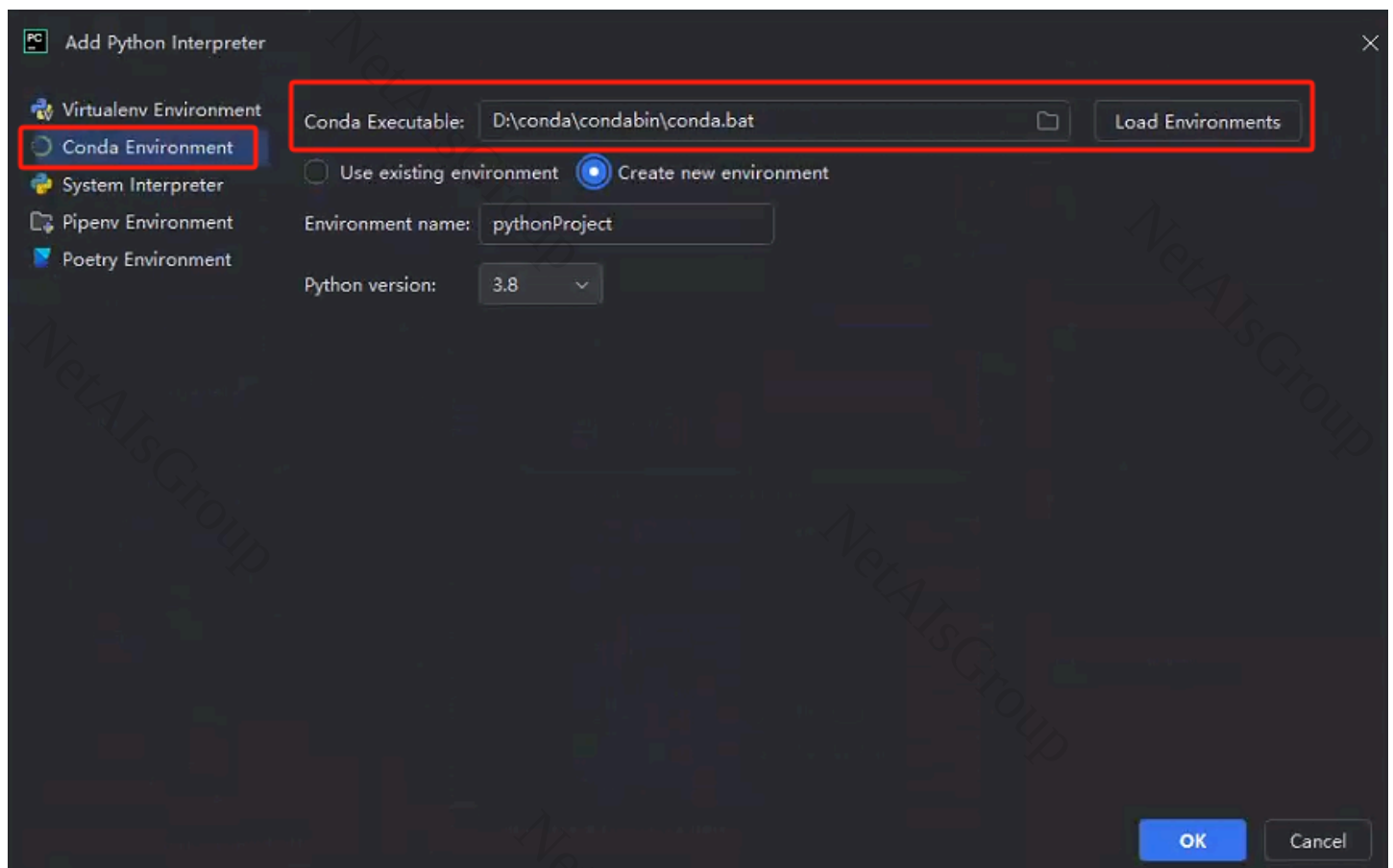


配置pycharm解释器

1. 打开 PyCharm，进入File -> Setting界面，找到Project：中的Python Interpreter选项。点击Add Local Interpreter，选择你希望使用的 Anaconda 环境作为解释器，并进行配置。



2. 找到conda Executable路径 然后load Environment。



可以使用现有环境，也可以创建新的环境。

三、Miniconda（服务器）

大家要学会使用集群或者服务器去跑代码，正常谁用自己电脑跑啊，不仅折寿，而且还慢，社团检测不算。

集群使用手册参考毕业师兄的飞书链接：

<https://knoozbeakj.feishu.cn/docx/doxcnOA1AeJl2K1wzuD5i2JOv3d> 密码：13Bj

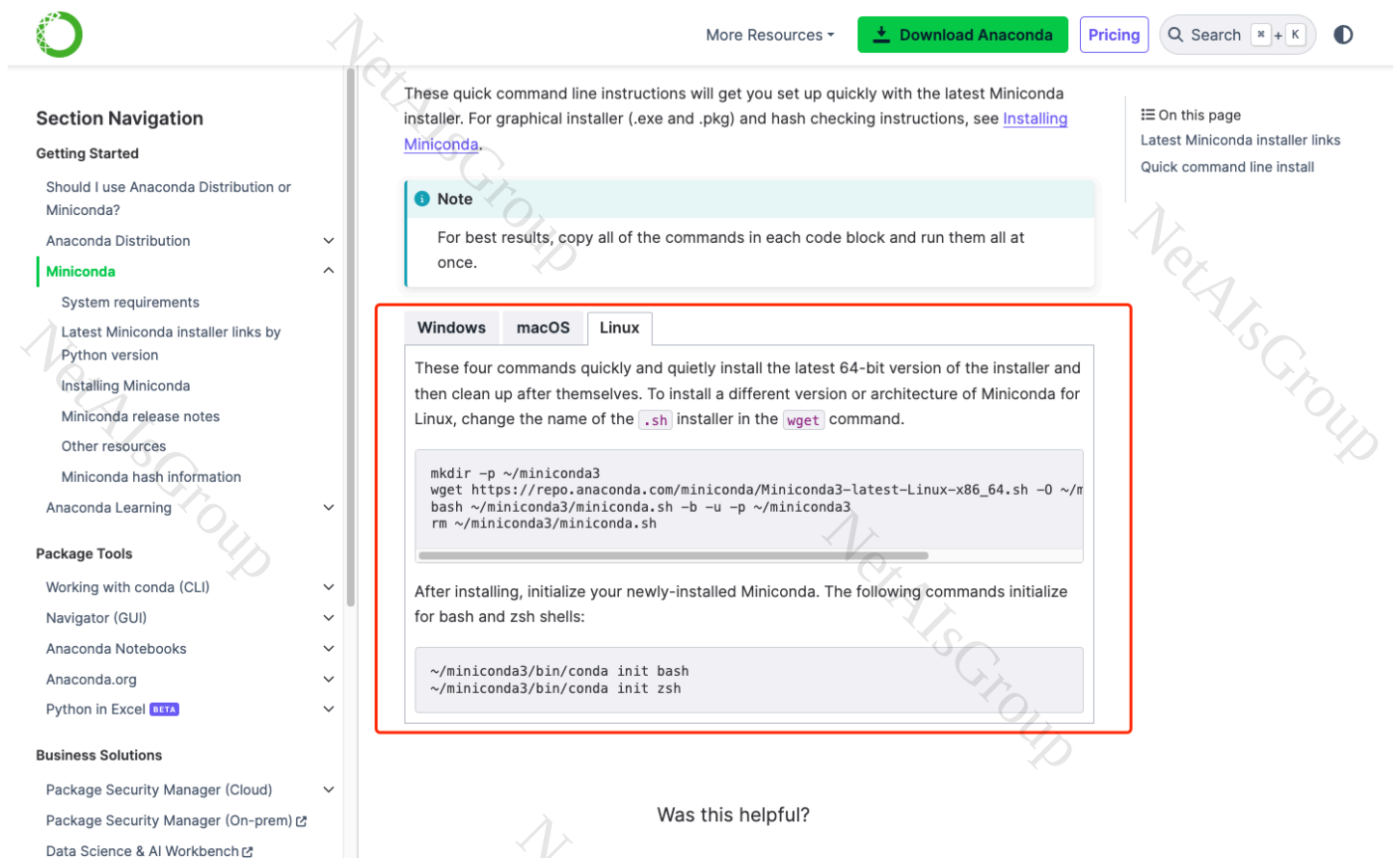
2025-9-13 补充：集群教程好几年前的，现在不需要自己尝试在虚拟机上搭建环境与上传了，大家可以去集群里面订阅别人发布的环境，比如订阅我发布的FastenV12（大概是这个名字）环境，发布者zm_211123030065，这个环境基本上包含了所需求的包；生信的同学可以订阅大师兄的包，找到分享者为wzy_XXX的包或者直接问师兄哪些包可以订阅。如果你想自己搞一个环境上传，那你可以继续往下学。

提示：这里补充Miniconda环境迁移至Docker以及如何迁移至其他Linux环境，如果你已经会了，请移步至第五节。

注意：你应该先安装一个Linux或者Centos虚拟机，同时你准备将自己的环境打包至集群，你才需要看这些步骤，否则请移步第四节。

Miniconda安装

1. 请直接按照官网教程输入指令进行安装。 <https://docs.anaconda.com/miniconda/>



The screenshot shows the Anaconda Miniconda installation page. The 'Linux' tab is selected, and the instructions for installing Miniconda3 are highlighted with a red box. The instructions are as follows:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/m
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

After installing, initialize your newly-installed Miniconda. The following commands initialize for bash and zsh shells:

```
~/miniconda3/bin/conda init bash
~/miniconda3/bin/conda init zsh
```

The page also includes a 'Section Navigation' sidebar on the left and a 'Was this helpful?' feedback section at the bottom.

Miniconda环境打包至Docker

注意：这里已经默认你有Miniconda以及Docker等必要工具，如果没有，自行百度。

1. 查看当前服务器下是否存在镜像continuumio/anaconda3: `sudo docker images`
2. 如果没有，下载镜像: `sudo docker pull continuumio/anaconda3`
3. 使用下载的镜像开一个容器: `sudo docker run -itd continuumio/anaconda3`
4. 查看容器是否运行: `docker ps -a`
5. 运行容器: `sudo docker exec -it 容器id bash`
6. 查找容器中的anaconda环境在哪儿: `whereis anaconda`，记住环境位置，一般是/opt/conda/envs
7. 新开一个终端，打开自己miniconda的envs目录，将miniconda的环境复制到容器环境中: `sudo docker cp ~/miniconda3/envs/graph 容器id:/opt/conda/envs`
8. 打开之前的终端，就可以正常在容器中激活cp的graph环境
9. 按照飞书上的docker部分内容在环境中安装htop, ssh等各种内容，以及修改ssh的config文件(一定要安装，不然不能上传集群)。逐行命令输入运行：

`apt-get update`

`apt-get install vim`

`apt-get install htop`

`apt-get install screen`

`apt-get install sudo`

`apt-get install openssh-server`

修改sshd配置文件

`vim /etc/ssh/sshd_config`



```
#
Port 22
#AddressFamily any
ListenAddress 0.0.0.0
ListenAddress ::
```

`/etc/init.d/ssh start`

最后，查看ssh是否运行

`ps -e | grep ssh`

10. 切换环境，输入pip list，此时无法使用报错如下：

`bash: /opt/conda/envs/py39/bin/pip: /public/kk/miniconda3/envs/py39/bin/python: bad interpreter: No such file or directory`

其中前一段/opt/conda/envs/py39/bin/pip是pip文件，往后的是报错信息

直接vim /opt/conda/envs/py39/bin/pip

修改第一行的错误文件地址，改为/opt/conda/envs/py39/bin/python，保存即可。

11. 退出容器：exit，将容器保存为镜像sudo docker commit 容器id 镜像名称
12. 查看保存的镜像：sudo docker images
13. 将保存的镜像保存为tar文件：sudo docker save 镜像名称 -o ~/docker/graph.tar
14. 删除容器，删除打包的镜像：sudo docker rm -f 容器id，docker rmi 镜像名称:latest

提示：你也可以不删镜像，省的下次还要打包一次，如果你不嫌内存占用太大就行。

15. 将打包后的容器上传到集群即可。
16. 注意，由于是整个conda打包，所以在集群中要先conda activate graph切换到需要的那个环境才行，base环境没东西的。
17. 如果误操作出现了悬空镜像，同意删除所有悬空镜像：sudo docker rmi \$(sudo docker images -f "dangling=true" -q)，例如<none>以及<none>标签

conda打包环境给另一个服务器

1. conda pack -n my_env -o out_name.tar.gz打包指定的环境为指定的压缩包
2. 将压缩包移动到另一个服务器上，并解压到conda的环境目录下tar -xzf my_env.tar.gz -C ~/miniconda3/envs/my_env

注意，注意：目标服务器中envs环境是没有my_env目录的，你需要手动新建一个。不能直接解压到envs目录中，这是无效的。

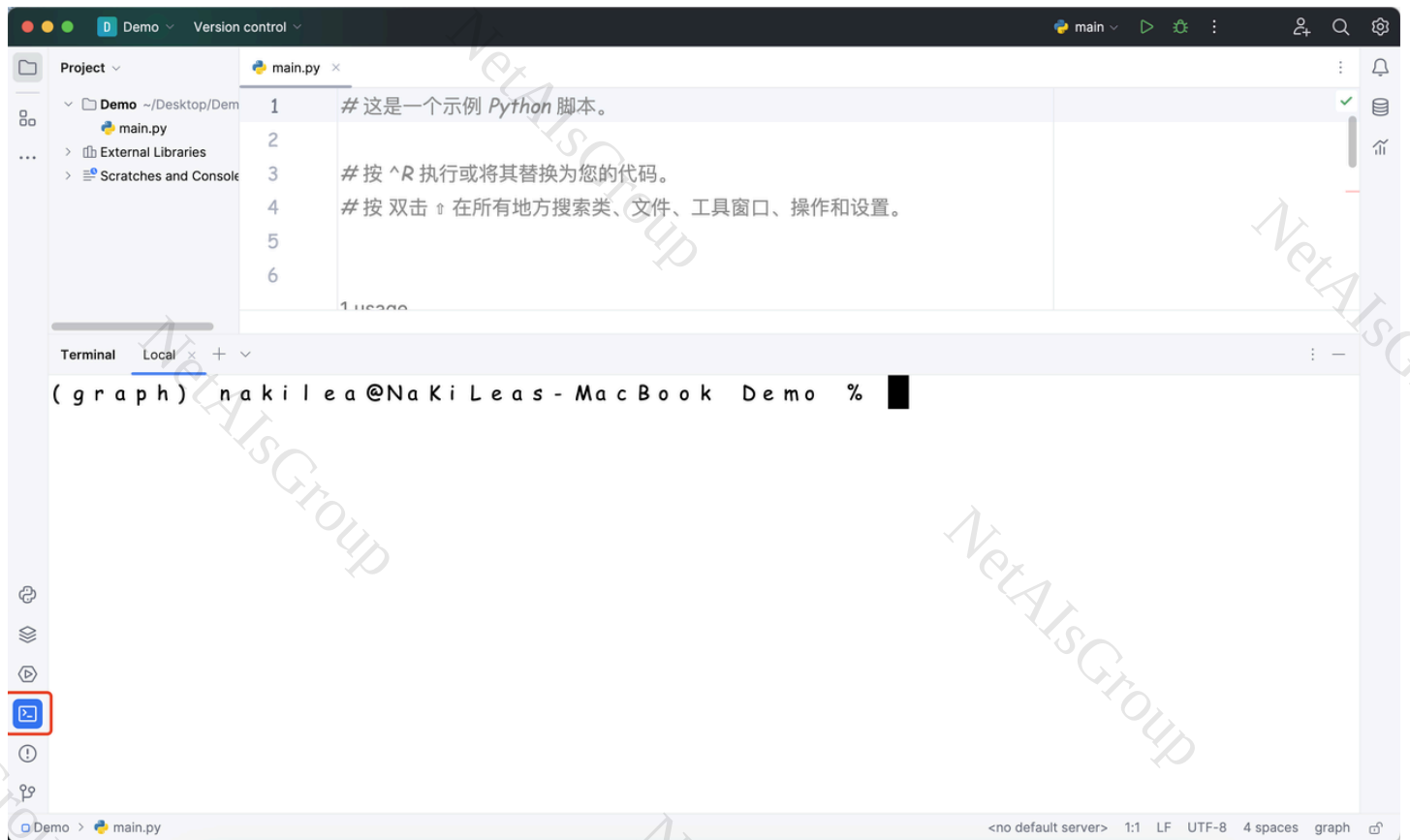
四、PyTorch以及Pyg安装

注意：所有的Python包请用pip install进行安装，不要使用conda install

Pycharm中进行环境安装

什么，你是Windows系统，都差不多的，举一反三。

1. 打开工程，点击左侧的终端



如果不想打开工程可以直接在终端中操作，Windows在cmd中操作，Mac就算了。

2. 安装python包，例如安装numpy，可以在终端中直接输入`pip install numpy`

安装PyTorch

1. torch建议2.6.0+cu118（从下面的网页链接里面找pip指令），安装时请注意Python版本。

2. 请选择对应系统对应版本的PyTorch，使用`nvidia-smi`查看自己电脑上的CUDA驱动版本。

<https://pytorch.org/>。如果安装其他版本，请点击的[install previous versions of PyTorch](#)。

NOTE: Latest PyTorch requires Python 3.8 or later.

PyTorch Build	Stable (2.4.0)			Preview (Nightly)	
Your OS	Linux		Mac	Windows	
Package	Conda	Pip		LibTorch	Source
Language	Python			C++ / Java	
Compute Platform	CUDA 11.8	CUDA 12.1	CUDA 12.4	ROCm 6.1	CPU
Run this Command:	pip3 install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118				

Previous versions of PyTorch >

安装Pyg

1. `print(torch.__version__)`可查看torch版本，比如我的就是这个版本。

2.0.0+cu118

2. 在<https://data.pyg.org/whl/>中选择指定torch版本的对应whl下载进行离线安装。
3. 简易安装，在官网可以选择对应版本的Pyg进行安装。

<https://pytorch-geometric.readthedocs.io/en/latest/install/installation.html>

 PyG is available for  Python 3.8 to  Python 3.12.

Note

We do not recommend installation as a root user on your system  Python. Please setup a virtual environment, e.g., via  Anaconda or Miniconda, or create a Docker image.

Quick Start

PyTorch	PyTorch 2.4.*	PyTorch 2.3.*	PyTorch 2.2.*
Your OS	Linux	Mac	Windows
Package	Pip	Conda	
CUDA	11.8	12.1	12.4
			CPU

Run: `pip install torch_geometric`

Optional dependencies:
`pip install pyg_lib torch_scatter torch_sparse torch_cluster
torch_spline_conv -f https://data.pyg.org/whl/torch-
2.4.0+cpu.html`

4. 安装torch_geometric: `pip install torch_geometric`

注意: torch_geometric属于必须安装项

五、梯子以及GPT注册

提示: 如果你已经会使用ChatGPT以及知道很多其他大模型, 请移步至第六节

梯子注册

注意: 如果你没有自己的梯子, 也不知道怎么整一个梯子, 同时你想偷懒, 请继续。

1. 优云, 包年七折, 自带新手教程, 很稳定, 注册: <https://youyun666.com/auth/register?code=dy0M>

优云666

首页 商店 我的账号 我的钱包 邀请注册 使用 节点列表 下载和教程 工单支持 节点设置 审计系统

入门版	标准版	高级版
¥19.99 / 每月	¥29.99 / 每月	¥39.99 / 每月
✓ 100GB使用流量 ✓ 2个设备同时使用 ✓ 最高60M速率 ✓ 工单客服支持 ✓ 支持SSR、V2ray节点 ✓ 国内中转节点 ✓ 解锁流媒体 ✓ 可用入门版节点 ✗ 可用标准版节点 ✗ 可用高级版节点 ✗ IPLC专线节点 ✗ 可用旗舰版节点(可用所有节点)	✓ 200GB使用流量 ✓ 3个设备同时使用 ✓ 最高150M速率 ✓ 工单客服支持 ✓ 支持SSR、V2ray节点 ✓ 国内中转节点 ✓ 解锁流媒体 ✓ 可用入门版节点 ✓ 可用标准版节点 ✗ 可用高级版节点 ✗ IPLC专线节点 ✗ 可用旗舰版节点(可用所有节点)	✓ 350GB使用流量 ✓ 5个设备同时使用 ✓ 最高300M速率 ✓ 工单客服支持 ✓ 支持SSR、V2ray节点 ✓ 国内中转节点 ✓ 解锁流媒体 ✓ 可用入门版节点 ✓ 可用标准版节点 ✓ 可用高级版节点 ✗ IPLC专线节点 ✗ 可用旗舰版节点(可用所有节点)
订阅 →	订阅 →	订阅 →
旗舰版	旗舰版大流量	
¥59.99	¥99.99	

2. CordCloud: <https://cordcloud.us/auth/login>。年包350g 63

3. 速子云 (比较贵): <https://get.tachvip.com/#/register?code=uDAttwT6>

4. 一元机场 (主打一个便宜): <https://ygjs.xfss3.top/#/register?code=rO1Bi7CO>, 1024G两年 77RMB

1024G-两年流量包 ¥77.00 / 每两年 立即购买 ✓ 每两年 1024G 流量 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✗ 新疆地区所有套餐都无法使用! 请勿购买! ✓ 1Gbps 带宽保证	IEPL专线300G套餐 ¥12.00 / 每月 立即购买 ✓ 每月 300G 流量 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✓ 1Gbps 带宽保证	IEPL专线750G流量版 ¥30.00 / 每月 立即购买 ✓ 每月 750G 绝对的够用 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✓ 1Gbps 带宽保证
一点点流量包月17G套餐 ¥1.00 / 每月 立即购买 ✓ 每月 17G 试用套餐 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✓ 1Gbps 带宽保证	差不多够用包月150G套餐 ¥7.00 / 每月 立即购买 ✓ 每月 150G 轻轻松松浏览 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✓ 1Gbps 带宽保证	绝对够用包月500G套餐 ¥20.00 / 每月 立即购买 ✓ 每月 500G 大流量套餐 ✓ 没有的宽带速率限制 ✓ 够齐全的流媒体解锁 ✓ 1Gbps 带宽保证

ChatGPT注册

1. 注册一个谷歌账户, 直接使用你的国内邮箱作为辅助邮箱即可。

2. <https://openai.com/chatgpt/>进入ChatGPT网页进行注册并使用, 这里会要求输入一个手机号, 手机号不能是香港等不支持地区的手机号。

注意: 你挂的梯子也不能是香港节点的。

3. 如果你已经有国外手机号，请自便。如果你没有，请注册<https://sms-activate.io/cn/freePrice#activation>。获取虚拟手机号，服务请选OpenAI。



你可以选择其中的免费虚拟手机号服务，也可以选择充值2美刀（最低）购买虚拟手机号的激活服务。

如果你不想注册SMS账户，你可以问我，我可以免费提供几个虚拟手机号。或者你们组团注册一个SMS账户，然后大家一起购买，一个虚拟手机号激活服务大概在几毛到一块多。

4. 使用虚拟手机号接收到的验证码完成ChatGPT的注册。

六、其他网站

镜像网站

1. 谷歌学术：https://xs.fropet.com/?security_verify_data=313932302c31303830

Chrome插件：

1. 划词翻译
2. EasyScholar（方便看期刊的水平）

校园正版软件平台

<http://mysoft.zjut.edu.cn/>，其中包含：

1. Windows专业版

2. Windows和Mac上office专业版
3. 福昕PDF

专利查询：

1. <https://pss-system.cponline.cnipa.gov.cn/conventionalSearch8>

七、入门教程

Python

1. 廖雪峰：<https://liaoxuefeng.com/books/python/introduction/index.html>

PyTorch

1. 视频教程（生肉）：https://www.youtube.com/watch?v=V_xro1bcAuA&t=11s。配套文案：
<https://www.learnpytorch.io/>

Git

1. 廖雪峰：<https://liaoxuefeng.com/books/git/introduction/index.html>

图机器学习

1. 斯坦福CS224W《图机器学习》课程(2021) by Jure Leskovec,
https://www.bilibili.com/video/BV1RZ4y1c7Co/?spm_id_from=333.337.search-card.all.click&vd_source=0830461b27407c7915f0d0fe3b4c1b0b
2. 图机器学习，宣琦，高等教育出版社（组里有很多本可以不买）

八、推荐系统入门

1. 协同过滤：<https://zhuanlan.zhihu.com/p/33187914>
2. 推荐系统入门笔记：<https://blog.csdn.net/baishuiniyaonulia/article/details/111030265?spm=1001.2014.3001.5506>

3. 推荐系统综述: https://blog.csdn.net/greedydstar/article/details/80813057?ops_request_misc=%257B%2522request%255Fid%2522%253A%2522169199876116800226548744%2522%252C%2522scm%2522%253A%252220140713.130102334..%2522%257D&request_id=169199876116800226548744&biz_id=0&utm_medium=distribute.pc_search_result.none-task-blog-2~all~top_positive~default-2-80813057-null-null.142^v92^controlT0_1&utm_term=%E6%8E%A8%E8%8D%90%E7%B3%BB%E7%BB%9F&spm=1018.2226.3001.4187

九、新手任务

提示: 你不是新手就不用看这个

GCN

论文:

SEMI-SUPERVISED CLASSIFICATION WITH GRAPH CONVOLUTIONAL NETWORKS

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ABSTRACT

We present a scalable approach for semi-supervised learning on graph-structured data that is based on an efficient variant of convolutional neural networks which operate directly on graphs. We motivate the choice of our convolutional architecture via a localized first-order approximation of spectral graph convolutions. Our model scales linearly in the number of graph edges and learns hidden layer representations that encode both local graph structure and features of nodes. In a number of experiments on citation networks and on a knowledge graph dataset we demonstrate that our approach outperforms related methods by a significant margin.

1 INTRODUCTION

We consider the problem of classifying nodes (such as documents) in a graph (such as a citation network), where labels are only available for a small subset of nodes. This problem can be framed as graph-based semi-supervised learning, where label information is smoothed over the graph via some form of explicit graph-based regularization (Zhu et al., 2003; Zhou et al., 2004; Belkin et al., 2006; Weston et al., 2012), e.g. by using a graph Laplacian regularization term in the loss function:

$$\mathcal{L} = \mathcal{L}_0 + \lambda \mathcal{L}_{\text{reg}}, \quad \text{with} \quad \mathcal{L}_{\text{reg}} = \sum_{i,j} A_{ij} \|f(X_i) - f(X_j)\|^2 = f(X)^\top \Delta f(X). \quad (1)$$

Here, $\mathcal{L}_0$ denotes the supervised loss w.r.t. the labeled part of the graph, $f(\cdot)$ can be a neural network-like differentiable function, λ is a weighing factor and X is a matrix of node feature vectors X_i . $\Delta = D - A$ denotes the unnormalized graph Laplacian of an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with N nodes $v_i \in \mathcal{V}$, edges $(v_i, v_j) \in \mathcal{E}$, an adjacency matrix $A \in \mathbb{R}^{N \times N}$ (binary or weighted) and a degree matrix $D_{ii} = \sum_j A_{ij}$. The formulation of Eq. 1 relies on the assumption that connected nodes in the graph are likely to share the same label. This assumption, however, might restrict modeling capacity, as graph edges need not necessarily encode node similarity, but could contain additional information.

In this work, we encode the graph structure directly using a neural network model $f(X, A)$ and train on a supervised target $\mathcal{L}_0$ for all nodes with labels, thereby avoiding explicit graph-based regularization in the loss function. Conditioning $f(\cdot)$ on the adjacency matrix of the graph will allow the model to distribute gradient information from the supervised loss $\mathcal{L}_0$ and will enable it to learn representations of nodes both with and without labels.

Our contributions are two-fold. Firstly, we introduce a simple and well-behaved layer-wise propagation rule for neural network models which operate directly on graphs and show how it can be motivated from a first-order approximation of spectral graph convolutions (Hammond et al., 2011). Secondly, we demonstrate how this form of a graph-based neural network model can be used for fast and scalable semi-supervised classification of nodes in a graph. Experiments on a number of datasets demonstrate that our model compares favorably both in classification accuracy and efficiency (measured in wall-clock time) against state-of-the-art methods for semi-supervised learning.

提示：自选一个代码完成任务，PyG代码的好处是已经封装好了，会比较好看。源码虽然完整，但是会比较杂乱。

初步要求

1. 自问自答：GCN是如何工作的？在CV领域，什么是CNN，CNN与MLP的区别在哪儿？GCN与MLP的区别在哪儿，与CNN的区别在哪儿？

提示：CNN理解可以参考<https://poloclub.github.io/cnn-explainer/>

2. 复现实验，记录Cora数据集上的节点分类任务准确率。
3. （可选）在服务器上完成实验。
4. 在集群上完成实验。
5. 自问自答：GCN是如何完成节点分类的？
6. 更改数据集为Citeseer，完成节点分类任务并记录。
7. 修改学习率，聚合层数，隐藏层大小等超参数，可视化分类准确率的对比效果，例如：



参考代码，（用ChatGPT生成的）：



 main.py

进阶要求

1. 随机删除Cora数据集中的 n 个节点，可视化删除后模型分类效果。提示：随机删除可以理解为，对应节点的特征置为0，同时邻接矩阵对应行列均置为0，该节点成为孤立节点：或者直接从特征矩阵与邻接矩阵中移除该节点对应的属性。
2. 往Cora数据集中随机增加 n 个节点，可视化增加后模型分类效果。提示：随机增加可以理解为，往特征矩阵与邻接矩阵中添加额外的节点属性，添加节点的特征可以是随机生成的，也可以是平均特征等等；添加节点的邻居可以是以随机抽取 m 个节点作为其邻居， m 在合理情况下应小于等于Cora数据集的平均度值。
3. 往数据集中添加节点可以扰动模型的分类效果，这就是节点注入攻击。考虑单节点注入攻击，同时该节点可以拥有 m 个邻居， m 为攻击预算，是超参数。当 m 为随机抽取 m 个邻居时，可视化攻击后模型分类效果。
4. 接3，当 m 为抽取度值最大的 m 个邻居时，可视化攻击后模型分类效果。
5. 修改完数据后，重新训练模型，此为中毒攻击，中毒攻击的数据修改一般针对训练集。修改完数据后，不重新训练模型，而是直接将数据送入之前训练好的模型中，观察模型分类结果，此为逃逸攻击，逃逸攻击的数据修改一般针对测试集。请分别针对中毒攻击和逃逸攻击完成删除节点，注入节点和单节点注入实验，并可视化结果。

HOW POWERFUL ARE GRAPH NEURAL NETWORKS?

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ABSTRACT

Graph Neural Networks (GNNs) are an effective framework for representation learning of graphs. GNNs follow a neighborhood aggregation scheme, where the representation vector of a node is computed by recursively aggregating and transforming representation vectors of its neighboring nodes. Many GNN variants have been proposed and have achieved state-of-the-art results on both node and graph classification tasks. However, despite GNNs revolutionizing graph representation learning, there is limited understanding of their representational properties and limitations. Here, we present a theoretical framework for analyzing the expressive power of GNNs to capture different graph structures. Our results characterize the discriminative power of popular GNN variants, such as Graph Convolutional Networks and GraphSAGE, and show that they cannot learn to distinguish certain simple graph structures. We then develop a simple architecture that is provably the most expressive among the class of GNNs and is as powerful as the Weisfeiler-Lehman graph isomorphism test. We empirically validate our theoretical findings on a number of graph classification benchmarks, and demonstrate that our model achieves state-of-the-art performance.

1 INTRODUCTION

Learning with graph structured data, such as molecules, social, biological, and financial networks, requires effective representation of their graph structure (Hamilton et al., 2017b). Recently, there has been a surge of interest in Graph Neural Network (GNN) approaches for representation learning of graphs (Li et al., 2016; Hamilton et al., 2017a; Kipf & Welling, 2017; Velickovic et al., 2018; Xu et al., 2018). GNNs broadly follow a recursive neighborhood aggregation (or message passing) scheme, where each node aggregates feature vectors of its neighbors to compute its new feature vector (Xu et al., 2018; Gilmer et al., 2017). After k iterations of aggregation, a node is represented by its transformed feature vector, which captures the structural information within the node's k -hop neighborhood. The representation of an entire graph can then be obtained through pooling (Ying et al., 2018), for example, by summing the representation vectors of all nodes in the graph.

Many GNN variants with different neighborhood aggregation and graph-level pooling schemes have been proposed (Scarselli et al., 2009b; Battaglia et al., 2016; Defferrard et al., 2016; Duvenaud et al., 2015; Hamilton et al., 2017a; Keames et al., 2016; Kipf & Welling, 2017; Li et al., 2016; Velickovic et al., 2018; Santoro et al., 2017; Xu et al., 2018; Santoro et al., 2018; Verma & Zhang, 2018; Ying et al., 2018; Zhang et al., 2018). Empirically, these GNNs have achieved state-of-the-art performance in many tasks such as node classification, link prediction, and graph classification. However, the design of new GNNs is mostly based on empirical intuition, heuristics, and experimental trial-and-error. There is little theoretical understanding of the properties and limitations of GNNs, and formal analysis of GNNs' representational capacity is limited.

^{*}Equal contribution.

[†]Work partially performed while in Tokyo, visiting Prof. Ken-ichi Kawarabayashi.

[‡]Work partially performed while at RIKEN AIP and University of Tokyo.

代码-源码: <https://github.com/weihua916/powerful-gnns>

代码-PyG: https://github.com/pyg-team/pytorch_geometric/blob/master/examples/mutag_gin.py

初步要求

1. 复现实验, 修改超参数, 可视化结果, 该步骤同GCN。
2. (可选) 在服务器上完成实验。
3. 在集群上完成实验。

进阶要求

暂无

推荐系统

推荐任务可以看作是图中特殊的链路预测任务, 即预测用户节点与哪些项目节点之间存在连接。推荐系统中的用户项目交互图是异构的二分图。

论文:

LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation

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ABSTRACT

Graph Convolution Network (GCN) has become new state-of-the-art for collaborative filtering. Nevertheless, the reasons of its effectiveness for recommendation are not well understood. Existing work that adapts GCN to recommendation lacks thorough ablation analyses on GCN, which is originally designed for graph classification tasks and equipped with many neural network operations. However, we empirically find that the two most common designs in GCNs — feature transformation and nonlinear activation — contribute little to the performance of collaborative filtering. Even worse, including them adds to the difficulty of training and degrades recommendation performance.

In this work, we aim to simplify the design of GCN to make it more concise and appropriate for recommendation. We propose a new model named LightGCN, including only the most essential component in GCN — neighborhood aggregation — for collaborative filtering. Specifically, LightGCN learns user and item embeddings by linearly propagating them on the user-item interaction graph, and uses the weighted sum of the embeddings learned at all layers as the final embedding. Such simple, linear, and neat model is much easier to implement and train, exhibiting substantial improvements (about 16.0% relative improvement on average) over Neural Graph Collaborative Filtering (NGCF) — a state-of-the-art GCN-based recommender model — under exactly the same experimental setting. Further analyses are provided towards the rationality of the simple LightGCN from both analytical and empirical perspectives. Our implementations are available in both TensorFlow¹ and PyTorch².

*Meng Wang is the corresponding author.

¹<https://github.com/kuandeng/LightGCN>

²<https://github.com/gusye1234/pytorch-light-gcn>

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SIGIR '20, July 25–30, 2020, Virtual Event, China

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ACM ISBN 978-1-4503-8016-4/20/07...\$15.00

<https://doi.org/10.1145/3397271.3401063>

CCS CONCEPTS

• Information systems → Recommender systems.

KEYWORDS

Collaborative Filtering, Recommendation, Embedding Propagation, Graph Neural Network

ACM Reference Format:

Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20)*, July 25–30, 2020, Virtual Event, China. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3397271.3401063>

1 INTRODUCTION

To alleviate information overload on the web, recommender system has been widely deployed to perform personalized information filtering [7, 45, 46]. The core of recommender system is to predict whether a user will interact with an item, e.g., click, rate, purchase, among other forms of interactions. As such, collaborative filtering (CF), which focuses on exploiting the past user-item interactions to achieve the prediction, remains to be a fundamental task towards effective personalized recommendation [10, 19, 28, 39].

The most common paradigm for CF is to learn latent features (a.k.a. embedding) to represent a user and an item, and perform prediction based on the embedding vectors [6, 19]. Matrix factorization is an early such model, which directly projects the single ID of a user to her embedding [26]. Later on, several research find that augmenting user ID with the her interaction history as the input can improve the quality of embedding. For example, SVD++ [25] demonstrates the benefits of user interaction history in predicting user numerical ratings, and Neural Attentive Item Similarity (NAIS) [18] differentiates the importance of items in the interaction history and shows improvements in predicting item ranking. In view of user-item interaction graph, these improvements can be seen as coming from using the subgraph structure of a user — more specifically, her one-hop neighbors — to improve the embedding learning.

To deepen the use of subgraph structure with high-hop neighbors, Wang et al. [39] recently proposes NGCF and achieves state-of-the-art performance for CF. It takes inspiration from the Graph Convolution Network (GCN) [14, 23], following the same



LightGCN-PyTorch-master.zip

42.77MB



代码-PyG: https://github.com/pyg-team/pytorch_geometric/blob/master/examples/lightgcn.py

提示: 集群上部署请使用PyG代码。

初步要求

1. 自问自答: 什么是协同过滤? 协同过滤在LGN中是如何体现的? 用户与项目的Embedding表示什么意思? 什么是BPR损失? BPR损失为什么能有效训练模型的推荐效果? BPR损失是如何生效的?
2. 可视化损失以及其他指标, 同时判断什么时候出现了过拟合。提示: 使用pyplot包画出曲线图
3. (可选) 在服务器上完成实验。
4. 在集群上完成实验。
5. 学习率、Embedding维度等调整, 可视化对比效果。 (新增) 新实验指标请使用Recall@10, Pre@10与NDCG@10。
6. (新增) 使用PyTorch的nn.Module类和PyG中的 `torch_geometric.nn.conv.GCNConv` 实现一个简单的卷积网络完成推荐任务。可视化对比LGN与这个模型之间的损失, 指标变化, 横坐标为迭代次数。提示: 主要流程可以参考代码-PyG中的流程。

进阶要求

7. 数据集换成Movielens, 同时训练集: 验证集: 测试集为8:1:1, 可视化训练效果。



ml-1m.zip

5.73MB



8. 寻找其他可以训练推荐系统的损失函数, 例如L2范数, 对比BPR损失, 哪个损失训练效果更好, 为什么有些损失函数无法训练高效的推荐模型。提示: BPR是排名损失, 存在正对与负对, 但是L2不需要负对, 反而需要评分信息。如果觉得训练太慢可以换小一点的数据集。
9. 使用MLP搭建传统双塔推荐模型, 然后使用表格记录LGN与该双塔变体模型的实验结果, 实验指标请使用Recall@10、Pre@10与NDCG@10。
10. 传统双塔推荐模型指分别处理用户Embedding和项目Embedding后合并。考虑一种情况, 使用LGN处理用户项目数据, 作为一个塔, 使用用户社交网络作为另一个塔, 两塔数据通过聚合层聚合, 聚合方法可以是拼接两塔数据再过MLP; 可以通过注意力机制加权聚合。提示: 用户社交网络简化为如果两个用户之间有共同的交互项目, 则他们之间存在一个连边。
11. 实验9中效果与LGN存在一定差距, 思考为什么会出现这种差距。提示: 考虑LGN的底层原理

12. 参考GCN任务中的攻击实验。在Movielens数据集中完成推荐系统中毒与逃逸攻击实验：启发式攻击请使用RandomAttack、AverageAttack、AoPAttack与BandwagonAttack；生成式攻击请使用RAPU。可视化攻击后的模型推荐指标变化。提示：推荐系统中的中毒攻击也叫先令(shilling)攻击，本次实验只考虑用户方面的数据修改，不修改项目数据。

十、科研上手捷径

这边讲述了一些科研中会遇到的问题及其解决方法。

但还是要边学边解决问题。

搜文献篇

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3. 期刊/会议官网搜索

第一步：找到对应的期刊/会议官网。

第二步：找到paper所在的地方（这步可能会有点难找）。

以ICML 2024 为例。

2024 Conference

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Sun Jul 21st through Sat Jul 27th

July 21 through July 27, 2024.

Announcements

Welcome to ICML 2024! [Registration](#) is now open for 2024. A blog post describing some of the elements of ICML this year is [here](#).

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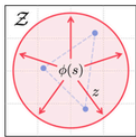
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[Boris van Breugel](#), [M van der Schaar](#)
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- has part: KDD Workshop on Causal Discovery, Prediction and Decision (CDPD)
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30th KDD 2024: Barcelona, Spain

Ricardo Baeza-Yates, Francesco Bonchi:
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29th KDD 2023: Long Beach, CA, USA

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第四步：用快捷键ctrl+F 利用关键词搜索论文。

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5. 知网搜索

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- 然后文献一般是按照Introduction、Methods、Results、Discussion和Conclusion五个部分组成
- 要把握每一个部分的作用，这样在阅读时才能抓住重点，尤其是需要进行PPT汇报时，很多同学不知道哪些部分该汇报，哪些部分可以略过。抓住了文献各部分的重点才能让我们自己写作的时候精简语言！

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文献的标题

Using citizen science to understand river water quality while filling data gaps to meet United Nations Sustainable Development Goal 6 objectives

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a, b等字母表示作者单位

HIGHLIGHTS

- Citizen scientist project to highlight pollution sources on a Dublin urban river
- High nitrates ($\text{NO}_3\text{-N}$) and phosphates
- Highlight突出本篇文章的个体贡献
- Aims to fill data gap for United Nations (UN) Sustainable Development Goal (SDG) 6
- Citizen scientist qualitative observations important in environmental reporting

GRAPHICAL ABSTRACT



图形摘要, 可以理解为简略的技术路线图, 以图片的形式说明文献主要研究问题

文章投稿信息, 初稿、返修稿、接收和见刊的时间

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Land use

River inputs

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文章关键字, 方便读者get文章的研究

领域和研究内容

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文章的通信作者, 一般是导师, 如果想要寻求文章作者帮助, 可以email通信作者

<https://doi.org/10.1016/j.scitotenv.2021.146953>

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1. Introduction

"Ambient fresh water quality is at risk globally". This is one of the key messages from the United Nations (UN) Sustainable Development Goal (SDG) 6 Synthesis Report on Water and Sanitation (UN, 2018). Target

正文部分一般分为Introduction、methods、results、discussion和conclusion五个部分

member states by existing and new national and regional legislation. For example, in the European Union (EU), the Water Framework Directive (WFD) (2000/60/EC) aims to maintain the quality of all fresh waters across member states by a) retaining "good status" in currently healthy water bodies, and b) raising the status of poorly performing waterbodies to "good" level. The status of waterbodies is monitored through national monitoring programmes, with management actions based on the catchments of water bodies (EPA, 2006).

The involvement of communities in water resource management at the catchment scale is also central to SDG 6, as Target 6b aims to support and strengthen their participation which can contribute to the conservation, restoration and sustainable use of freshwater ecosystems and their services (UN, 2018). The WFD also aims to have citizen participation, stating that public and stakeholder participation in water resource management is required. It thus aims to enhance resource management and involve individuals and groups in a democratic way. Evaluation is also required of any participatory projects to assess whether objectives are being achieved and to identify how improvements can be made (Carr et al., 2012). Thus clear guidance is needed so that data generated is fit-for-use by national agencies, and therefore included in the SDG reporting process (Fritz et al., 2019) as well as for the WFD.

Citizen science (CS) (i.e. the involvement of non-professional scientists in scientific research and data gathering (Bonney et al., 2009), pre-

Introduction一般需要说明研究问题的必要性、描述研究主体的定义、梳理目前该问题的研究现状以及目前仍存在什么不足之处

to have broader environmental, social, economic, and political benefits (Hecker et al., 2018), with the extent to which these are realised dependent upon the type of citizen science deployed (Van Noordwijk et al., 2021). The UN recognises that citizen science has the potential to increase the knowledge base that underpins progress towards the SDGs by increasing the frequency and geographic range of available data, making better use of local knowledge, and educating the public (UN Environment, 2019). The EU has also identified the need for complementary data while monitoring EU environment policy, stating that citizen science offered an alternative and cost-effective way to collect environmental data which may be useful in providing early warnings about environmental trends and specific problems (EC, 2020).

Against a background of increasing concern for the status of fresh water across the world, the Water Institute of Dublin City University (DCU) launched BACKDROP in May 2019. This project aimed to train a team of citizen scientists with limited prior experience of freshwater research to understand and regularly measure water quality along different parts of the river Liffey in Dublin city, Ireland. Dublin, along with most cities worldwide, is experiencing increasing urbanisation and land-use change which has impacted on freshwater ecosystems, their biodiversity and recreational use. The river Liffey, which flows through

the centre of Dublin city on its final course to the sea, is subject to run-off from urban surfaces and through storm water drains. This input into fresh water can have a detrimental effect on the water quality as increased nutrients from anthropogenic activity can damage freshwater ecosystems (Paul and Meyer, 2001; Sonoda et al., 2001; UN Water, 2018). There is also anecdotal evidence of residential discharges from misconnected sewer pipes into areas along the river's course, similar to what has been reported in other cities (Ellis and Butler, 2015). The Irish Environmental Protection Agency (EPA), which monitors water quality nationwide in line with the WFD, has monitoring stations along the course of the river Liffey (Fig. 1). However as with all statutory monitoring networks, some areas may be overlooked, due to resourcing, or may not be sampled at a time when pollution incidents are occurring. For the BACKDROP project, citizen scientists, in particular the community of urban paddle boarders and other river users (kayakers, rowers), collected data along the urban Liffey from the west of the city to its point

基于对该问题已有研究的梳理，尤其是已有研究的不足之处，提出本研究的主要工作

This paper focusses specifically on the utility of data collected by the citizen scientists in the BACKDROP project, and asks the following questions:

1. Do urban drainage systems have localised effects on water quality?
2. Does the Liffey experience localised water quality responses to rainfall events as a result of urban drainage systems?
3. To what extent can the spatially and temporally granular data collected by citizen scientists be used to fill gaps in data and information in the statutory monitoring programme?

Using data collected by citizen scientists over the course of nine months from the urban stretch of the river Liffey in Dublin, Ireland, this study aims to answer these questions.

2. Methods

2.1. Site description

The river Liffey rises 125 km from Dublin city in the Wicklow mountains and drains a catchment area of 1616 km² containing a population of 1.26 million people. This catchment comprises 17 sub-catchments with 77 river water bodies, six lakes, six transitional and five coastal water bodies, and 16 groundwater bodies (EPA, 2018a). This study is concerned with the last 16 km of the Liffey's journey, from Lucan in the west of the city to where the river enters the Irish Sea at the East Link Bridge (Fig. 1). The main inputs in this section are the rivers

Methods部分则是说明本研究所用的数据来源、所用指标以及统计分析方法，同时预先分析的结果可以在方法部分说明，作为选择某种特定统计分析方法的依据

from good to poor (EPA, 2020b). The Liffey flows through suburban parkland for 7 km from Lucan to Chapelizod (between CS sites 1 and 3, Fig. 1) where land use changes to pockets of suburban and industrial development for another 3 km to Islandbridge (between CS sites 3 and 9, Fig. 1). For the remaining 6 km the river continues through an urban landscape. Anecdotal evidence of residential discharges into areas along the river's course where land use changes to suburban with an increased housing stock as a result of a building boom in the past 20 years makes this a particular area of interest for this study. The suburban

sanitation (EPA, 2018b). There are no national environmental quality standards for nitrate in Ireland, but average nitrate concentration values less than 0.9 mg/l $\text{NO}_3\text{-N}$ (4 mg/l NO_3) are considered as indicative of high quality water by the EPA. Values between 0.9 mg/l and 1.8 mg/l $\text{NO}_3\text{-N}$ (8 mg/l NO_3) are considered by the EPA to be indicative of good water quality (EPA, 2020e).

Average phosphate concentrations of less than 0.025 mg/l P and less than 0.035 mg/l P have been established in Ireland as environmental quality standards (EQS), which are legally binding to support the achievement of high and good ecological status as required by the WFD (European Communities Environmental Objectives (Surface Waters) Regulations, 2009; EPA, 2018b). The resultant nitrate and phosphate limits for the study are outlined in Table 1 with associated FWW parameter categories and midpoint values. For the purposes of this study, nitrate values of <1.0 mg/l $\text{NO}_3\text{-N}$ are deemed to be indicative of high quality water, values 1.0–2.0 mg/l $\text{NO}_3\text{-N}$ are interpreted as indicative of good quality, and values >2.0 mg/l $\text{NO}_3\text{-N}$ indicative of lower quality. For phosphates, values <0.025 mg/l P are described as indicative of high quality water, values of 0.025–0.035 mg/l P as indicative of good quality water while values >0.035 mg/l P were deemed indicative of lower quality water.

3. Results

3.1. Spatial patterns of water quality

Fig. 2 shows a box plot of the midpoint values of all samples of nitrates (a) and phosphates (b) recorded at 19 locations along the Liffey. The site locations are as per Fig. 1. This data represents 101 samples

Results部分则是对文章的结果进行说明，但需注意只需要描述数值结果，不需要分析为什么会出现这个结果

ples, good quality in 35.6% ($n = 36$) and of lower quality in 35.6% ($n = 36$). There were spatial differences in nutrient concentrations across the sample locations, shown in Fig. 2. These differences can be understood by examining the three distinct types of land use through

which the river flows: parkland; pockets of suburban and industrial development; and urban tidal landscape (Fig. 3).

The parkland section (sites 1–2) was less accessible, therefore fewer samples were obtained (5 in total, compared to monthly samples at other sites). In this stretch of the river, there were no measurements indicative of low water quality. However, the lack of sampling along this stretch makes it difficult to determine whether these results are representative of the overall water quality for this section over the study period. The sites within the suburban and industrial section (sites 3–9) were more frequently sampled, giving a wider range of data for both nitrates and phosphates. Within this stretch, high nitrate levels indicating low water quality were recorded in 6 samples (16%). High phosphate levels, indicative of low water quality, were recorded in 22 samples (21.7%). 61% of all high phosphate samples indicating low water quality were found within this stretch, and 60% of nitrate samples which indicate low water quality located here. Sample numbers in the urban tidal landscape were higher ($n = 66$) due to greater accessibility. Here, the river runs through the centre of Dublin city (Fig. 1). In this stretch of the river, there were 4 samples with high nitrate levels (lower water quality) and 13 phosphate results indicating lower water quality.

The data suggests that the suburban/industrial section of the study site 8 on Fig. 1) has a lower water quality. This will be discussed further

3.2. Impacts of rainfall on water quality

By examining the hydrometric data for the river Tolka at the Botanic Gardens and the rainfall data for the same time period at the monitoring station at the same location, it was found that rainfall of up to 5 mm had no impact on the water flow in the Tolka, while flow increased by 2–3 m^3 the day following a rainfall event of up to 10 mm. Water flow at least doubled the day following a rainfall event of greater than 10 mm. This suggests that surface runoff from rainfall is a contributor to the rivers in this area when daily precipitation is more than 5 mm.

By applying these criteria to the data collected by the citizen scientists on the Liffey, the day preceding low quality water results were checked for significant rainfall that may have influenced water flow, and therefore the potential quality across the study area. Of the 9.9% of samples ($n = 10$) in the low quality category for nitrates (>2.0 mg/l $\text{NO}_3\text{-N}$) (Fig. 4a) six 描述结果时，需要标注出来这个数值结果是在哪个图里或是哪个表里体现 the

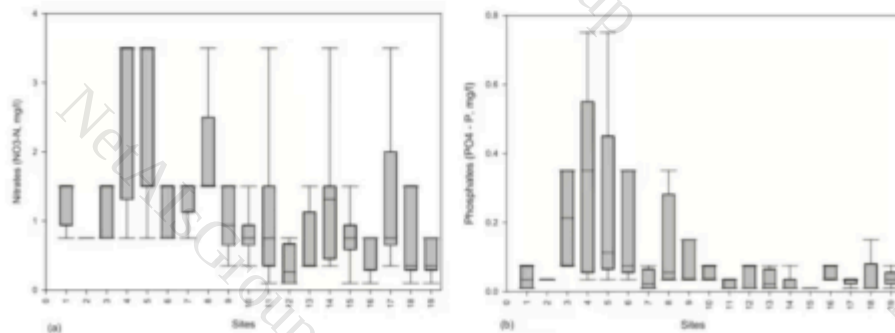


Fig. 2. Nitrate and phosphate results (with result ranges) from the citizen scientist sample sites along the river Liffey.

Four sites (CS sites 4, 5 and 6, Fig. 1, all within the Chapelizod / Islandbridge area) recorded both nitrate and phosphate levels indicative of low water quality on the same sampling occasion. One of these sites, site 4, recorded nutrient levels indicative of low water quality on three distinct occasions. It is noteworthy that these three sites are adjacent to each other, suggesting a localised issue of pressures on water quality at this location around Chapelizod. The citizen scientists reported seeing water discharged into the river at these points, and on three sampling occasions, recorded in the notes: 'discharge smells of detergent' / 'cloudy discharge from pipe directly into river' / 'Unknown cloudy discharge from beneath the water level'. The significance of these notes and findings will be discussed further below.

3.4. Complementarity of citizen science and agency data

EPA sampling took place at 5 locations which correlated with BACKDROP citizen scientist sampling locations (Fig. 1). 39 samples were tested for nitrate and 38 for phosphate by the EPA at these locations during the period covered by the BACKDROP project. The correlation of EPA results with BACKDROP results in the brackish water transitional section of the Liffey, and upstream, suggests that the methodology used in FWV is robust also for this section of the Liffey (Fig. 5). However, the data collected by the citizen scientists in this study showed a greater range of phosphates in particular than those recorded by the EPA at the Chapelizod site.

35.9% of the EPA samples ($n = 14$) were indicative of low water quality because of excessive nitrates (>2.0 mg/l $\text{NO}_3\text{-N}$). Of these

samples, 7 occurred during some or significant rain and 7 after little or no rain, and all were from locations upstream of the weir at Islandbridge. For phosphates, the EPA recorded no phosphate levels which were indicative of lower water quality (Fig. 5).

4. Discussion

Citizen science has the potential to produce data at a much higher spatial frequency than traditional monitoring techniques, and for this reason it is commonly cited as a means to increase data coverage and to fill gaps in regulatory monitoring (Quinlivan et al., 2020b). In this

Discussion是对研究的总结，第一段需简述研究利用什么数据针对什么研究问题得到了什么研究结果

use across the city, and at a resolution greater than that carried out by agency monitoring. This allowed trends in water quality related to these land use changes to be examined. The data showed that nutrients, particularly phosphate, were elevated as the river flowed through suburban and industrial land, particularly in the Chapelizod area. This is

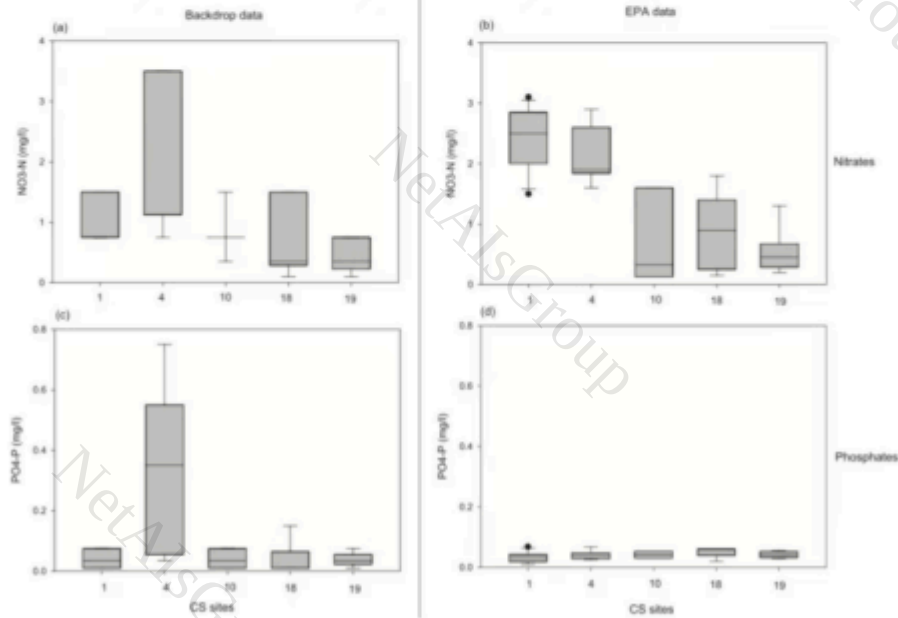


Fig. 5. Ranges of data from five locations sampled by both EPA and Backdrop citizen scientists over the same period. Locations are as per Fig. 1. Nitrates BACKDROP, (b) nitrates EPA, (c) phosphates BACKDROP and (d) phosphates EPA. Backdrop samples CS site 1 ($n = 3$), 4 ($n = 5$), 10 ($n = 7$), 18 ($n = 6$), 19 ($n = 5$). EPA samples CS site 1 ($n = 14$), 4 ($n = 7$), 10 ($n = 3$), 18 ($n = 7$), 19 ($n = 8$).

likely due to phosphate inputs from urban discharges, which are common in urban areas (Ternus et al., 2011). It also showed decreases in nutrients further downstream, towards the highly urbanised city centre. Here, tidal dilution as well as inputs from rivers and canals along this stretch may also be a factor, but this has not yet been formally assessed (Caccia and Boyer, 2005; EPA, 2019a,b).

Research has shown that water quality can also be impacted by seasonal variations in nitrate and phosphate which also relate to land cover (Mouri et al., 2011; Pratt and Chang, 2012; Shi et al., 2017; Shupe, 2017). Because citizen scientists in BACKDROP measured monthly, it was possible to investigate the interactions between weather, land-use, and water quality at a more granular level. Indeed, the data revealed differences in the impacts of individual rainfall events on nutrients in the upstream areas versus the downstream areas. Upstream of Islandbridge, where land cover was either parkland or suburban, poor water quality was associated generally with periods of lower rainfall. However, in the estuarine stretch, where the river runs through the city centre and is subject to runoff from hard urban surfaces, a number of high nutrient measurements occurred in association with rainfall events.

It is important to note that the use of citizen science does not always

can inform this type of remedial work, and can add valuable data if similar projects are rolled out in other local authority areas.

Interestingly, the qualitative observations of citizen scientists described above all relate to locations where land use changes and parkland gives way to pockets of suburban and industrial development. Both the qualitative observations and areas of high phosphates measured by BACKDROP citizen scientists also correspond to areas where intensive housing development took place along the course of the Liffey in the early 2000s, during what was termed the Celtic Tiger period (Fig. 6). Urbanisation generally tends to decrease water infiltration into the soil and concentrate flows in storm drains and channels as the percentage of impervious surface is increased, with a resultant impact on the ecology of nearby rivers and streams (James and Lecce, 2013). Furthermore, during the Celtic Tiger period, Ireland experienced rapid economic growth and a building boom and strategic planning practices were absent or poorly implemented; many housing developments consisted of suburban sprawl on greenfield sites (Winston, 2007). Urban land expansion had an annual growth rate of 2.5% in Ireland between 2000 and 2012, among the highest rates in Europe (Ahrens and Lunn, 2019). This unprecedented growth took its toll how-

在总结完主要研究结果后各段文字的组织形式如下:

- 1、讨论研究结果和既往结果的异同点
- 2、讨论相同或不同的主观和客观原因
- 3、讨论结果的应用价值和方法的启发

and degradation which can have an environmental impact with nutrient losses (nitrogen and phosphorus) through run-off and leaching, especially with excessive rainfall (Bock and Easton, 2020).

adding value to their body of data (Winston et al., 2017; Starkey et al., 2017), and the BACKDROP study further supports this.

4.2. Significance of qualitative data from citizen scientists

In addition to quantitative measurements of water quality, various qualitative and photographic observations from BACKDROP citizen scientists measuring locations upstream of Islandbridge indicate instances of pollution. Photographs showed cloudy, frothy discharge from pipes into the river, while citizen scientists noted evidence of eutrophication and floating mats of algal growth in these locations, some of which correlated with high nitrate and phosphate results during periods without rainfall. Many studies have cited the presence of algal blooms as an indicator of the degradation of water quality based on measurements of nutrient (nitrate and/or phosphate) concentrations (Cunha et al., 2017; Thornhill et al., 2017). Frothy discharge from pipes, meanwhile, is indicative of drainage system misconnections; indeed, as noted above, some participants recorded witnessing a smell of detergent at some locations. Household appliances that are wrongly connected to separate surface water sewers can potentially lead to pollution of receiving waters with elevated concentrations of nutrients (phosphate, ammonia, nitrates) and non-compliance with statutory water quality standards (Ellis and Butler, 2015). The main pollutant loading contribution to misconnections arising from washing machines and dishwashers is $\text{PO}_4\text{-P}$. Even with substantial dilution in the order of 50–100:1 to conform to ecological criteria, the receiving water would be of poor quality status (Revitt and Ellis, 2016).

The EPA have cited urban domestic misconnections as a concern for urban freshwater quality in Ireland (EPA, 2017). Dublin local authorities have estimated that 8% of houses are misconnected with other sewers, and have suggested that this figure may be as high as 20% in some areas (EPA, 2019b). In 2020 one of Dublin's local authorities, South Dublin Council, began a project (Dublin Urban Rivers Life) to map and remedy these. While the study area is not within the catchments covered by the Dublin Urban Rivers Life, the methodology in this study

4.3. Citizen scientists' contribution to monitoring schemes

Research has shown that citizen science has potential to augment professional water quality monitoring by generating additional data on previously under-reported freshwater ecosystems (Hadj-Hammou et al., 2017), and also fill the data gaps as suggested by the UN and EU (McGoff et al., 2017; Scott and Frost, 2017). In its recommendations in 'Best Practices in Citizen Science for Environmental Monitoring', the European Commission has suggested that citizen science could be a cost-effective way to complement environmental reporting in the priority areas of the Green Deal, and initiatives to fight environmental pollution could be identified and supported as part of the zero pollution ambition (EC, 2020). The report also recommended that both the EU and its member states could provide funding and support for these initiatives to contribute to the knowledge base required, while evaluating their policy and societal impacts (EC, 2020).

This study not only shows that citizen science data can detect the same environmental trends as data collected by statutory agencies, but also demonstrates that citizen science data can capture additional relevant information. The data collected by BACKDROP citizen scientists aligned well with statutory (EPA) data both spatially and temporally (Fig. 5). Both datasets showed a lowering of nitrate and phosphate concentrations and a reduction in their range as sampling locations move to the mouth of the river and waters are diluted by the tides. Additionally, there were some nuances in the data that were not visible in the EPA dataset but were revealed by the citizen scientists. In particular, the range of data for BACKDROP samples is wider in the Chapelizod area of the study region, where citizen scientists took measurements at different times and in different locations to the EPA. Here, the range of nutrient concentrations recorded by citizen scientists was greater than those recorded by the EPA (Fig. 5), and citizen scientists are therefore possibly detecting pollution instances that are not being picked up

Discussion差不多倒数第二段的位置还需要简要说明研究的局限性，和作者为了解决局限性做出的努力

its use as a research tool, concluded that programmes which partnered with academic institutions and whose participants had sufficient training using standardised protocols with an accessible dataset were more likely to be utilised.

As part of its recommendations in 'Best Practices in Citizen Science for Environmental Monitoring' (EC, 2020), the European Commission has many suggestions for public authorities at EU level and in member

Discussion最后一段可以简述一下结果的应用价值以及未来的研究方向

organisations like FWW, which are available on an open platform, continue to monitor water quality and bring the findings to the notice of relevant environmental bodies.

5. Conclusions

This citizen science project assessed water quality along the river Liffey to fill the data gaps in environmental monitoring for UN SDG Indicator 6.3.2, while detecting any significant detrimental nutrient inputs to the river that may have come from storm drains and residential

Conclusion总结本研究的发现以及研究价值或者对具体问题的回答

was also analysed to assess the complementarity of CS data and the findings concluded that the range of data from the BACKDROP project spatially and temporally enhanced national agency monitoring. An expansion of the programme to better distribute sampling points and continued monitoring would also strengthen the case for use of the data by the appropriate national agencies. The use of the tried and tested FWW methodology by a CS programme partnered with an academic institution and carried out by trained citizen scientists has the potential to assist in progressing SDG Indicator 6.3.2 upwards from Tier 2 to Tier 1. The results produced from this study demonstrate the value of citizen scientists in assessing water quality which cannot be underestimated.

Credit是指每一位作者对研究的具体贡献

CRediT authorship contribution statement

Susan Hegarty: Conceptualization, Data curation, Visualization, Formal analysis, Writing – review & editing, Supervision. **Anna Hayes:** Writing – original draft, Visualization, Formal analysis. **Fiona Regan:** Writing – review & editing, Funding acquisition. **Isabel Bishop:** Resources, Writing – review & editing. **Ruth Clinton:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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研究需要感谢的合作单位、个人或者是项目等的支持

Watch. DCU Water Institute also contributed funding to this project. The anonymous reviewers made very helpful comments to improve

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References

最后就是研究的参考文献，需要按照期刊要求组织参考文献的格式！

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作者之间或作者与期刊编辑是否有潜在的需要说明的利益冲突

第一步：明确阅读的目的与核心问题

- 我为何要读它？
 - 是为了了解某个特定知识点？学习一种新的实验方法或分析思路？还是为了借鉴其写作框架和论证逻辑？
- 我想从中获得什么？
 - 阅读的全过程都要抓住自己的阅读重点，不要读着读着就“散了”。

第二步：构建文献的“第一印象”

- **借助他人智慧:**高质量的文献(尤其是需要精读的经典之作)往往已有前辈或领域大V进行过解读。可以在微信公众号、知乎、学术论坛等平台搜索相关解读,快速了解文章的核心贡献、主要思路和其他人视角下的重点。
- **AI 助手辅助梳理:**
 - **推荐工具:**可以尝试 Gemini、Deepseek、chatGPT等。不同 AI各有千秋,建议都体验一下,找到最适合自己逻辑和需求的。例如, Gemini 在信息整合和遵循指令方面表现稳健;Deepseek 在逻辑推理上可能有惊喜;ChatpT 则在语言流畅性和联想能力上较强(chatgpt有的时候会像脱野马~般发散开,但对于头脑风暴很有用)。
 - **构建个人精读笔记大纲:**在请求 AI 协助前,最好先根据自己的理解和需求,搭建一个初步的笔记框架。
 - **参考 Prompt:**“你好,我正在进行文献精读。请基于我提供的文献[粘贴文献标题/摘要/DQ!/(但最好直接附上文件pdf,并参考我以下的笔记大纲结构,为我生成一份初步的文献精读笔记。我的大纲包括:[列出你的大纲要点,如研究背景、科学问题、主要方法、核心发现、主要结论、亮点与不足等]。
 - **绘制思维导图/流程图:**结合上述信息,亲自动手梳理文章的整体思路和分析脉络。将主要模块、实验流程、逻辑关系等画出来,能默写或复述一遍更佳。

文献精读的核心步骤

第一步:快速定位核心信息 skimming & scanning

- **标题:**精准概括研究主题。
- **摘要:**浓缩的精华。重点关注其提出的研究背景、核心问题、主要方法、关键结果和 主要结论。
- **引言:**
 - 引人入胜的“故事开篇”。通常包含以下逻辑层次:
 - i. 研究领域现状/问题严重性
 - ii. 已有研究进展与不足
 - iii. 本文研究的切入点/重要性/假说理解这部分,能让你明白作者“为什么“要做这项研究,并学习其“讲故事”的技巧。。读完后,即可初步填充笔记中的【文章背景】和【研究问题】
- **讨论-初读:**快速浏览讨论部分的首段和末段,通常这里会重申主要发现并点明其意义和局限性。与摘要内容相互印证,加深对文章核心贡献的理解。

阶段性小结:此时精读笔记应该整理好这些

标题:
年份/期刊/影响因子:
关键词:
文章背景:
研究问题:
方法:
关键发现:

第二步:精读结果部分

- 分段解析:通常,结果部分会按照 Figure/Table(图/表)或实验组别来组织,将其划分为若干小节进行精读。
- 关注重点(因人而异,以下为参考):
 - 实验设计与分析流程:
 - 每个Figure/Table背后对应怎样的实验设计?数据是如何采集和分析的?
 - 关注关键步骤、参数设置、使用的统计方法和软件工具。
 - 先根据Result部分的描述总结,后续再到Methods部分查找更详尽的补充。
 - 核心数据与图表解读
 - 先看图/表本身:
 - 仔细观察图例、坐标轴、单位、样本分组、显著性标记(*,p-value)等。
 - 在不看正文描述的情况下,理解图表传递的核心信息和初步结论,。“这张图想告诉我什
么?”
 - 再读图/表对应的正文描述:
 - 找到对该图/表的解释。
 - 一个小技巧:可以直接在文章中搜索“Fig. “或” Figure”,这样可以快速定位到与图表相关的描述段落,这些段落通常包含了关键的结论。
 - 总结该图/表的核心结论:用自己的话,清晰、简明地总结这个图/表证明了什么。
 - 逻辑串联:思考当前结果与上一个结果之间的逻辑关系,它们是如何共同支撑一个小论点的。
 - 数据验证与意义阐释:作者是如何确保结果可靠性的?这个结果说明了什么问题?它有什么潜在
- 阅读策略:
 - 原文为主,翻译为辅:尽量直接阅读英文原文,遇到难懂的长句或段落,可借助翻译工具辅助理解,但务必对照原文,避免翻译引入的偏差。
 - 主动输出,边读边记:不要只是被动输入。一边阅读,一边用自己的语言将理解的关键信息、实验逻辑、数据结论等整理到Zotero Better Notes或其他笔记软件中。

- 标记疑问:遇到不理解的术语、缩写、统计方法或逻辑跳跃, 立即在原文或笔记中标记出来(例如: “CLDN-1是什么?”“为何这里用这个统计方法?”)。一部分疑问可能随着阅读深入自然解开, 剩下的则在通读全文后集中查阅资料解决。

第三步:深入讨论与方法

- 精度讨论
 - 结果的深入解读:作者如何将各个Results串联起来, 解释其研究结果的整体意义?(例如, “ABCA12功能缺失导致屏障破坏 “是对多个实验结果的综合解读)
 - 回应引言中的问题:研究结果是否有效地回答了引言中提出的科学问题或验证了假说?
 - 与前人研究的联系与区别:作者如何将自己的研究成果置于现有知识体系中?有哪些创新和突破?
 - 研究的局限性:诚实地指出研究中存在的不足(如样本量小、未考虑某些变量、方法本身的局限等)。这不仅体现了科研的严谨性, 也是未来研究可以深入的方向。
 - 未来展望:基于本研究, 作者对未来研究方向有何建议?
- 查阅方法
 - 针对在阅读Results时遇到的实验细节、参数设置、试剂来源、统计分析等疑问, 回到Methods部分寻找答案。
 - 对于你想学习和借鉴的方法, 仔细研读其操作步骤和关键参数。

第四步:关注补充材料

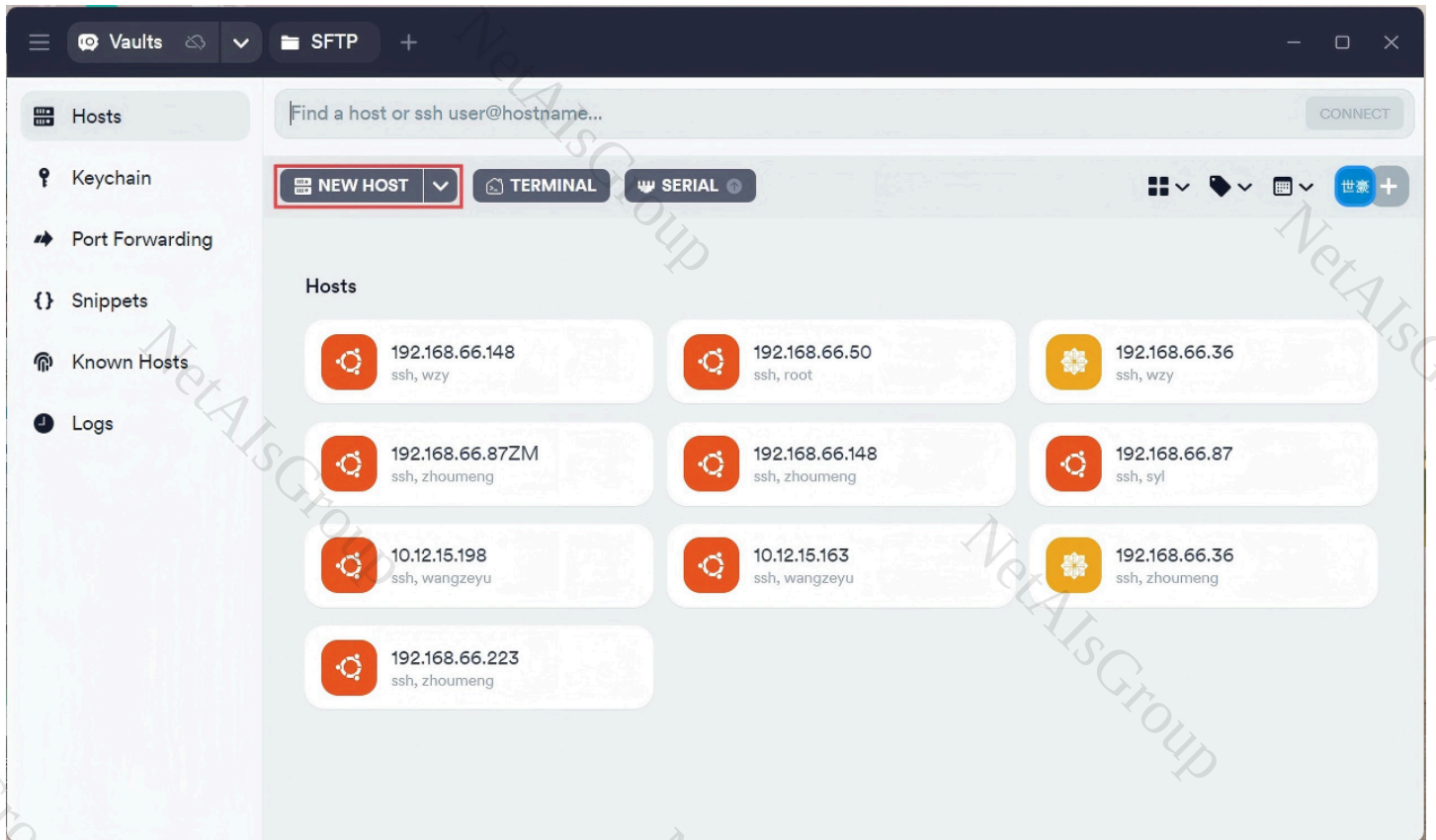
- 补充材料往往包含了主刊中无法展示的额外数据、详细方法、阴性结果等。对于理解文章的完整性和严谨性非常有价值, 有时甚至能发现意外的“宝藏”。

服务器篇

我们平时跑的代码需要的计算资源都很大, 用自己的电脑跑是行不通的, 所以一般都是使用集群或者服务器来跑代码, 集群的使用已经讲过了, 这里是服务器的使用。

1. 服务器

首先, 下载Termius软件来使用服务器。(下载地址: <https://termius.com/>)



CONNECT

New Host
Personal vault ▾

Address



General





 **Default**

Share this host 

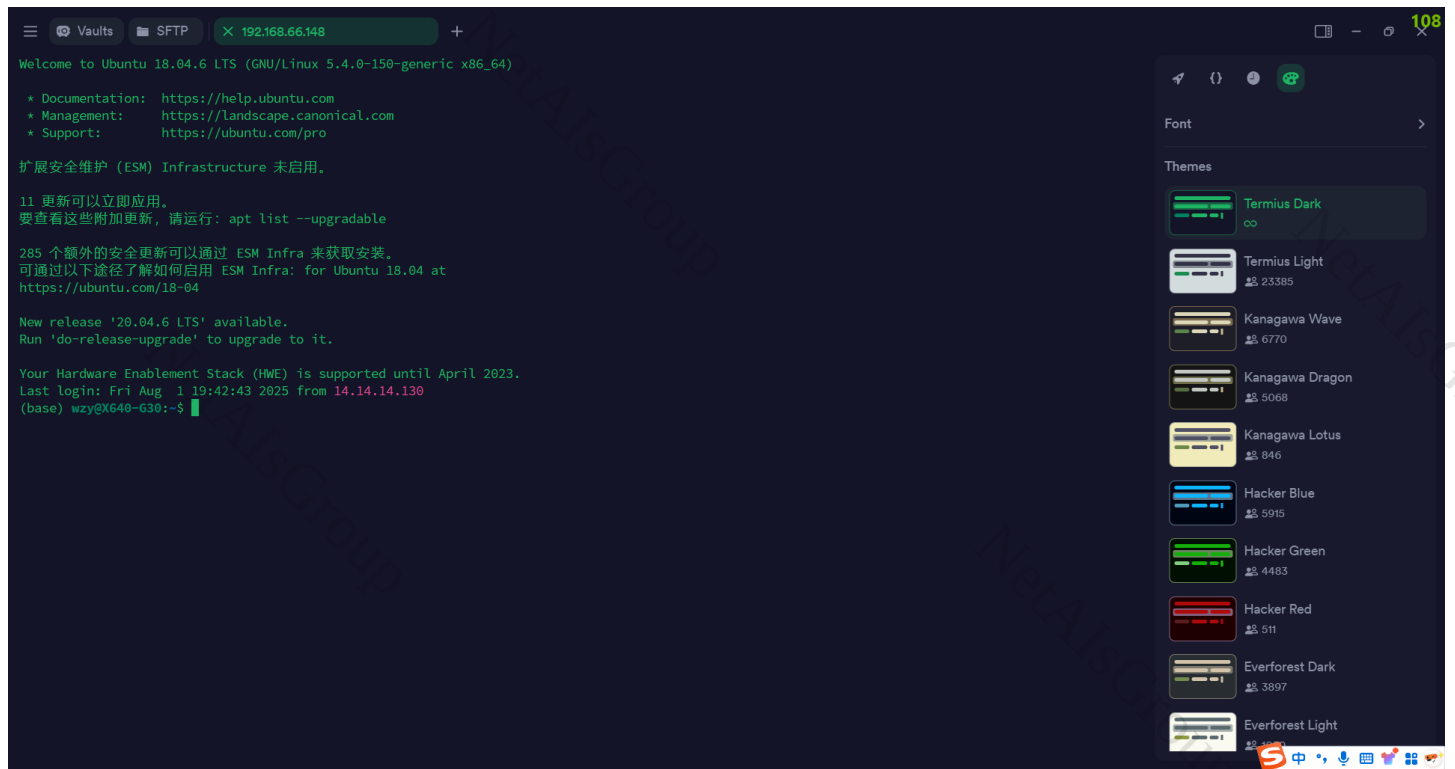
SSH on **port**

Credentials





像上图一样添加新的服务器，输入服务器的ip、端口号和账户和密码（这些后面需要直接找师兄师姐要）。



这便是服务器进来的样子，之后我们就可以输入指令来跑我们的代码了。

1.1 服务器中常用指令

进入指定目录：`cd` 回退文件目录：`cd ..` 查看当前目录：`ls`

启动新的screen会话：`screen -S 名字`（名字是自己起的字母数字什么都行）

查看已有的screen：`screen -ls`

进入已有的screen：`screen -r 名字`

删除已有的screen：`screen -X -S 名字 quit`

从screen中退出：`Ctrl+A+D`

查看环境列表：`conda env list` 激活环境：`conda activate 环境名`

查看显卡的使用情况：`nvidia-smi` 进程查看：`htop`

（日常使用服务器跑代码，这些指令基本够用了，如果用到别的指令，网上都可以搜索到）

1.2 使用服务器跑代码

在跑代码前，需要确定你的代码在你使用账号的文件中，然后就可以找到你代码的路径去跑代码。

192.168.66.148

Filter

Actions

<

>

public

>

wzy

Name	Date Modified	Size	Kind
..			
Yaoqiang drwxr-xr-x	5/22/2025, 2:55 PM	--	folder
requirements.txt -rw-rw-r--	10/16/2024, 7:29 PM	4.23 kB	txt
zhangying drwxr-xr-x	5/22/2025, 10:20 PM	--	folder
83 drwxrwxr-x	9/29/2022, 5:19 PM	--	folder
save_path drwxrwxr-x	10/14/2024, 1:13 AM	--	folder
tm_pytorch2.tar -rwxrwxr-x	1/10/2025, 8:28 PM	23.12 GB	tar
create_user.sh -rw-rw-r--	4/23/2024, 10:26 PM	306.00 Bytes	sh
syd drwxr-xr-x	1/25/2025, 4:46 PM	--	folder
more_itertools-10.5.0-py3-none-any.whl -rw-rw-r--	11/6/2024, 1:28 PM	59.52 kB	whl
rdkit-Release_2024_09_4.tar.gz -rw-rw-r--	3/26/2025, 1:33 PM	78.25 MB	gz
111.csv -rw-rw-r--	4/9/2024, 9:15 PM	0 Bytes	csv
qsh drwxr-xr-x	7/31/2025, 2:26 AM	--	folder
pysmi_dependence drwxrwxr-x	10/14/2024, 1:14 AM	--	folder
miniconda3 drwxrwxr-x	7/14/2025, 9:54 PM	--	folder

这是一个其中一个服务器的账号的文件夹，可以像这里面其他人一样创建一个你的文件夹，把你的代码都放到你的文件夹里。

然后，先创建一个screen，简单说就是为了让你关了连接服务器的软件，你的代码也可以一直运行。

```
(base) wzy@X640-G30:~$ screen -S 123
```

之后在你的screen中选择能跑你代码的环境。

```
(base) wzy@X640-G30:~$ conda env list
# conda environments:
#
base                  *  /public/wzy/miniconda3
jty_py38              /public/wzy/miniconda3/envs/jty_py38
tensor_1.14          /public/wzy/miniconda3/envs/tensor_1.14
torch_py39            /public/wzy/miniconda3/envs/torch_py39
wzy_py38              /public/wzy/miniconda3/envs/wzy_py38

(base) wzy@X640-G30:~$ conda activate jty_py38
(jty_py38) wzy@X640-G30:~$
```

具体你的代码要用哪个环境，到时候问师兄师姐吧。

下面就可以找到要跑代码的路径去跑了，下面举个例子。

```
(jty_py38) wzy@X640-G30:~$ cd qsh/
(jty_py38) wzy@X640-G30:~/qsh$ ls
DBGME-main  DBGME-main-1  DL-MIA-KDD-2022-main  G  test
(jty_py38) wzy@X640-G30:~/qsh$ cd test/
(jty_py38) wzy@X640-G30:~/qsh/test$ python test.py
```

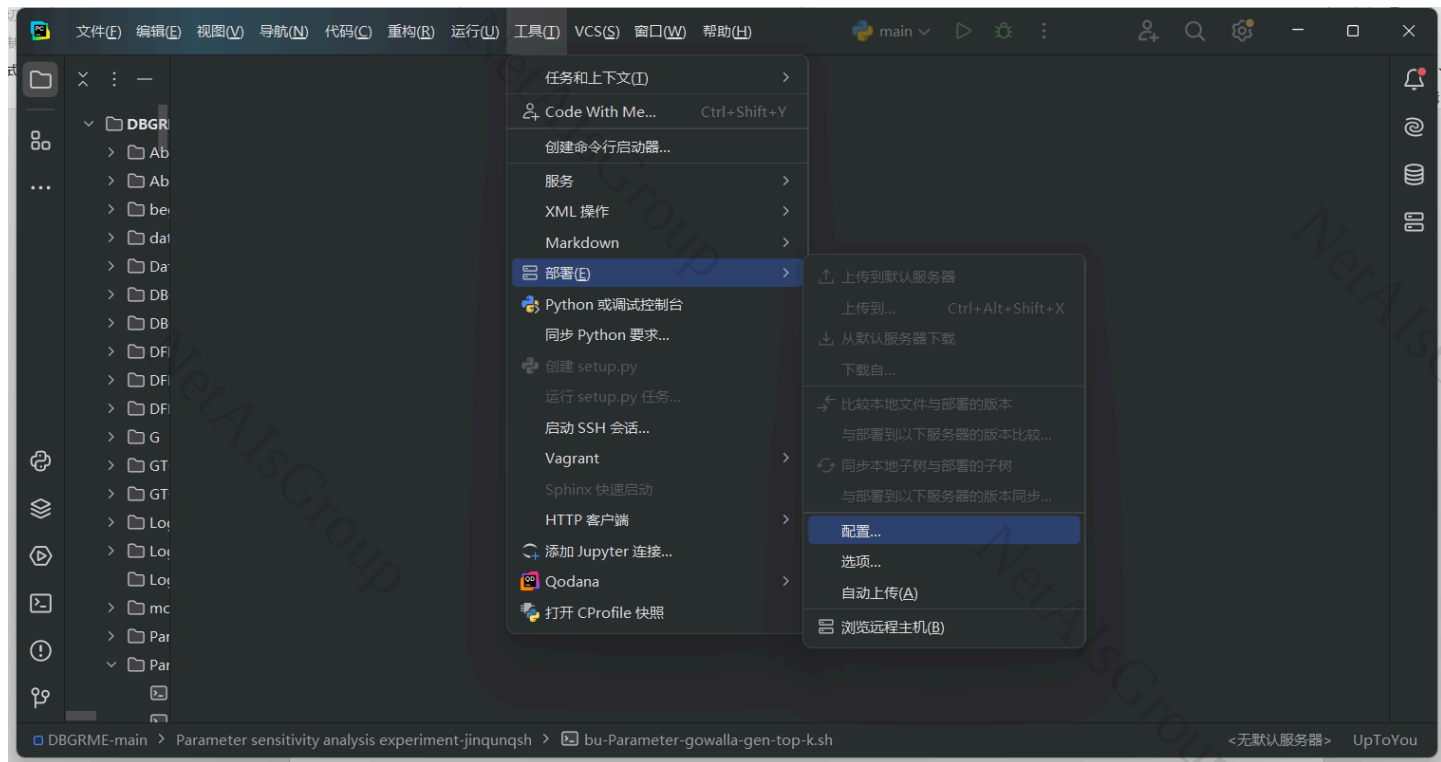
是.py文件就使用python指令，是.sh文件就是用sh 指令。这就是一个使用服务器跑代码的简单过程。

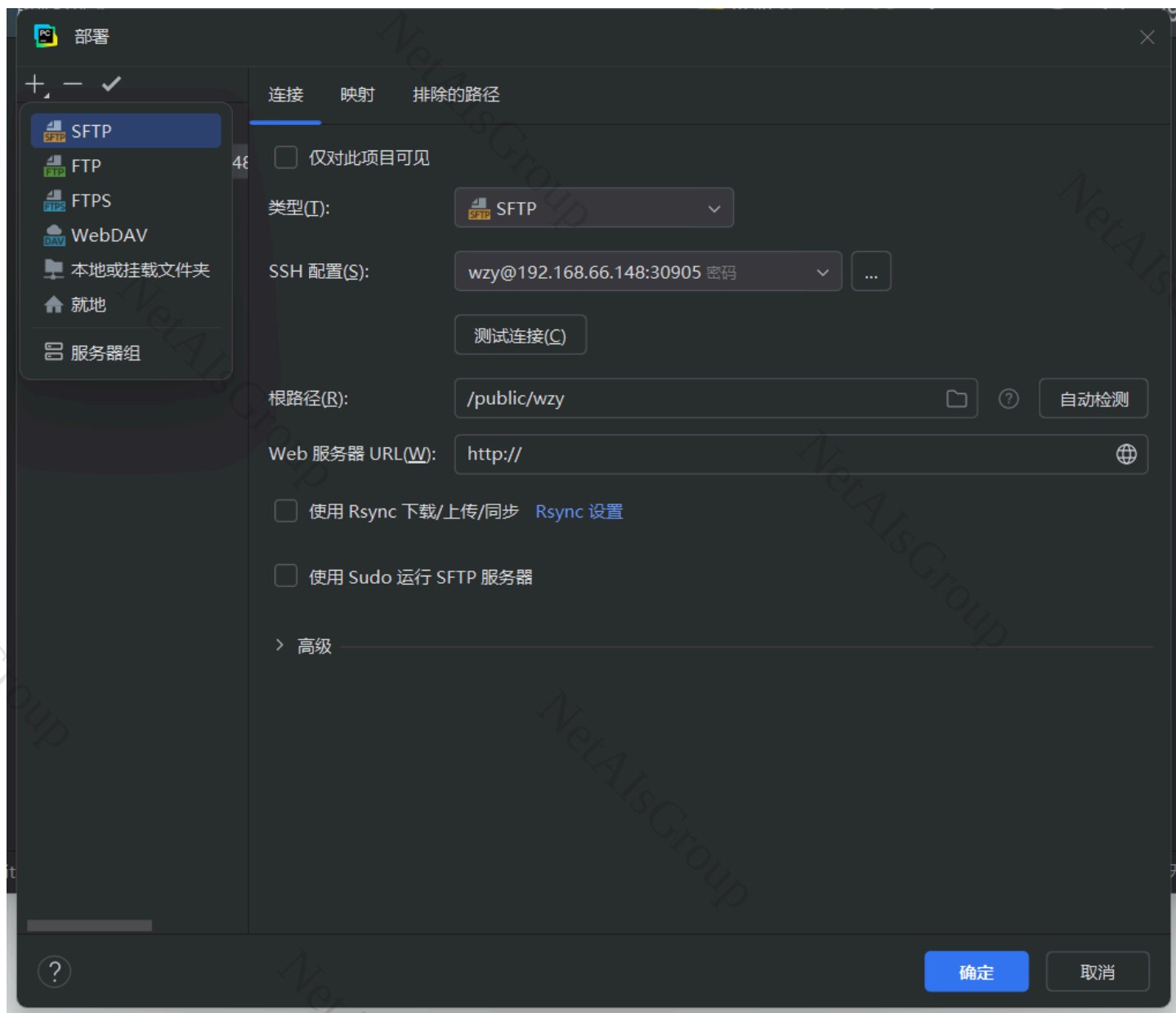
2. 使用pycharm连接服务器进行debug

在我们编写代码时，经常要使用debug对代码进行调试，但我们用到的代码都会需要很大的计算量，所以使用自己的电脑进行debug是不太行的，这时候就可以借助服务器进行debug。下面是pycharm连接服务器和使用服务器debug的教程。

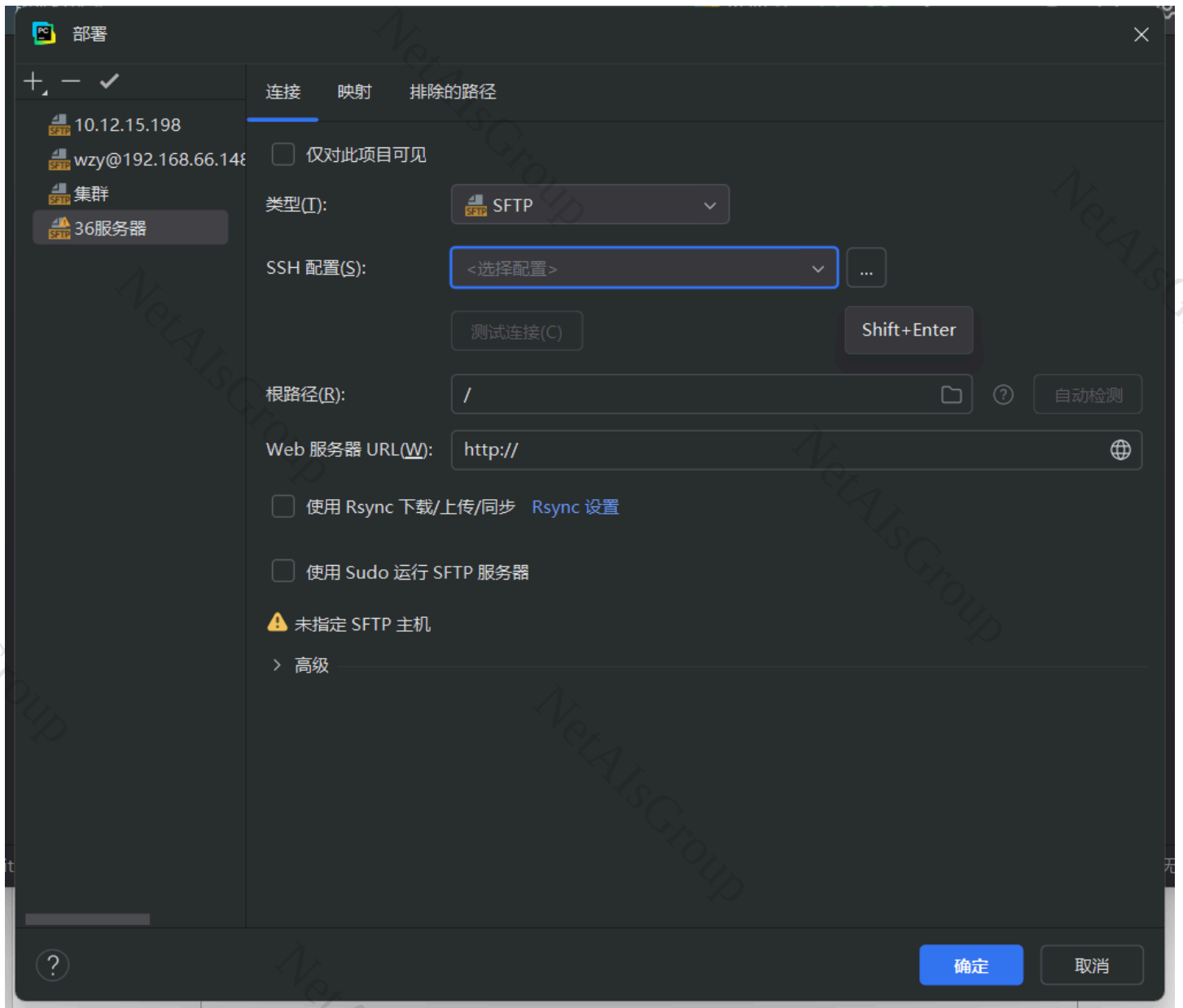
2.1 pycharm连接服务器nvidia-

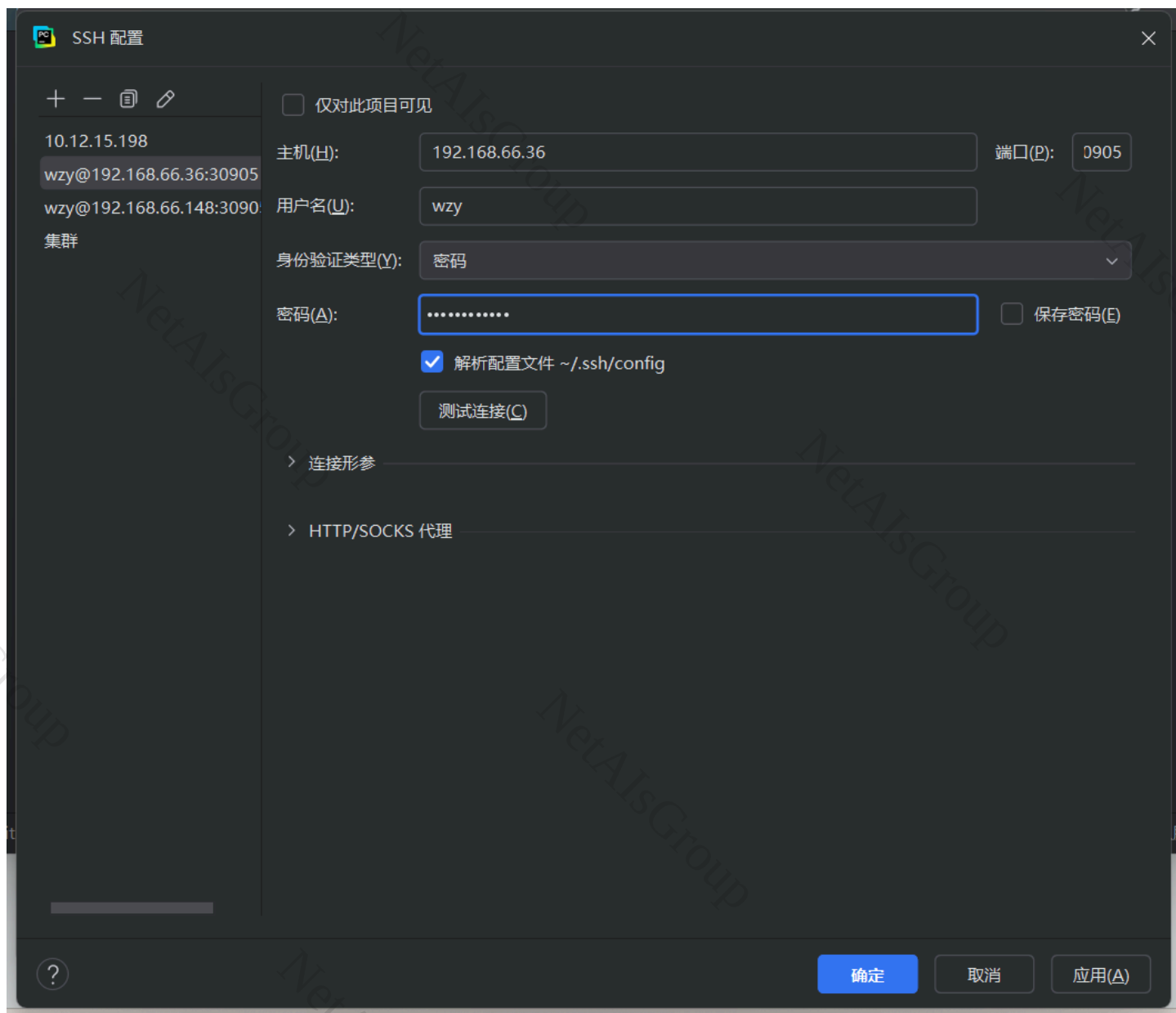
这个其实和pycharm 连接集群是一样的过程。这里我举个例子。





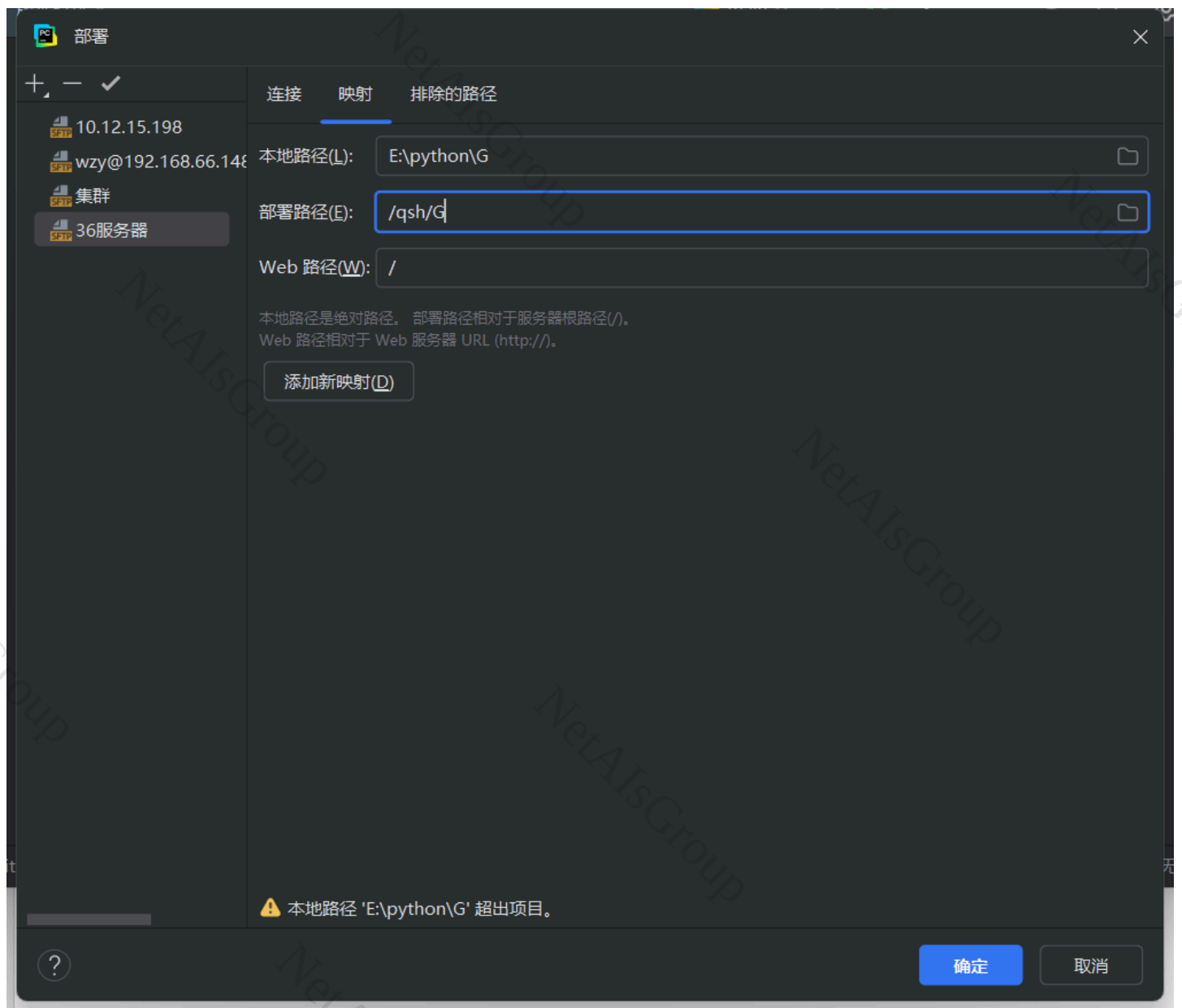
然后选择配置，没有配置要先创建一个配置。





配置输入你想要连接的服务器的ip、端口号和账号密码就好了。

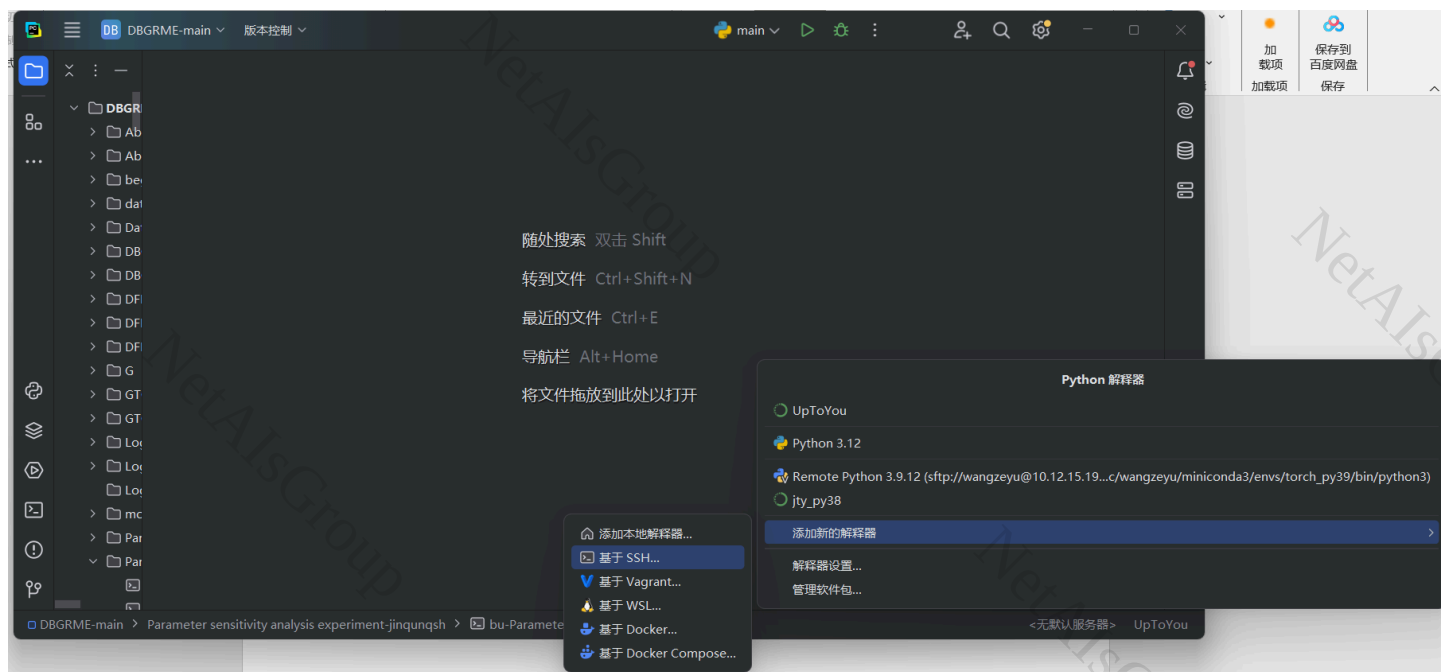
配置之后选择你的根目录，自动检测一下就好了，也就是你使用的这个账号在服务器文件的目录。之后选择映射路径，就是你当前pycharm中打开的项目在服务器中对应的文件路径。



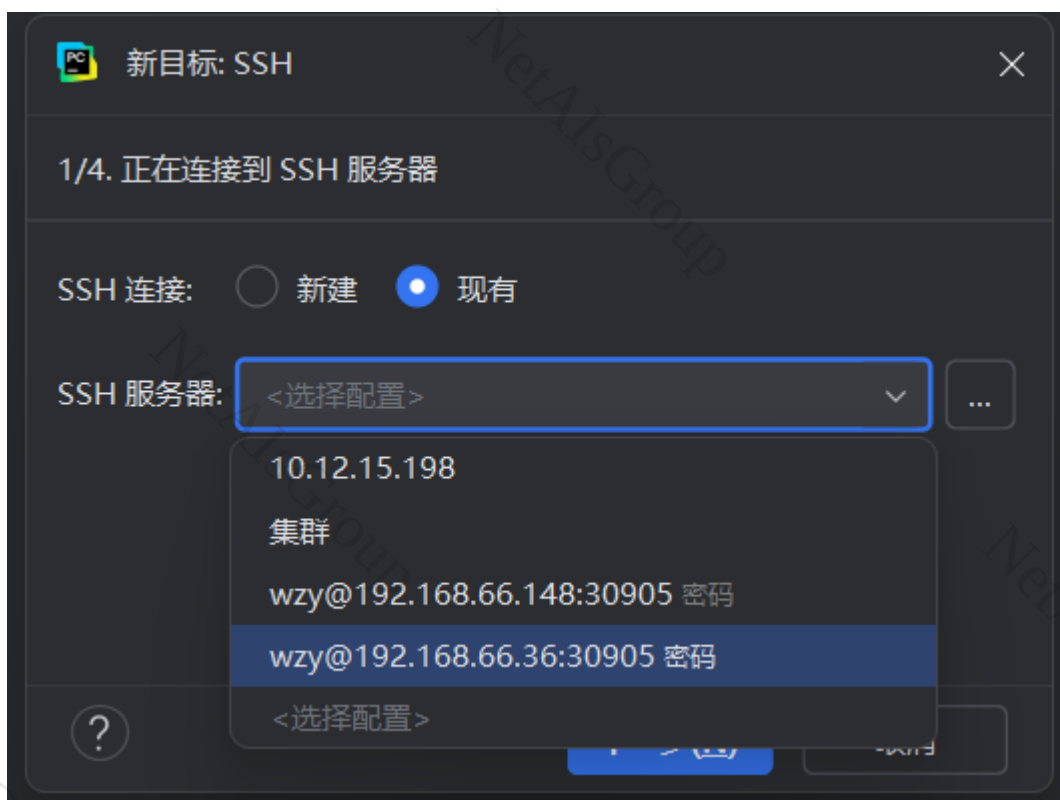
确定就连接完成了，之后就可以直接在pycharm中将打开的本地项目上传到对应的服务器路径项目文件中，也可以在pycharm中打开服务器中的代码文件进行修改之类的。

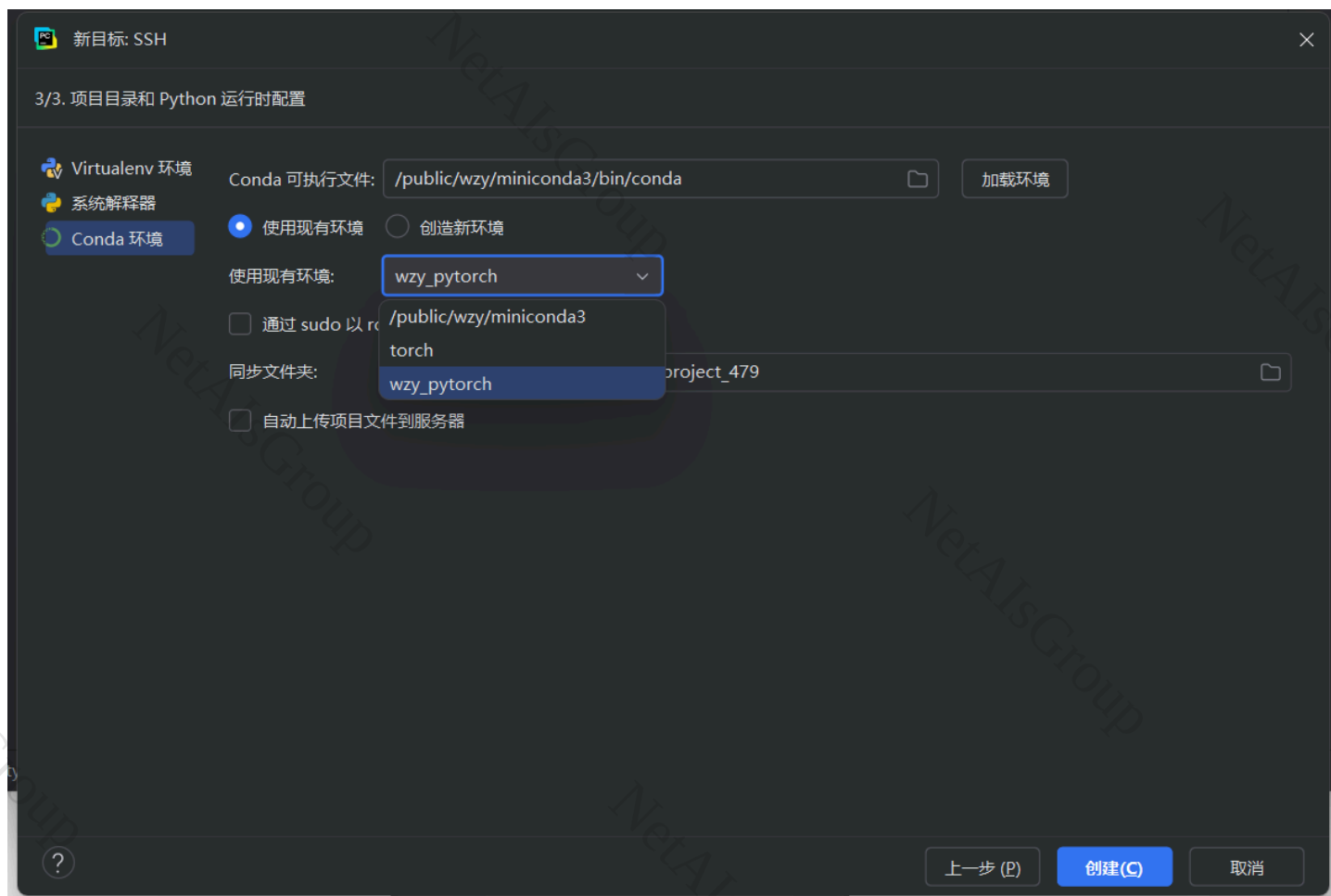
2.2 使用服务器debug

首先点击pycharm右下角，添加新的解释器，基于SSH添加。

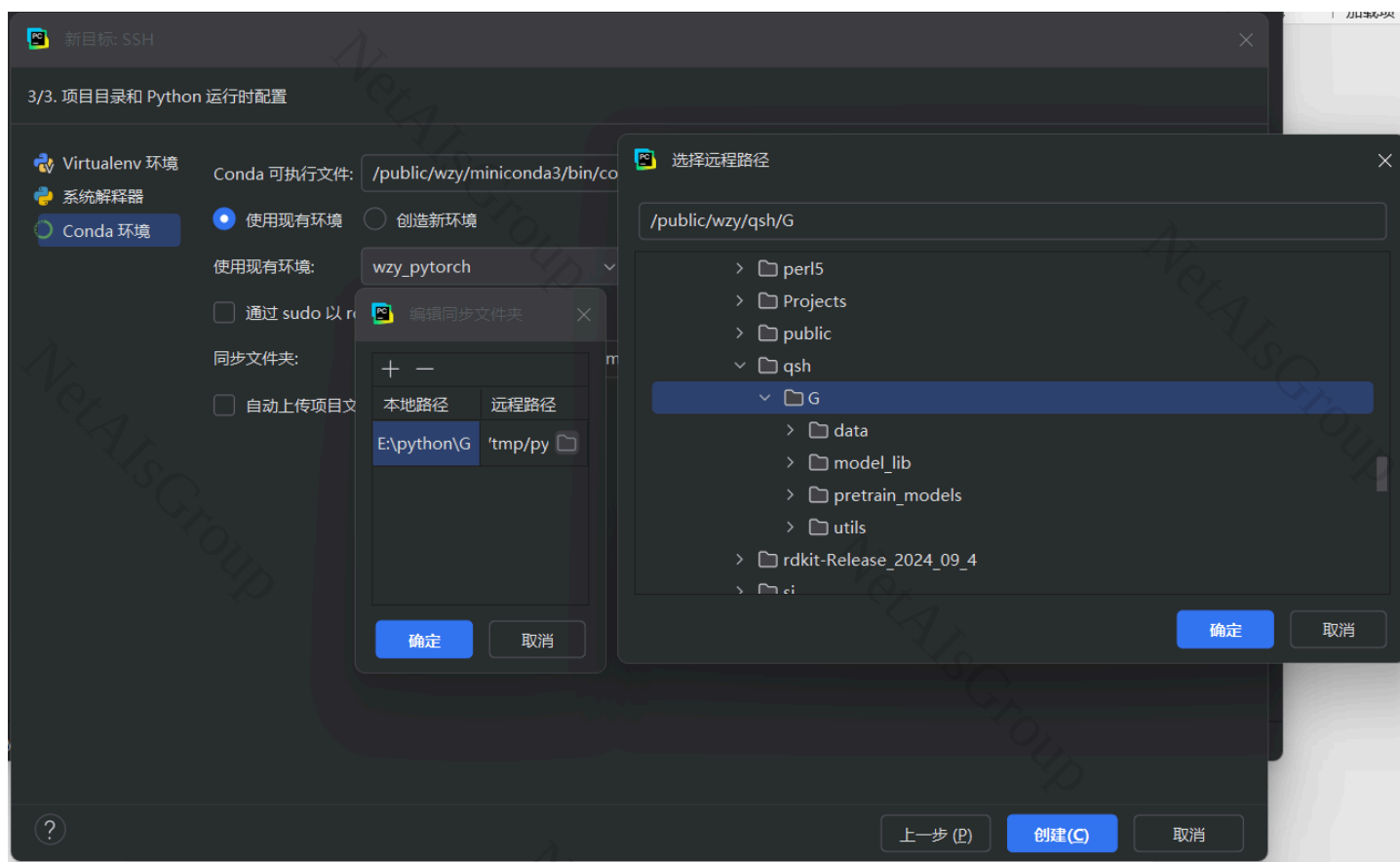


如果上一步你已经连接了服务器就可以直接选择现有的，没有连就点新建输入服务器ip、端口号和账号密码。之后就选择要使用的环境。





然后同步本地文件和对应服务器中文件的路径，这里多说一点，路径是一定要对的上的，而且两边的文件要都一致的，就是你本地改了，服务器对应的文件也要同步一下。



确定之后，等pycharm自己更新完解释器，你就可以直接对你的代码进行debug了，这样使用的就是服务器的计算资源。

科研工具篇

国内外大模型

1. Deepseek: <https://chat.deepseek.com>
2. ChatGPT(需要梯子):<https://chatgpt.com>
3. Grok(需要梯子):<https://grok.com/>
4. Kimi:<https://www.kimi.com/kimiplus-square>
5. 通义千问: <https://www.tongyi.com/qianwen>
6. Aminer(看文献很方便):<https://www.aminer.cn>

翻译

1. 小绿鲸英文文献阅读器: <https://www.xljsci.com/index.html>
2. Zotero: <https://www.zotero.org/download>
3. 一译网页: <https://yiyibooks.cn>

整理笔记

Typora: <https://typoraio.cn>

公式识别

Simpletex: https://simpletex.cn/ai/latex_ocr

Chrome插件

1. 火山翻译
2. EasyScholar (方便看期刊的水平)

推荐顶会篇

这部分整理了推荐系统及相关的会议，推荐系统的论文一般会发到推荐系统 (RecSys)、数据挖掘 (SIGKDD / SIGMOD / ICDM / WSDM / CIKM / SIGIR)、人工智能综合 (AAAI / IJCAI) 及交叉综合 (WWW) 等方向的几个会议。

此外，机器学习 (NeurIPS/ICML/ICLR)、计算机视觉 (CVPR/ICCV/ECCV)、自然语言处理 (ACL/EMNLP/NAACL)等基础任务的研究和实践通常会对搜广推有较大的借鉴意义，信息安全（S&P、USENIX Security、CCS）中的安全性与鲁棒性研究，对推荐系统领域也具有重要参考价值。搜广推领域的很多论文，其思想都是源于这些研究，如果有精力，浏览一下这些论文对于开拓思路大有裨益。

领域	会议简称	CCF 评级	清华评级	会议全称	链接
机器学习	NeurIPS	A	A	Annual Conference on Neural Information Processing Systems	https://neurips.cc/
	ICML	A	A	International Conference on Machine Learning	https://icml.cc/
	ICLR	A	A	International Conference on Learning Representations	https://iclr.cc/
数据挖掘	SIGKDD	A	A	ACM Knowledge Discovery and Data Mining	https://www.kdd.org/
	SIGMOD	A	A	ACM Conference on Management of Data	https://sigmod.org/
	ICDM	B	B	International Conference on Data Mining	https://www.ic-icdm.org/
	ICDE	A	A	IEEE International Conference on Data Engineering	https://www.icde.org/
	WSDM	B	A	ACM International Conference on Web Search and Data Mining	https://www.wsdm-conference.org/2025/
	CIKM	B	B	ACM International Conference on Information and Knowledge Management	https://cikm2025.org/
	SIGIR	A	A	International Conference on Research on Development in Information Retrieval	https://sigir2025.dei.unipd.it/
人工智能综合	AAAI	A	A	AAAI Conference on Artificial Intelligence	https://aaai.org/
	IJCAI	A	B	International Joint Conference on Artificial Intelligence	https://www.ijcai.org/
		B	B		https://recsys.acm.org/

推荐系统	RecSys			Conference on Recommender Systems	
交叉/综合	WWW	A	A	International World Wide Web Conferences	https://www2025.thewebconf.org/
计算机视觉	CVPR	A	A	IEEE Conference on Computer Vision and Pattern Recognition	https://cvpr.thecvf.com/
	ICCV	A	A	International Conference on Computer Vision	https://iccv.thecvf.com/
	ECCV	B	A	European Conference on Computer Vision	https://eccv.ecva.net/
	BMVC	C	B	British Machine Vision Conference	https://www.bmva.org/bmvc
自然语言处理	ACL	A	A	Annual Meeting of the Association for Computational Linguistics	https://2025.aclweb.org/
	EMNLP	B	A	Conference on Empirical Methods in Natural Language Processing	https://2025.emnlp.org/
	NAACL	C	B	The Annual Conference of the North American Chapter of the Association for Computational Linguistics	https://naacl.org/
信息安全	CCS	A	A	ACM Conference on Computer and Communications Security	https://www.sigsec.org/ccs.html
	USENIX Security	A	A	USENIX Security symposium	https://www.usenix.org/conferences
	S&P	A	A	IEEE Symposium on Security and Privacy	https://www.ieee-security.org/TC/SP-Index.html

生信顶会篇

如何迅速找到顶会论文?

进入这个网址: <https://github.com/naganandy/graph-based-deep-learning-literature>

GitHub - naganandy/graph-based-deep-learning-lit...

https://github.com/naganandy/graph-based-deep-learning-lit...

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graph-based-deep-learning-literature

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LICENSE.md Rename LICENSE t... 5 years ago

README.md Update README... 2 weeks ago

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Graph-based Deep Learning Literature

The repository primarily contains [links to conference publications](#) in graph-based deep learning.

The repository also contains links to:

links to conference publications in graph-based deep learning

deep-learning

graph

neural-networks

graph-convolutional-networks

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Learning On Graphs Conference - 2024 | 2023 | 2022

Machine Learning Conferences

NeurIPS - 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018 | 2017 | 2016 | 2015

ICML - 2025 | 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018 | 2017 | 2016

ICLR - 2025 | 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018 | 2017 | 2016 | 2014

Data Mining Conferences

KDD - 2025 (V.1) | 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018

ICDM - 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018

No packages published

Contributors 11

Languages

Jupyter Notebook 100.0%

在该网址中，收录了NeurIPS、ICML、ICLR、KDD等顶会近几年的文章

以NeurIPS-2024为例：



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2658807 · 3 months ago

579 lines (356 loc) · 69.8 KB

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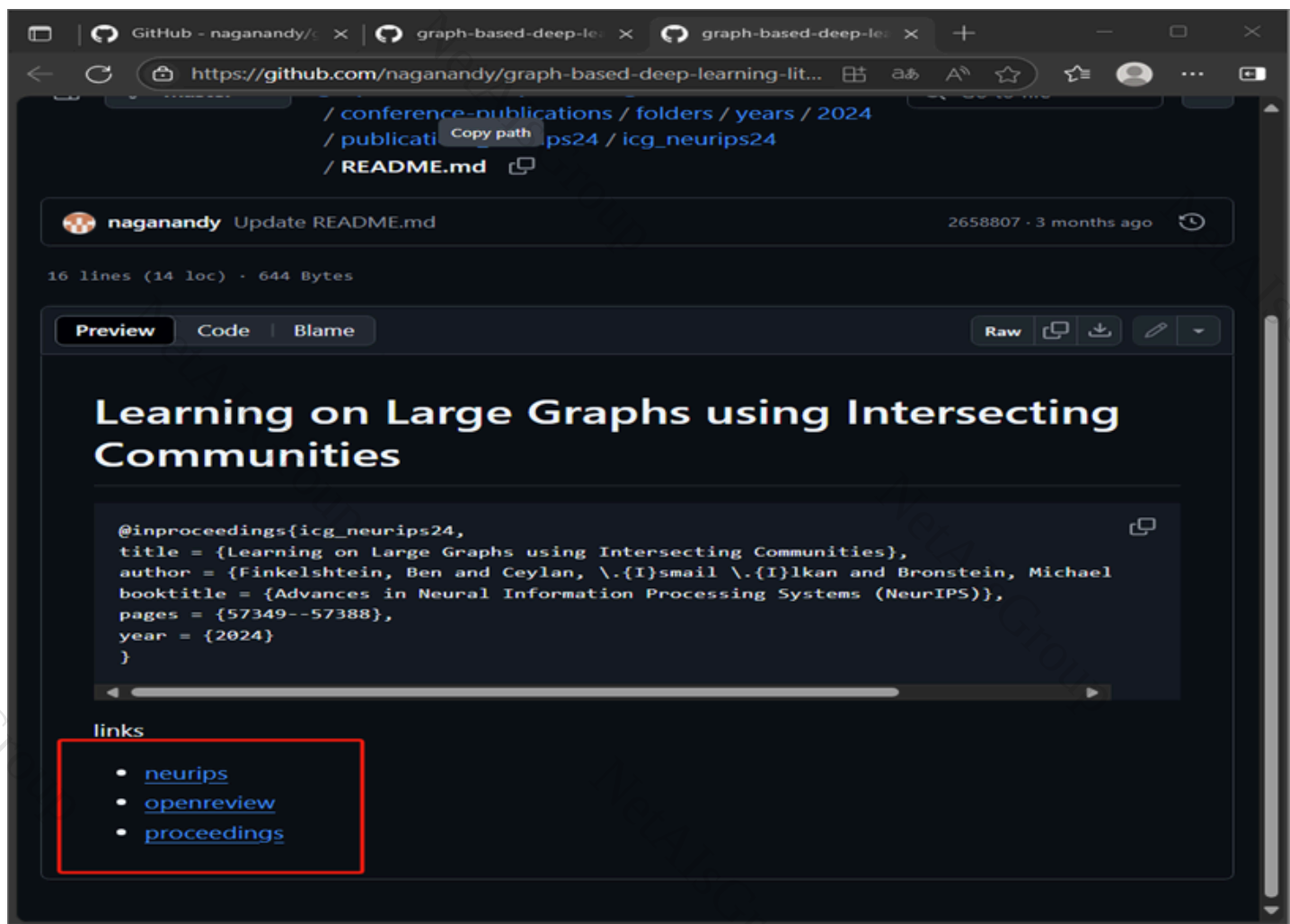




Publications in NeurIPS 2024

Scalability

- Learning on Large Graphs using Intersecting Communities
- The Importance of Being Scalable: Improving the Speed and Accuracy of Neural Network Interatomic Potentials Across Chemical Domains
- DFA-GNN: Forward Learning of Graph Neural Networks by Direct Feedback Alignment



生信方向的顶刊：BIB、JMC、JBHI（nature之类加上其子刊的不必多说）

Tips: 期刊的话，直接期刊里找相关论文方便点；如果你想找的顶会不在上述网址内，也可以直接进入顶会官网去找；组会上读顶会文章好一点，期刊论文可以自己看。

另外：ctrl+f可以帮助你官网找论文时搜索关键字，应用如下：

435	12:30pm-2:30pm	12812	Using an Optimal MIP-Based Ensemble Weighting Approach	Michael
436	Friday February 28, 12:30pm-2:30pm	4855	Dynamic Expansion Diffusion Learning For Lifelong Generative Modelling	Fei Ye, Adrian G. Bors, Kun Zhang
437	Friday February 28, 12:30pm-2:30pm	5773	Improving Private Random Forest Prediction using Matrix Representation	Arisa Tajima, Joie Wu, Amir Houmansadr
438	Friday February 28, 12:30pm-2:30pm	15127	Equal Merit Does Not Imply Equality: Discrimination at Equilibrium in a Hiring market with Symmetric Agents	Serafina Kamp, Benjamin Fish
439	Friday February 28, 12:30pm-2:30pm	15940	Improved Rates of Differentially Private Nonconvex-Strongly-Concave Minimax Optimization	Ruijia Zhang, Mingxi Lei, Meng Ding, Zihang Xiang, Jinhui Xu, Di Wang
440	Friday February 28, 12:30pm-2:30pm	11070	Evolutionary Large Language Model for Automated Feature Transformation	Nanxu Gong, Chandan K. Reddy, Wangyang Ying, Haifeng Chen, Yanjie Fu
521	Friday February 28, 12:30pm-2:30pm	10496	Heterogeneous Graph Neural Network on Semantic Tree	Mingyu Guan, Jack W Stokes, Qinlong Luo, Fuchen Liu, Purvanshi Mehta, Elnaz Nouri, Taesoo Kim
522	Friday February 28, 12:30pm-2:30pm	10847	MORE: Molecule Pretraining with Multi-Level Pretext Task	Yeongyeon Son, Dasom Noh, Gyoungyoung Heo, Gyoung Jin Park, Sunyoung Kwon
523	Friday February 28, 12:30pm-2:30pm	12682	Expressive Power of Temporal Message Passing	Przemysław Andrzej Wałęga, Michael Rawson

524	Friday February 28, 12:30pm-2:30pm	13721	Pre-training Graph Neural Networks on Molecules by using Subgraph-conditioned Graph Information Bottleneck	Van Thuy Hoang, O-Joun Lee
525	Friday February 28, 12:30pm-2:30pm	3590	Open-set Cross-network Node Classification via Unknown-excluded Adversarial Graph Domain Alignment	Xiao Shen, Zhihao Chen, Shirui Pan, Shuang Zhou, Laurence Tianruo Yang, Xi Zhou
526	Friday February 28, 12:30pm-2:30pm	679	GLAD: Improving Latent Graph Generative Modeling with Simple Quantization	Van Khoa Nguyen, Yoann Boget, Frantzeska Lavda, Alexandros Kalousis
527	Friday February 28, 12:30pm-2:30pm	6158	Through the Dual-Prism: A Spectral Perspective on Graph Data Augmentation for Graph Classifications	Yutong Xia, Runpeng Yu, Yuxuan Liang, Xavier Bresson, Xinchao Wang, Roger Zimmermann
528	Friday February 28, 12:30pm-2:30pm	7296	On Oversquashing in Graph Neural Networks Through The Lens of Dynamical Systems	Alessio Gravina, Moshe Eliasof, Claudio Gallicchio, Davide Bacciu, Carola-Bibiane Schöntlieb
529	Friday February 28, 12:30pm-2:30pm	8771	Crossfire: An Elastic Defense Framework for Graph Neural Networks under Bit Flip Attacks	Lorenz Kummer, Samir Moustafa, Wilfried N. Gansterer, Nils Morten Kriege
530	Friday February 28, 12:30pm-2:30pm	1749	HeterGP: Bridging Heterogeneity in Graph Neural Networks with Multi-View Prompting	Di Jin, Fengyu Yan, Xiaobao Wang, Dongxiao He, Longbiao Wang, Jianwu Dang
531	Friday February 28, 12:30pm-2:30pm	7915	Beyond Homophily: Graph Contrastive Learning with Macro-Micro Message Passing	Yiyuan Chen, Donghai Guan, Weiwei Yuan, Tianzi Zang
532	Friday February 28, 12:30pm-2:30pm	12064	GraSP: Simple yet Effective Graph Similarity Predictions	Haoran Zheng, Jieming Shi, Renchi Yang
533	Friday February 28, 12:30pm-2:30pm	5712	Inverse Reinforcement Learning by Estimating Expertise of Demonstrators	Mark Belliaev, Ramtin Pedarsani
534	Friday February 28, 12:30pm-2:30pm	10732	Markov Balance Satisfaction Improves Performance in Strictly Batch Offline Imitation Learning	Rishabh Agrawal, Nathan Dahlin, Rahul Jain, Ashutosh Nayyar
591	Friday February 28, 12:30pm-2:30pm	15089	Data Augmentation for Instruction Following Policies via Trajectory Segmentation	Niklas Hoepner, Ilaria Tiddi, Herke van Hooft
	Friday February 28,		MimiQ: Low-Bit Data-Free Quantization of Vision Transformers with Encouraging Inter-Head Attention	Kanghyun Choi, Hyeyoon Lee, Dain Kwon, Sunjong Park, Kyuyeun Kim, Noseong Park, Jonghyun

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期刊名: ISSN: 请完整填写 研究方向: 影响因子: SCI收录: 不限 大类学科: 不限 小类学科: 不限 中国科学院分区: 不限 是否OA期刊: 不限 结果排序: 相关性 查询

近期推荐: 全流程投稿协助套餐服务 SCI论文AI润色+人工QC服务 Springer Nature 特刊征稿 已发表SCI? 是时候来Springer出书了!

同行评审状态查询 Manuscript Number * Last Name * First Name * 出版社: Elsevier 查询 重置

📄 ⚡ 投稿难? 全流程投稿协助套餐, 直达Accept! 点此了解详情 →

十大热门期刊: [查看热门领域] [查看全部领域]

Nature Communications 中国科学院分区大类: 综合性期刊 1区 中国科学院分区小类: 综合性期刊 1区

Journal of Alloys and Compounds 中国科学院分区大类: 材料科学 2区 中国科学院分区小类: 冶金工程 2区 中国科学院分区小类: 材料科学: 综合 3区

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Renewable Energy LetPub评分 本刊评论 期刊详情

三 评论分类 全部评论

怎么寻找专利呢?

进入如下网站: <https://pss-system.cponline.cnipa.gov.cn/conventionalSearch>

专利检索及分析 | 常规检索
Patent Search and Analysis

自动识别 检索要素 申请号 公开号 申请人 发明人 发明名称

数据范围 分子性质预测

检索

检索模式：自动识别

1. 支持二目逻辑运算符and、or。
2. 多个检索词之间用空格间隔，如：智能 手机。
3. 系统默认二目逻辑运算符是and，如输入“智能 手机”，系统按照“智能 and 手机”进行检索。
4. 日期支持间隔符“-”、“.”，支持如下格式：YYYY-MM-DD、YYYY.MM.DD、YYYYMMDD、YYYYMM、YYYY。
5. 支持半角0算符，如输入国产 (智能 手机)，系统优先执行“智能 AND 手机”，然后将所得结果集与“国产”进行AND运算。
6. 如果检索条件中包含空格、保留关键字或运算符，需使用半角双引号，如：“WILLIAMS AND LANE INC”。
7. 中国专利号长度12位或14位可以正常检索，例如201920517482.6或201920517482；如果长度是13位，需删除最后一位检索，否则会检索不到数据，例如2019205174826，请输入201920517482检索。



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专利类型：发明 实用新型 外观设计 有效专利：有效 无效

申请日：开始日期 - 结束日期 公开日：开始日期 - 结束日期 授权日：开始日期 - 结束日期

确定 重置

检索结果统计 127 条数据

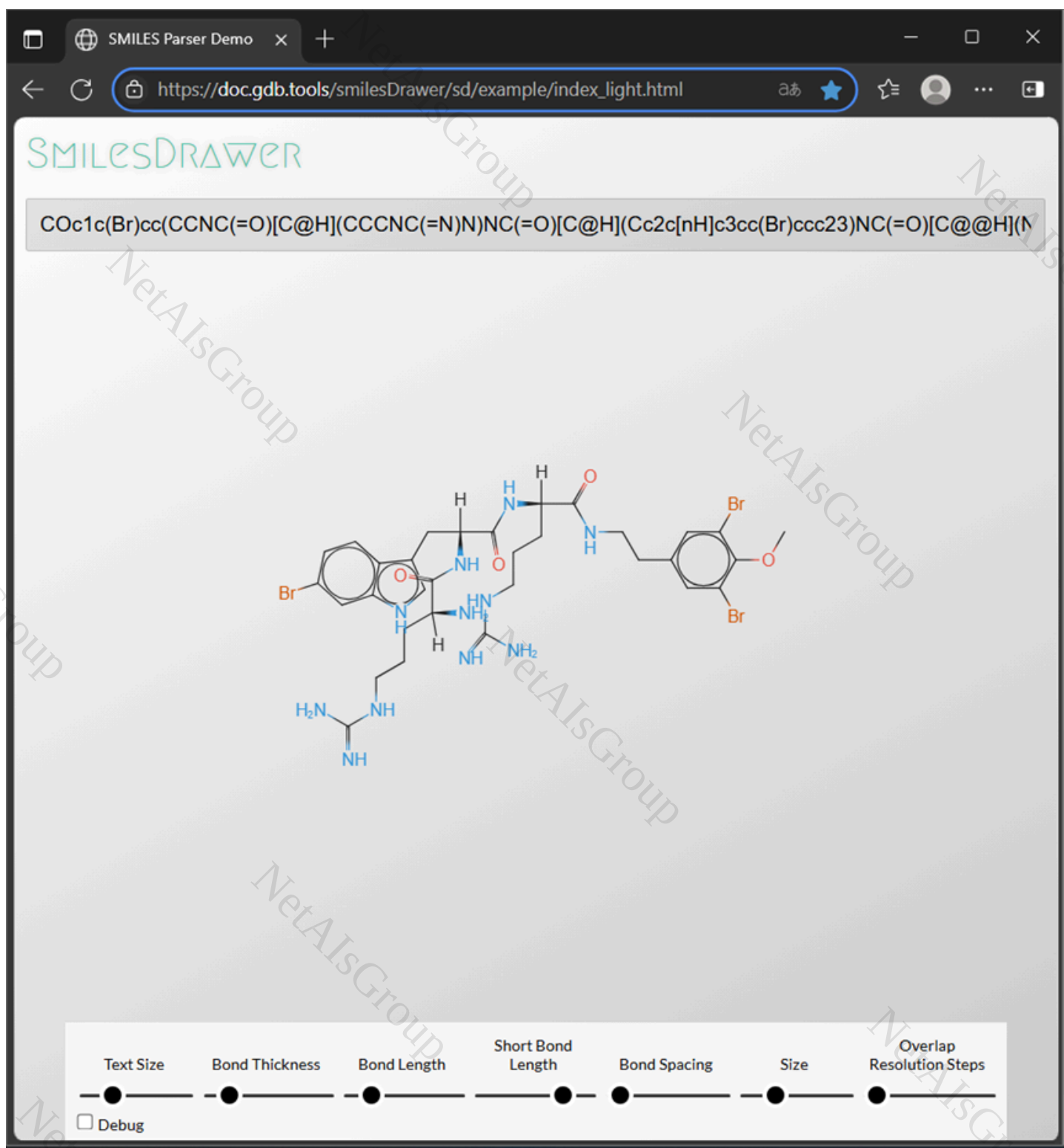
预览 批量收藏 加入批量下载库 加入分析库 跟踪 打印

图文 列表 多图 申请日排序

公开号	申请号	申请日	发明名称	申请人
☐ CN120183541A	CN202510653686.2	2025.05.21	多模态分子性质预测方法、装...	上海张江数学研究院
摘要 主权权利要求 著录项目 IPC分类 CPC分类 法律状态 同族 引证 被引证 本申请涉及一种多模态分子性质预测方法、装置、设备、存储介质和计算机程序产品。所述方法包括：将目标数据集输入分子表示学习模型，得到目标分子的一维序列特征；将所述目标数据集输入消息传递神经网络模型，得到所述目标分子的二维图像特征；根据所述一维序列特征以及所述二维图像特征进行特征融合并对所述目标分子进行药物性质预测，输出所述目标分子的药性性质预测结果，所述药性性质预测结果包括所述目标分子的可视化的多个基团的注意力权重结果，所述注意力权重结果描述对应所述基团对所述药性性质预测结果的贡献程度。采用本方法能够提高分子性质预测的预测精度和预测可靠性。 预览 收藏 加入批量下载库 加入分析库 跟踪 打印				
☐ CN120032755A	CN202510512021...	2025.04.23	一种星际化合物分子性质预测...	之江实验室
摘要 主权权利要求 著录项目 IPC分类 CPC分类 法律状态 同族 引证 被引证 本申请涉及一种星际化合物分子性质预测方法、装置、设备和介质，其中，星际化合物分子性质预测方法包括：通过获取星际分子的结构表征信息；基于所述星际分子的结构表征信息与所述星际分子的实际分子性质信息，对模型进行训练，获得性质预测模型；将待预测星际分子的结构表征信息输入至所述性质预测模型中，获得所述待预测星际分子的性质，本申请通过构建星际化合物分子结构与性质信息关系的模型，可以高效准确的预测星际化合物分子的性质。 预览 收藏 加入批量下载库 加入分析库 跟踪 打印				

生信方向的一个实用网站：分子SMILES式的可视化

https://doc.gdb.tools/smilesDrawer/sd/example/index_light.html



十一、配置环境时遇到的bug

1.问题：CentOS 7 yum无法使用解决方法Could not retrieve mirrorlist
<http://mirrorlist.centos.org/?release=7&arch=>

解决方案：

<https://www.cnblogs.com/kohler21/p/18331060#:~:text=%E5%87%BA%E7%8E%B0%E8%BF%99%E4%B8%AA%E9%94%99%E8%AF%AF%E6%98%AF%E5%9B%A0%E4%B8%BA%E4%BD%BF%E7%94%A8%E7%9A%84%20CentOS%207%20%E4%BB%93%E5%BA%93%E5%B7%B2%E7%>

BB%8F%E8%A2%AB%E5%BD%92%E6%A1%A3%EF%BC%8C%E5%BD%93%E5%89%8D%E7%9A%84%E9%95%9C%E5%83%8F%E5%9C%B0%E5%9D%80%E6%97%A0%E6%B3%95%E6%89%BE%E5%88%B0%E6%89%80%E9%9C%80%E7%9A%84%E6%96%87%E4%BB%B6%E3%80%82%20CentOS%207%20%E7%9A%84%E5%AE%98%E6%96%B9%E6%94%AF%E6%8C%81%E5%B7%B2%E7%BB%8F%E7%BB%93%E6%9D%9F%EF%BC%8C%E9%83%A8%E5%88%86%E4%BB%93%E5%BA%93%E5%B7%B2%E8%A2%AB%E7%A7%BB%E8%87%B3%E5%BD%92%E6%A1%A3%E5%BA%93%E3%80%82%20%E8%BF%99%E5%AF%BC%E8%87%B4%E4%BA%86%E4%BD%A0%E7%9A%84,%E6%97%A5%E4%B9%8B%E5%90%8E%E5%B7%B2%E7%BB%8F%E5%81%9C%E6%AD%A2%E7%BB%B4%E6%8A%A4%E3%80%82%20%E5%9B%A0%E6%AD%A4%EF%BC%8C%E4%BD%BF%E7%94%A8%E6%9C%80%E6%96%B0%E7%9A%84%20CentOS%207%20%E5%AE%98%E6%96%B9%E4%BB%93%E5%BA%93%E5%8F%AF%E8%83%BD%E4%BC%9A%E9%81%87%E5%88%B0%E9%97%AE%E9%A2%98%E3%80%82%20%E4%B9%8B%E5%90%8E%E4%BE%9D%E6%AC%A1%E6%89%A7%E8%A1%8C%EF%BC%9A%20vi%20CentOS-Base.repo

2.问题： Docker运行hello-world镜像失败或超时

https://blog.csdn.net/m0_51193794/article/details/141396742